

COMPREHENSIVE REVIEW

Recent advances in examining the factors influencing the efficacy of biocides against *Listeria monocytogenes* biofilms in the food industry: A systematic review

Michael Arthur^{1,2}  | Edmund Larbi Afari³  | Elena-Alexandra Alexa² |
 Mei-Jun Zhu³ | Michael T. Gaffney⁴ | Jesus Maria Frias Celayeta⁵  |
 Catherine M. Burgess¹ 

¹Food Safety Department, Teagasc Food Research Centre, Ashtown, Dublin, Ireland

²School of Food Science and Environmental Health, Technological University Dublin, Dublin, Ireland

³School of Food Science, Washington State University, Pullman, Washington, USA

⁴Horticulture Development Department, Teagasc Food Research Centre, Ashtown, Dublin, Ireland

⁵Sustainability and Health Research Hub, Technological University Dublin, Dublin, Ireland

Correspondence

Catherine M. Burgess, Food Safety Department, Teagasc Food Research Centre, Ashtown, Dublin D15 DY05, Ireland.

Email: kaye.burgess@teagasc.ie

Funding information

Teagasc; Department of Agriculture, Food and the Marine, Ireland

Abstract

Controlling *Listeria monocytogenes* and its associated biofilms in the food industry requires various disinfection techniques, including physical, chemical, and biological treatments. Biocides, owing to their ease of use, cost-effectiveness, dissolvability in water, and efficacy against a wide range of microorganisms, are frequently selected options. Nonetheless, concerns have been raised about their efficacy in controlling *L. monocytogenes* biofilm, as laboratory-based and commercial studies have reported the persistence of this bacterium after cleaning and disinfection. This review systematically examined scientific studies, sourced from the Web of Science, Scopus, and PubMed databases between January 2010 and May 2024, that investigated the effectiveness of the most commonly used biocides in the food industry against *L. monocytogenes* biofilms. A total of 92 articles which met the screening criteria, were included, with studies utilizing biocides containing sodium hypochlorite, quaternary ammonium compounds, and peroxyacetic acid being predominant. Studies indicated that several key factors may potentially influence biocides' efficacy against *L. monocytogenes* biofilms. These factors included strain type (persistent, sporadic), serotype, strain origin (clinical, environmental, or food), surface type (biotic or abiotic), surface material (stainless steel, polystyrene, etc.), incubation time (biofilm age) and temperature, presence of organic matter, biocide's active agent, and the co-culture of *L. monocytogenes* with other bacteria. The induction of the viable but nonculturable (VBNC) state following disinfection is also a critical concern. This review aims to provide a global understanding of how *L. monocytogenes* biofilms respond to biocides under different treatment conditions, facilitating the development of effective cleaning and disinfection strategies in the food industry.

This is an open access article under the terms of the [Creative Commons Attribution-NonCommercial-NoDerivs](https://creativecommons.org/licenses/by-nc-nd/4.0/) License, which permits use and distribution in any medium, provided the original work is properly cited, the use is non-commercial and no modifications or adaptations are made.

© 2024 The Author(s). *Comprehensive Reviews in Food Science and Food Safety* published by Wiley Periodicals LLC on behalf of Institute of Food Technologists.

KEYWORDS

biocide, disinfection, efficacy, *Listeria monocytogenes* biofilm, VBNC

1 | INTRODUCTION

The increase in foodborne diseases has a significant global impact on both public health and economies (Havelaar et al., 2015). *Listeria monocytogenes* is one of the most important foodborne pathogens that pose risks to human health due to its virulence, high mortality rate, propensity for cross-contamination during processing, and its ability to grow at refrigeration temperatures (Camargo et al., 2017; Possas et al., 2022; Santos et al., 2019). It has been detected in a variety of ready-to-eat (RTE) finished products, including milk and dairy products, meat and poultry products, fresh produce, and seafood (Barroso et al., 2019; Buchanan et al., 2017; McLauchlin et al., 2020; Stessl et al., 2022). *L. monocytogenes* causes listeriosis, a rare but fatal disease that particularly affects vulnerable populations such as children, the elderly, pregnant women, and individuals with a weakened immune system (Sinclair et al., 2022; Zhang, Wang, et al., 2021). Listeriosis can lead to severe outcomes, including hospitalizations, deaths, and miscarriages (Dumitru et al., 2021). Global reports on listeriosis outbreaks indicate that the consumption of foods contaminated with *L. monocytogenes* remains the primary transmission route to humans (Aryal & Muriana, 2019; CDC, 2014; ECDC, 2022; Estrada et al., 2020; Guan et al., 2021; Stoller et al., 2019; Townsend et al., 2021; Véghová et al., 2017). For instance, between 2017 and 2018, the largest recorded listeriosis outbreak, including 1060 cases and 216 deaths in South Africa, was linked to the consumption of polony, an RTE meat product contaminated with *L. monocytogenes* (Olanya et al., 2019; Tchatchouang et al., 2020). Moreover, fresh produce such as cantaloupes, apples, frozen vegetables, salads, bean sprouts, and mushrooms, as well as processed foods including ice cream, have been implicated in multistate listeriosis outbreaks in the United States (CDC, 2012, 2014, 2015, 2016; Conrad et al., 2022; FDA, 2023). More recently, a *L. monocytogenes*-associated outbreak involving peaches, nectarines, and plums was linked to 11 cases, 10 hospitalizations, and one death in the United States (CDC, 2023). In the European Union (EU), the European Food Safety Authority (EFSA) reported that listeriosis outbreaks affected 94 people in 2021, with 38 cases hospitalized and nine deaths (EFSA & ECDC, 2021). Meat products, fruit juices, and vegetables were identified as the implicated foods in these outbreaks (EFSA & ECDC, 2021). The annual epidemiological report on listeriosis by the European Centre for Disease Prevention and Control (ECDC) reported 2268 confirmed listeriosis cases in 2021 across the 30 EU member states

and European Economic Area (EEA) (ECDC, 2022). Due to the hospitalization and mortality rates, listeriosis remains a significant concern among foodborne diseases under EU surveillance (Mazaheri et al., 2021).

Ubiquitous in the natural environment, *L. monocytogenes* has multiple avenues to infiltrate food-processing facilities, and once introduced, it exhibits remarkable resilience, persisting for extended periods, spanning from days to years. For example, over a 4-year period (2010–2014), *L. monocytogenes* was detected at a prevalence of 30.6% on floors, drains, walls, washbasins, and protective overshoes within one meat-processing facility in Slovakia (Véghová et al., 2017). This enduring presence can be attributed to the bacteria's adaptability to a wide range of temperatures, salt concentrations, pH, and osmotic stresses (Caurio et al., 2022; Mafuna et al., 2021; Zoz et al., 2017). Additionally, its ability to form biofilms on surfaces contributes significantly to prolonged persistence and stress tolerance in the food-processing environment (Doijad et al., 2015; Lee et al., 2019; Ramires et al., 2021). The extracellular polymeric substance (EPS) produced by cells within biofilms not only enhances microbial attachment to the surface and to each other but also protects the biofilm matrix against physical, chemical, and biological treatment, making complete elimination challenging (Sadekuzzaman et al., 2017). Effective removal of *L. monocytogenes* biofilms in the food industry can however be achieved through proper cleaning and disinfection procedures. Many outbreaks associated with *L. monocytogenes* have been directly linked to inadequate cleaning and disinfection practices on surfaces in the food-processing environment. For instance, Hanson et al. (2019) reported that, a listeriosis outbreak in Canada linked to the consumption of chocolate milk was caused by poor equipment design and unclean processing equipment. Similarly, inadequate cleaning and disinfection of contact surfaces were identified as potential routes of contamination of ice cream with *L. monocytogenes* in the United States between 2015 and 2016 (Lindsay et al., 2022).

Controlling *L. monocytogenes* biofilms in the food industry utilizes various disinfection techniques, including physical, chemical, and biological treatments (Hua & Zhu, 2024a). Physical treatments, such as ultraviolet (UV) light, light-emitting diodes (LEDs), ultrasonication, and dry heat, have demonstrated efficacy against *L. monocytogenes* biofilms, showing 2- to 5-log reductions (Bae & Lee, 2012; Ban & Kang, 2016; Bernbom et al., 2011; Harada & Nascimento, 2021; Tolvanen et al., 2009). Biological treatments such as bacteriophages and protective

cultures including lactic acid bacteria (LAB) have been recognized for their effectiveness in reducing *L. monocytogenes* biofilms by 2–7 log units on various surfaces, ranging from stainless steel (SS) to food products (Arachchi et al., 2013; Dinçer, 2024; Gómez et al., 2016; Montañez-Izquierdo et al., 2012; Sadekuzzaman et al., 2017). Factors such as in-process sanitation, which can reduce the bacterial load during processing (Bosilevac et al., 2024), and the long-term effectiveness of these methods may impact the overall feasibility and efficiency of cleaning protocols (Kübler et al., 2015). However, physical and biological methods may be cost-prohibitive and impractical for cleaning large-scale food-processing equipment (Hao et al., 2022). As a practical alternative, biocides are frequently used in the food industry due to their user-friendliness, cost-effectiveness, water solubility, and efficacy against a wide range of microorganisms, including *L. monocytogenes* in both planktonic and biofilm states (Dygico et al., 2021; Hua et al., 2019; Meireles et al., 2017; Skowron et al., 2019). Commercially available biocides come in various formulations, incorporating one or more active biocidal ingredients designed to target bacterial cells through mechanisms such as the disruption of cell structure, metabolic pathways, or lowering cellular pH (Donaghy et al., 2019; Pietrysiak et al., 2019). Biocides commonly used in the food industry include sodium hypochlorite (NaOCl), chlorine dioxide (ClO₂), hydrogen peroxide (H₂O₂), peroxyacetic acids (PAAs), organic acids, sodium hydroxide, ozone, and quaternary ammonium compounds (QACs), such as benzalkonium chloride (BAC) (Clayton et al., 2021; Dhowlaghar et al., 2018; Dygico et al., 2021; Meireles et al., 2016; Olszewska et al., 2016). While biocides are designed to reduce or eliminate harmful microorganisms, including *L. monocytogenes*, their residues may remain on surfaces or in the environment at sublethal concentrations (Duze et al., 2021). This can contribute to the development of biocide-tolerant bacterial strains, which may also result in an increased risk of cross-protection against certain antibiotics (Davin-Regli & Pages, 2012; Gnanadhas et al., 2013). Moreover, the residues of certain biocides such as chlorine-based disinfectants have been reported to exhibit mutagenic and carcinogenic properties (Clayton et al., 2019, 2021; Gil et al., 2015; Meireles et al., 2017), raising significant concerns for food safety and public health.

In the European Union, a biocidal product is defined as “any substance or mixture, in the form in which it is supplied to the user, consisting of, containing, or generating one or more active substances, with the intention of destroying, deterring, rendering harmless, preventing the action of, or otherwise exerting a controlling effect on, any harmful organism by any means other than mere physical or mechanical action.” It is worth noting that the term “biocide” covers a broad category encompassing disinfectants, sanitizers, detergents, and any chemical agent used to inac-

tivate or impede the growth of microorganisms (Donaghy et al., 2019). Although these terms are often used interchangeably in scientific literature, they exhibit distinctions based on the extent of microbial reduction on food contact surfaces (FCS) and nonfood contact surfaces (NFCS), in compliance with different regulations (Bland et al., 2022). In the United States, the Environmental Protection Agency (EPA) specifies that disinfectants must achieve a 6-log microbial reduction on both FCS and NFCS, whereas sanitizers must achieve a 5-log reduction on FCS. On NFCS, sanitizers must attain a 3-log microbial reduction within a 5-min contact time (EPA, 2012, 2018). It should be noted that water hardness or organic soiling may impact the overall effectiveness of sanitizers under practical conditions (EPA, 2012, 2018). In Europe, biocides are regulated under EU regulation 528/2012 and are classified into product types (PTs) based on their intended use (European Commission, 2012). Among the 22 biocidal products (BPs), PT1 to PT5 are classified as disinfectants. Particularly, PT4 includes disinfectants used in the food and feed sector and should result in 4- to 5-log bacterial reduction within 5 min to 1 h on soiled and cleaned FCS and NFCS for all biocidal products (ECHA, 2018). While these criteria pertain to planktonic cells, it is suggested they should theoretically also apply to biofilms, as biofilms consist of cells that similarly pose a risk to food safety (Tuytschaever et al., 2023). However, it is well established that cells within biofilms exhibit significantly higher tolerance to biocides (Barroso et al., 2019; Paluszak et al., 2021; Yun et al., 2012). These regulations and guidelines play a crucial role in assessing the efficacy of each BP in experimental designs. However, the focus of this review is not to evaluate the efficacy of biocides based on the degree of microbial reduction, but rather to explore the factors accounting for a biocide’s effectiveness against *L. monocytogenes* biofilms. Consequently, in the current review, the use of the terms “sanitizers,” “disinfectants,” and “biocides” is not associated with the degree of bacterial reduction.

The efficacy of a biocide against biofilms is dependent on several factors including, biocide concentration, contact time, temperature, the presence of organic matter, the age and distribution of biofilms on the surface, and surface material (Bas et al., 2017; Bonneville et al., 2021; Ibusquiza et al., 2011; Kocot et al., 2021; Nilsson et al., 2011). While many studies highlight the significance of chemical action, temperature, and time in biofilm control, the role of mechanical action is less prominently addressed (Dygico et al., 2021; Hua et al., 2019; Korany et al., 2018). This could be due to the inherent challenges of replicating mechanical cleaning processes in laboratory settings or the focus on identifying chemical agents that are effective without the need for mechanical action. Nevertheless, optimal treatment conditions, mechanical action such as brushing, and the appropriate biocide dosage have been reported to

be effective against biofilms, although some studies have observed a reduction in the activity of some biocides in the presence of organic matter (Dhowlaghar et al., 2018; Hua et al., 2019; Korany et al., 2018). For instance, Hua et al. (2019) reported that, irrespective of surface (SS, low-density polyethylene [LDPE], polyvinyl chloride [PVC], polyester, and rubber) and biocide type (QAC, NaOCl, PAA, and ClO_2), the presence of food residues (apple juice or milk) resulted in a reduction of biocidal activity against *L. monocytogenes* biofilms. To achieve complete inactivation of *L. monocytogenes* biofilms on FCS, Dhowlaghar et al. (2018) recommended the use of commercial biocides containing several active agents, such as QAC, PAA, and H_2O_2 . However, the safety and compatibility of combining different active agents in commercial biocides needs to be taken into consideration. In another study, Aryal and Muriana (2019) investigated the efficacy of QAC-, chlorine-, and H_2O_2 -based biocides individually and in combination against 7-day-old *L. monocytogenes* biofilms, with each applied at its manufacturer-recommended concentration and exposure time. Their findings indicated that, when used individually, QAC- and chlorine-based biocides were the least effective disinfectants, achieving ~2-log reduction within 2 h. However, a biocide containing QAC, H_2O_2 , and diacetin resulted in ~6-log reduction, reaching levels below the limit of detection (LOD) within a 1-min contact time. Such studies illustrate the need for careful selection of appropriate biocides and the design of effective cleaning and disinfection schedules that are fit for purpose in a specific production site.

This review aims to summarize the existing scientific literature on studies investigating the efficacy of commonly used disinfectants in the food industry in controlling *L. monocytogenes* biofilms. Specifically, the authors intend to highlight the major factors affecting the efficacy of biocides against *L. monocytogenes* biofilms using a systematic review approach. This approach is frequently recommended for assessing past and current findings, identifying evidence-based research gaps, and providing a framework for guiding new research activities (Sargeant & O'Connor, 2020). As mentioned earlier, the effect of disinfectants on biofilms of *L. monocytogenes* is intricate and variable, influenced by numerous factors. A number of studies have reported on these factors, but their findings often contradict each other, with only a few well-established trends existing across studies, which will be discussed in this review. Currently, no existing review has comprehensively addressed the key factors associated with the effectiveness of biocides in controlling *L. monocytogenes* biofilms. Addressing this gap is essential for developing sustainable strategies to control this pathogenic bacterium and its biofilms on relevant surfaces in the food industry. Furthermore, given the huge annual investment surpassing 5 trillion euros in the implementation of hygienic systems

in the European food industry (Rodríguez-López et al., 2018), it is important to provide a global overview on how *L. monocytogenes* biofilms respond to different biocides and diverse treatment conditions. To set the context, examining the prevalence of *L. monocytogenes* in the food-processing environment is relevant as this evidence will highlight the need for effective and sustainable cleaning and disinfection strategies to combat pathogens and their associated biofilms.

1.1 | Prevalence of *L. monocytogenes* on contact surfaces in food-processing facilities

Food processing entails significant contact of products with various surfaces, especially during washing, cutting, packing, transportation, and storage. These procedures may pose significant risks to pro, because contaminated FCS have been identified as potential vehicles for product contamination (Castro-Ibáñez et al., 2017). Generally, bacterial attachment and biofilm formation occur on various FCS, including conveyor belts, knives, trolleys, shredders, cutting boards, and slicing machines (Faille et al., 2018). The persistence of *L. monocytogenes* on conveyor belts and shredders in the food industry has been well documented (Buchholz et al., 2012; Chaitiemwong et al., 2010). Also, NFCS such as floors, drains, walls, and door handles have been reported to support biofilm formation and to be potential sources of *L. monocytogenes* contamination in the food industry (Bolocan et al., 2015; Dygico et al., 2020; Muhterem-Uyar et al., 2015; Rodríguez-López et al., 2018; Zhang, Wang, et al., 2021). To highlight the prevalence of *L. monocytogenes* on FCS and NFCS in the food-processing environment and the importance of implementing effective sanitization strategies, a summary of relevant studies is provided in Table 1.

These studies reported prevalence of *L. monocytogenes* on FCS and NFCS in various food-processing plants, ranging from 1.6% to 66% (Table 1). Even after cleaning and disinfection procedures, *L. monocytogenes* has been detected at lower prevalence on contact surfaces in the food-processing environment. Gómez et al. (2015) examined the occurrence of *L. monocytogenes* in RTE meat products and their processing plants in Spain over 12 months. *L. monocytogenes* was detected on SS working tables, cutting boards, and conveyor belt surfaces with a prevalence of 84% and 16%, respectively, for samples collected before and after cleaning and disinfection. Environmental conditions such as temperature, substrate properties, and the presence of nutrients play significant roles in biofilm formation and persistence (Klopper et al., 2023; Manville et al., 2023).

L. monocytogenes exhibits a notable ability to persist on both FCS and NFCS within food-processing environ-

TABLE 1 Summary of studies on the prevalence of *Listeria monocytogenes* on food and nonfood contact surfaces in the food-processing environment.

Country	Study duration (months)	Type of facility	Implicated surfaces	Prevalence		References
				No. of samples tested (+ve samples)	Serotype	
China	12	Pork retail market	FCS: meat mincers, cutting tables, and weighing scales	176 (11.4)	1/2b (43.1%), 1/2c (35.5%), 1/2a (19.1%), 4b (1.1%), and 3a (1.1%)	Luo et al., 2017
China	1	Meat-processing plant	FCS: conveyor apparatus, cutting boards, knives, and weighing tools	12 (66.7)	NM	Zhang, Que, et al., 2021
Greece	NM	Sausage-processing plant	FCS: grinders, mixers, mincing machines, stuffing machines, knives, and tables NFCs: walls	34 (11.7)	NM	Gounadaki et al., 2008
Ireland	12	Dairy, meat, and fresh-cut vegetables plant	NFCs: drains, floor, storage shelf, and cutting areas	1574 (4.4)	1/2a (41%), 4b/4e (27%), 1/2b (17%), and 1/2c (15%)	Leong et al., 2014
Ireland	24	Seafood Meat Dairy Vegetable	NFCs: drains, floor, storage shelf, and walls FCS: cutting areas	709 (9.17) 279 (6.09) 855 (5.03) 370 (8.92)	NM	Alvarez-Ordóñez et al., 2018
Ireland	36	Dairy-, meat-, seafood-, and vegetable-processing plant	NFCs: drains and floors FCS: shelves	4667 (3.77)	NM	Leong et al., 2017
Italy	24	Dairy plant	NFCs: drainage, filters, and floors FCS: internal fridge and wooden shelves	547 (4.6)	NM	Parisi et al., 2013
Italy	24	Cheese making plant	FCS: raw milk filters, tables, shelves, and trolleys NFCs: walls, drainage, washing machines, and floors	409 (12.7)	1/2a (61.7%), 4b (23.7%), 1/2b (7.3%), 3a (3.6%), 1/2c (2.5%), and 3b (1.1%)	Spanu et al., 2015
Malaysia	12	Food premises, food-processing plants, hypermarkets, wet market, and a night market	FCS: working benches, food-processing machines, cutting boards, conveyor belts, mixing bowls, and surfaces of food product NFCs: trolleys, racks and rack stands, pallets, walls, and wiping clothes	93 (11.83) 59 (6.78)	NM NM	Chen et al., 2021

(Continues)

TABLE 1 (Continued)

Country	Study duration (months)	Type of facility	Implicated surfaces	Prevalence		References
				No. of samples tested (+ve samples)	Serotype	
Poland	48	Vegetable-processing facility	FCS: conveyor belt, post blanching, and cooling NFCs: unit rung feeder, floor, central drain, freezing tunnel, and vibratory conveyor	213 (41.3)	1/2a, 1/2b, and 4b	Pappelbaum et al., 2008
Slovakia	48	Meat-processing facility	NFCs: floor, drains, walls, washbasins, and protective overshoes	196 (30.6)	NM	Véghová et al., 2017
Spain	24	Meat-processing plant	FCS: machines, knives, blades, tables, cutting boards, containers, and conveyor belts	110 (22.7)	NM	Gómez et al., 2015
Spain	12	Poultry-processing facility	NFCs: drains, walls, and floors FCS: knives, cutting table, conveyor belt, and mincing machines	250 (45.6)	1/2c-3c (68.8%), 1/2a-3a (27.8%), 4b-4d-4e (2.3%), and 1/2b-3b-7-1.1%	Melero et al., 2019
United States	24	Blue crab-processing plant	NFCs: floor drains, scrap barrels, raw crab cooler, scrap receptacle, cooked crab tray, raw crab receiving dock, gloves, and table	624 (2.1)	4b, 2b, 1/2a, and 1/2b	Pagadala et al., 2012
United States	NM	Mushrooms	NFCs: shovel, hoses, door and floor surfaces, wooden trays, water drains, and baskets FCS: mixer, hopper, conveyor, and knives	184 (1.6)	4a	Viswanath et al., 2013
United States	13	Mushroom-processing facility	NFCs: Equipment bolts and nuts, floors, drains, and walls FCS: rollers	255(18.8)	1/2c (93.6%), 1/2b (3.5%), and 1/2a (2.9%)	Murugesan et al., 2015
United States	8	Fruit- and vegetable-packing houses	NFCs: forklift wheels, drains, dump tank legs, cold room floors, and squeegees	1588 (3.2)	NM	Estrada et al., 2020
United States	12	Fresh produce-handling facilities	NFCs: forklift tire, side of conveyor, pallet, bins, entry points, floors, and coolers	350 (4.3)	31% (1/2a, 3a) and 69% (4 b, 4 d, 4e)	John et al., 2020
United States	12	Produce-packing houses and fresh-cut facilities	NFCs: Drains, squeegees, sandwich junctures, carts, and forklifts	2564 (3.0)	NM	Sullivan & Wiedmann, 2020
United States	24	Fruit-packing houses	NFCs: Floors, bins, dump tanks, spray bars, drip pans, drains, and forklifts	1437 (17.5)	NM	Simonetti et al., 2021

Abbreviations: FCS, food contact surfaces; NFCs, nonfood contact surfaces; NM, not mentioned.

ments. A recent study by Daeschel et al. (2022) reported that *L. monocytogenes* persisted for 17 years on FCS and NFCS in a seafood-processing environment. The persistence of *L. monocytogenes* on contact surfaces has been comprehensively reviewed by Hua and Zhu (2024a). To ensure products' safety and mitigate the risk of cross-contamination with *L. monocytogenes*, studies cited in Table 1 highlighted the importance of implementing an effective and appropriate cleaning and disinfection strategy of FCS and NFCS within the food-processing plant (Chen et al., 2021; Leong et al., 2014; Murugesan et al., 2015).

2 | METHODOLOGY

2.1 | Literature search: Database sources and keywords

This review commenced with the question, *What factors influence biocide efficacy against L. monocytogenes biofilm in the food industry?* A systematic review was conducted to answer this question. The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines (Liberati et al., 2009) and the framework developed by Sargeant and O'Connor (2020) for systematic reviews were employed in this study. Three databases were used for the literature search to retrieve extensive literature and reduce biases (Bramer et al., 2017; Vassar et al., 2017). The databases included the Web of Science (WOS), Scopus, and PubMed. The keywords used for the literature search in this review were adapted from a study by Meireles et al. (2016) with modifications. The following search string was used: ("*Listeria monocytogenes*" AND "biofilm") AND ("biocide" OR "disinfectant" OR "detergent" OR "sanitizer" OR "chlorine" OR "chlorine dioxide" OR "sodium hypochlorite" OR "organic acid" OR "hydrogen peroxide" OR "peracetic acid" OR "peroxyacetic acid" OR "quaternary ammonium compound" OR "benzalkonium chloride"). The predefined search strategy was performed on May 10, 2024, and the results were exported in RIS format for screening. It should be noted that the biocidal products used for the literature search represent the commonly used biocides in the food industry, as reported in many studies (Aryal & Muriana, 2019; Belessi et al., 2011; Bland et al., 2022; Bonneville et al., 2021; Boucher et al., 2021; Cruz & Fletcher, 2012; Das & Lalitha, 2022; Dygico et al., 2021; Fagerlund et al., 2017; Henriques & Fraqueza, 2017; Hua et al., 2019; Korany et al., 2018; Kyoui et al., 2016; Nakamura et al., 2013; Poimenidou et al., 2016; Rodríguez-López et al., 2018; Rodríguez-Melcón et al., 2023; Skowron et al., 2019).

2.2 | Screening of literature: Inclusion and exclusion criteria

A three-stage screening process (Castellini & Graffigna, 2022) was used in this study to select relevant documents. In the first stage, articles were de-duplicated using the Rayyan Systematic Review (RSR) platform (<https://rayyan.qcri.org/welcome>). A second round of screening was performed by a careful reading of the title and abstract of each document to include eligible studies based on the study objectives, as shown in Table 2. In instances of uncertainties and clarity of study, an article was deemed relevant for the final screening stage. In the third and final screening stage, the full-text articles were thoroughly analyzed to include only relevant articles. The second and third screening stages were repeated twice to ensure that the screening quality was not compromised by either including irrelevant articles or deleting relevant studies.

2.3 | Data extraction

All eligible documents were extracted and categorized into the following themes:

- Factors that affected the efficacy of biocides against mono-species *L. monocytogenes* in biofilms;
- Mixed-species biofilms with *L. monocytogenes* and its response to disinfection;
- Challenges encountered during disinfection with biocides.

3 | RESULTS AND DISCUSSION

3.1 | Literature search and study characteristics

Figure 1 shows the flow diagram for article selection. In total, 1067 articles were retrieved from the WOS, Scopus, and PubMed databases. Following de-duplication, screening of abstracts and titles, and full-text reading of articles, 92 original research articles were included in this review. The reasons for the inclusion/exclusion of articles are shown in Figure 1.

3.2 | Data extraction and analysis

The included articles were divided into three categories to facilitate data extraction and analysis. These were articles that focused on biocide efficacy in controlling mono-

TABLE 2 Inclusion and exclusion criteria used for document screening.**Inclusion criteria**

- Articles published between January 2010 and May 2024;
- Articles published in English;
- Articles in which *Listeria monocytogenes* biofilms or mixed-species biofilms containing *L. monocytogenes* were investigated;
- The use of at least one biocide such as QAC, NaOCl, PAA, or H₂O₂ to control *L. monocytogenes* biofilms on surfaces relevant to the food-processing environment;
- Articles in which disinfectant concentration, treatment time, and biofilm formation parameters such as incubation temperature and time (biofilm age) were reported.

Exclusion criteria

- Reviews, editorials, and conference papers;
- Articles wherein plant extracts such as essential oils and other phytochemicals were used;
- The use of physical treatment including UV and ultrasound;
- Articles that focused on other *Listeria* species such as *L. innocua*;
- Biological treatments such as the use of other bacteria and phage.

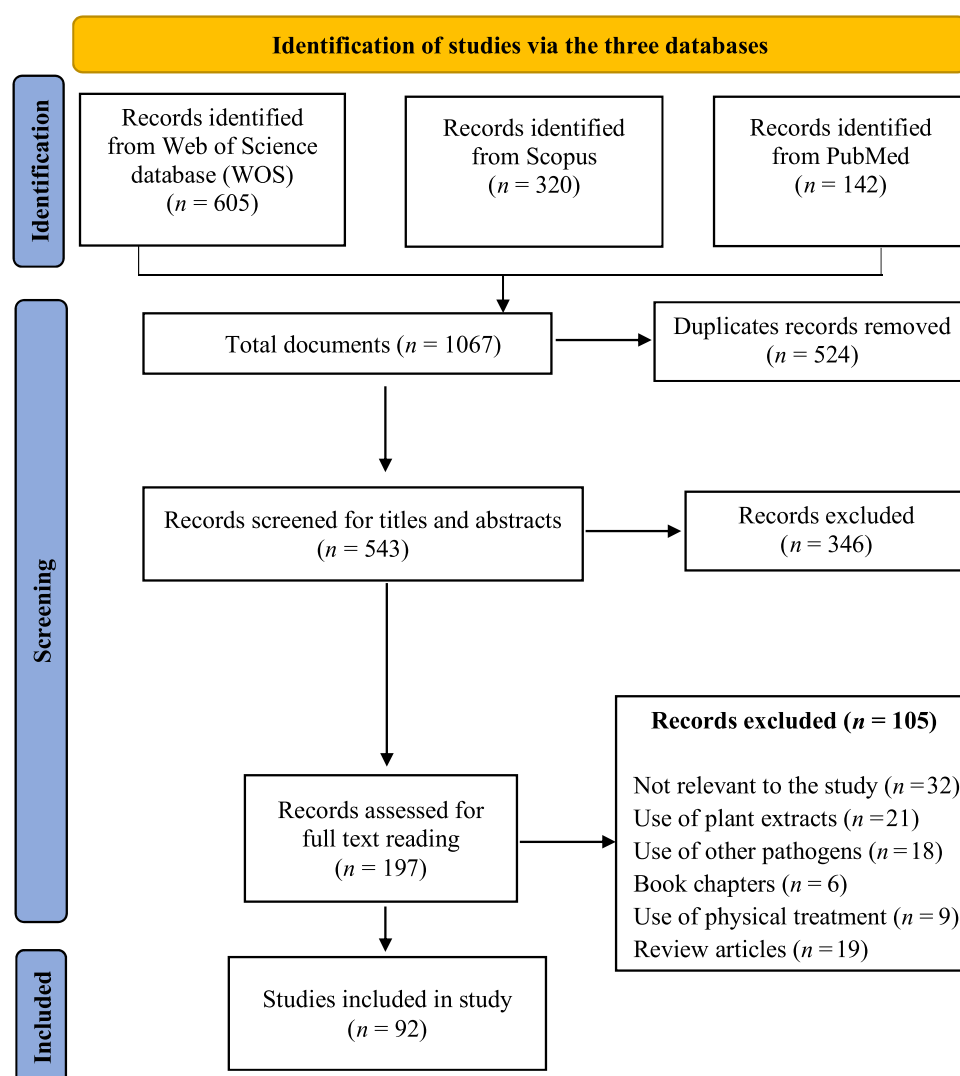
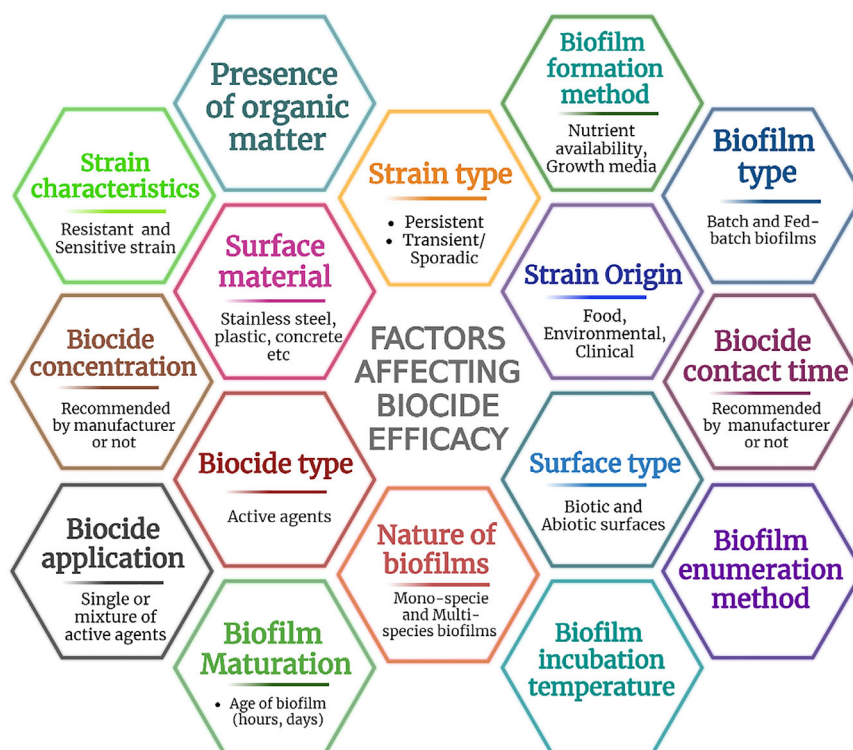
**FIGURE 1** Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow diagram for literature search and screening process.

FIGURE 2 Factors affecting biocide efficacy in controlling *Listeria monocytogenes* biofilms (figure designed on <https://www.biorender.com/>).



species *L. monocytogenes* biofilms ($n = 70$), multispecies biofilms containing *L. monocytogenes* ($n = 17$), and challenges associated with the use of biocides for disinfection ($n = 6$). Summaries of studies that reported numerical reductions are presented in Tables 3 and 4. This was to ensure uniformity of results across the studies. However, the studies not included in the tables but meeting the inclusion criteria ($n = 24$) contributed valuable insights and were thoroughly discussed in the review. The summary of the included studies is presented in Tables S1 and S2. The extracted data were analyzed using the narrative synthesis approach, where each factor presented in Figure 2 was thoroughly examined.

3.3 | Defining efficacy/effectiveness of biocides

The efficacy or effectiveness of a biocidal product refers to its demonstrated capacity to “destroy, render harmless, prevent the action of, or otherwise exert a controlling effect on any harmful organism by chemical or biological means” (European Commission, 2009). The efficacy of a biocide typically involves evaluating the biocide’s ability to achieve the intended level of microbial reduction, inhibition, or elimination within a specified timeframe and under relevant conditions. While regulatory bodies have defined the degree of microbial reduction that must be achieved for a biocidal product to claim effectiveness

(Bland et al., 2022; EPA, 2018; European Commission, 2012), drawing direct comparisons based on the degree of microbial reduction across various studies can present challenges, due to the unavailability of biocide validation data by manufacturers and the variability in reduction thresholds reported in research studies. For instance, Barroso et al. (2019) and Olszewska et al. (2016) considered a disinfectant effective if it removed 4 log CFU/cm² of the bacterial biofilm, citing EN 13697 (Swedish Institute for Standards, 2001). On the contrary, Boucher et al. (2021) reported that citric acid and QAC biocides, applied at manufacturers’ recommended concentrations, were ineffective against *L. monocytogenes* biofilms because the disinfectants could not reduce the viable biofilm cell numbers by >5 log CFU/cm². The 5-log reduction threshold did not cite any established standards or regulatory guidelines. This is an important point to consider as it gives rise to conflicting observations among studies. Therefore, instead of focusing on the level of microbial reduction as a sole parameter for the effectivity of a biocide, the factors underpinning the efficacy of a biocide to achieve a bacterial reduction are discussed in the subsequent sections. It is important to note that evaluation of a biocide’s efficacy encompasses various factors, such as the biocide type, concentration, exposure duration, biofilm age, biofilm formation parameters, and bacterial strain type and characteristics. All these elements collectively determine a biocide’s ability to achieve the desired level of microbial reduction.

TABLE 3 Summary of studies on biocide efficacy against mono-species *Listeria monocytogenes* in biofilms.

References	Surface type	Serotype	Biofilm growth media	Biofilm incubation temperature (°C)	Age of biofilm (h)	Treatment conditions (biocide type, concentration, and contact time)	Log reductions	Applied at MRC/MRCT
Andrade et al., 2020	Polystyrene	1/2a, 1/2b, 1/2c, and 4b	BPW	30	120	- QAC, 150 ppm, 5 min - BAC, 10 ppm, 5 min	- QAC: 2.4 log CFU/mL - BAC: 3 log CFU/mL	Yes
Aryal & Muriana, 2019	Polystyrene	NM	BHI	30	168	- QAC, 200 ppm, 15 min - NaOCl, 200 ppm, 15 min - PAA, 500 ppm, 5 min	- QAC: 5 log CFU/mL - NaOCl: 1–2 log CFU/mL - PAA: >7.5 log CFU/mL	Yes
Aryal et al., 2023	Bell pepper	1/2a and 4b	TSBYE	4, 25	24, 48, 72	NaOCl, 100 ppm, 5 min	<1.18 log CFU/g	Yes
Avila-Novoa et al., 2023	SS and PP	NM	TSB	37	240	- CA, 2 mg/mL, 10 min - LA, 2 mg/mL, 10 min - GA, 2 mg/mL, 10 min	- CA: 5.46 log CFU/cm² - LA: 4.48 log CFU/cm² - GA: 4.13 log CFU/cm²	NM
Ban et al., 2012	SS and PVC	NM	TSB	25	144	- LA, 2%, 30 s—SS - LA, 2%, 30 s—PVC	0.9 log CFU/coupon 0.9 log CFU/coupon	NM
Ban & Kang, 2016	SS	NM	TSB	25	144	- NaOCl, 100 ppm, 30 s - QAC, 100 ppm, 30 s - H₂O₂, 2%, 30 s	- NaOCl: 1.25 log CFU/coupon - QAC: <1 log CFU/coupon - H₂O₂: 1.2 log CFU/coupon	Yes
Barroso et al., 2019	SS	NM	TSBYE	11 and 25	168, 48	- PAA, 1000 ppm, 5 min - BAC, 2560 ppm, 10 min	- PAA: 4 log CFU/cm² - BAC: 4 log CFU/cm²	MIC
Belessi et al., 2011	SS	4b	TSBYE	5 and 20	288, 456	- PAA, 2%, 1 min - NaOCl, 0.465%, 6 min - QAC, 1%, 6 min	- PAA: 2–2.9 log CFU/cm² - NaOCl: 1–3 log CFU/cm² - QAC: 1.5–4.8 log CFU/cm²	Yes
Bonneville et al., 2020	SS	1/2a	TSBYE	25	48	BAC, 1280 mg/L, 5 min	~3 log CFU/cm²	MIC
Boucher et al., 2021	SS and HDPE	1/2a, 1/2b, 1/2c, 3a, and 4b	TSB	4	24	- QAC, 200 ppm, 2 min - CA, 0.660%, 1 min	QAC or CA: <5 log CFU/cm²	Yes
Cabeça et al., 2012	SS	NM	TSB	37	120	- QAC, 0.5% (w/v), 10 min - PAA, 0.5% (w/v), 10 min - NaOCl, 1.5% (w/v), 10 min	- QAC: 1.4 log CFU/cm² - PAA: 5.1 log CFU/cm² - NaOCl: 5.2 log CFU/cm²	Yes

(Continues)

TABLE 3 (Continued)

References	Surface type	Serotype	Biofilm growth media	Biofilm type	Biofilm incubation temperature (°C)	Age of biofilm (h)	Treatment conditions (biocide type, concentration, and contact time)	Log reductions	Applied at MRC/MRCT
Chaitiemwong et al., 2014	SS	1/2a, 1/2b, and 4e	BHI	Three-strain cocktail ^a	30	336	• QAC, 740 mg/L, 5 min	• 5 log CFU/mL	Yes
Chaves et al., 2023	SS	1/2a, 1/2b, 1/2c, and 4b	LB	Single-strain biofilm ^a	37	24	• QAC > 5 ppm, 3 min > 10 ppm, 3 min > 20 ppm, 3 min • ClO ₂ > 5 ppm, 3 min > 10 ppm, 3 min > 20 ppm, 3 min	• QAC > 1.8–3.9 log CFU/cm ² > 1.3–3.1 log CFU/cm ² > 3.3–6.9 log CFU/cm ² • ClO ₂ > 2.5–6.9 log CFU/cm ² > 3.0–6.9 log CFU/cm ² > 6.0–7.2 log CFU/cm ²	NM
Chen et al., 2015	SS	1/2c and 4b	BHI	Five-strain cocktail ^b	21	72	• QAC, 150 ppm, 5 min • NaOCl, 100 ppm, 5 min • H ₂ O ₂ , 2%, 5 min • LA, 3%, 5 min	• QAC: 4.4 log CFU/coupon • NaOCl: 0.7 log CFU/coupon • H ₂ O ₂ : 0.3 log CFU/coupon • LA: 4.2 log CFU/coupon	Yes
Das & Lalitha, 2022	Glass	NM	TSBYE	Single-strain biofilm ^a	30	48	• NaOCl > 10 ppm, 5 min > 220 ppm, 5 min • BAC > 50 ppm, 30 s > 100 ppm, 30 s > 150 ppm, 30 s > 200 ppm, 30 s	• NaOCl > 4.5 log CFU/cm ² > 5.9 log CFU/cm ² • BAC > 1.28 log CFU/cm ² > 2.4 log CFU/cm ² > 2.98 log CFU/cm ² > 5.9 log CFU/cm ²	NM
de Oliveira et al., 2012	SS	4d	TSB	Single-strain biofilm ^a	37	48	• QAC, 0.5%, 20 min • NaOCl, 0.01%, 20 min • H ₂ O ₂ , 3%, 20 min	• QAC: 5.64 log CFU/cm ² • NaOCl: 0.9 log CFU/cm ² • H ₂ O ₂ : 5.64 log CFU/cm ²	Yes
Dhowlaghar et al., 2018	SS	NM	TSBYE	Six-strain cocktail ^a	22	96	• QAC > 103 ppm, 1 min > 103 ppm, 3 min • PAA > ~29 ppm, 1 min > ~29 ppm, 3 min • NaOCl > 114 ppm, 1 min > 114 ppm, 3 min	• QAC > 2.3 log CFU/coupon > 4.2 log CFU/coupon • PAA > 4 log CFU/coupon > 6.2 log CFU/coupon • NaOCl > 3.6 log CFU/coupon > 4.8 log CFU/coupon	Yes

(Continues)

TABLE 3 (Continued)

References	Surface type	Serotype	Biofilm growth media	Biofilm incubation temperature (°C)	Age of biofilm (h)	Treatment conditions (biocide type, concentration, and contact time)	Log reductions	Applied at MRC/MRCT
Dygico et al., 2021	SS	NM	BHIYE	25	72	- NaOCl, 150 ppm, 20 min - QAC > 960 ppm, 5 min > 1000 ppm, 15 min - PAA, 245 ppm, 5 min - BAC, 0.125 g/L, 60 min	- NaOCl: 4.7 log CFU/cm² - QAC > 4.9 log CFU/cm² > 5.1 log CFU/cm² - PAA: 4.4 log CFU/cm² ~7 log CFU/well	Yes
Gómez et al., 2012	Polystyrene	1/2a, 1/2b, 1/2c, 4a, and 4b	TSB	30	24	BAC, 0.015%, 30 min	3.4 log CFU/cm²	Yes
Harada & Nascimento, 2021	SS	1/2b, 1/2c, and 4b	TSB	25	168	QAC, 0.015%, 30 min	3.4 log CFU/cm²	Yes
Henriques & Fraqueza, 2017	Polystyrene	1/2a, 1/2b, 1/2c, 4a, and 4b	BPW	30	120	- BAC > 1.2 mg/mL, 5 min > 1 mg/mL, 5 min > 0.8 mg/mL, 5 min > 0.5 mg/mL, 5 min > 0.25 mg/mL, 5 min > 0.1 mg/mL, 5 min - NaOCl > 1.1 mg/mL, 5 min > 1 mg/mL, 5 min > 0.8 mg/mL, 5 min > 0.5 mg/mL, 5 min > 0.25 mg/mL, 5 min > 0.1 mg/mL, 5 min	- BAC > 4.07–7.48 log CFU/mL > 2.5–7.34 log CFU/mL > 1.52–6.31 log CFU/mL > 1.19–6.31 log CFU/mL > 0.66–3.16 log CFU/mL > 0.64–3.54 log CFU/mL - NaOCl > 2.78–7.48 log CFU/mL > 2.66–7.48 log CFU/mL > 1.23–5.98 log CFU/mL > 1.17–4.09 log CFU/mL > 1.05–3.79 log CFU/mL > 1–3.42 log CFU/mL	Yes

(Continues)

TABLE 3 (Continued)

References	Surface type	Serotype	Biofilm growth media	Biofilm incubation temperature (°C)	Age of biofilm (h)	Treatment conditions (biocide type, concentration, and contact time)	Log reductions	Applied at MRC/MRCT
Hua et al., 2019	SS, LDPE, PVC, PET, and rubber	NM	MWB	22	168	- QAC, 400 ppm, 1 min - NaOCl, 200 ppm, 1 min - ClO₂, 5 ppm, 1 min - PAA, 160 ppm, 1 min	- QAC: 2.3–2.9 log CFU/coupon - NaOCl: 2.2–2.6 log CFU/coupon - ClO₂: 1.5–2.3 log CFU/coupon - PAA: 3.5–4.3 log CFU/coupon	NM
Kim et al., 2023	SS	NM	TSB	25	120	- PAA 100 ppm, 5 min 100 ppm, 10 min 100 ppm, 15 min 100 ppm, 20 min - H₂O₂ 2%, 5 min 2%, 10 min 2%, 15 min 2%, 20 min - CA 2%, 5 min 2%, 10 min 2%, 15 min 2%, 20 min	- PAA 0.49 log CFU/cm² 0.54 log CFU/cm² 0.93 log CFU/cm² 1.26 log CFU/cm² - H₂O₂ 0.47 log CFU/cm² 0.53 log CFU/cm² 0.94 log CFU/cm² 1.13 log CFU/cm² - CA 0.82 log CFU/cm² 0.89 log CFU/cm² 0.78 log CFU/cm² 0.97 log CFU/cm²	NM
Kim & Park, 2023	SS and HDPE	NM	TSB	25	48	- ClO₂ (gas)—SS—48 h biofilm 30 ppm, 5 min 30 ppm, 10 min 30 ppm, 15 min 30 ppm, 20 min - ClO₂ (gas)—HDPE—48 h biofilm 30 ppm, 5 min 30 ppm, 10 min 30 ppm, 15 min 30 ppm, 20 min	- ClO₂ (gaseous)—SS 2.16 log CFU/cm² 3.71 log CFU/cm² 4.96 log CFU/cm² > 7.02 log CFU/cm² - ClO₂ (gaseous)—HDPE 1.63 log CFU/cm² 2.99 log CFU/cm² 4.04 log CFU/cm² > 7.13 log CFU/cm²	NM

(Continues)

TABLE 3 (Continued)

References	Surface type	Serotype	Biofilm growth media	Biofilm incubation temperature (°C)	Age of biofilm (h)	Treatment conditions (biocide type, concentration, and contact time)	Log reductions	Applied at MRC/MRCT
Kim & Park, 2023	SS and HDPE	NM	TSB	25	120	• ClO₂ (gas)—SS—120 h biofilm • ClO₂ (gas)—HDPE—120 h biofilm	• ClO₂ (gaseous)—SS • ClO₂ (gaseous)—HDPE	NM
						• 30 ppm, 5 min • 30 ppm, 10 min • 30 ppm, 15 min • 30 ppm, 20 min	• 2.27 log CFU/cm² • 2.87 log CFU/cm² • 3.57 log CFU/cm² • 4.47 log CFU/cm²	
						• 30 ppm, 5 min • 30 ppm, 10 min • 30 ppm, 15 min • 30 ppm, 20 min	• 1.86 log CFU/cm² • 2.54 log CFU/cm² • 2.90 log CFU/cm² • 3.64 log CFU/cm²	
Korany et al., 2018	Polystyrene	1/2a, 1/2b, and 4b	MWB	22	48, 168	• QAC, 400 ppm, 1 min • 48 h biofilm • 168 h biofilm • PAA, 80 ppm, 1 min • 48 h biofilm • 168 h biofilm • NaOCl, 200 ppm, 1 min • 48 h biofilm • 168 h biofilm	• QAC • 3.1 log CFU/well • 2.2 log CFU/well • PAA • 3.5 log CFU/well • 3.3 log CFU/well • NaOCl • 2.8 log CFU/well • 1.7 log CFU/well	NM
Lake et al., 2024	PVC and SS	1/2a, 1/2b, and 4b	TSBYE	20	48	• H₂O₂ + PAA, 3%, 5 min • PVC • SS	• H₂O₂ + PAA • 1.9 log CFU/cm² • <1.5 log CFU/cm²	Yes
Lee, Barancelli, et al., 2017	SS and polystyrene	4b and 1/2b	TSB	35	48	• PAA • 0.5%, 60 min • 0.5%, 120 min • 0.5%, 180 min	• PAA • 2.1–2.3 log CFU/well • 2.1–2.4 log CFU/well • 1.8–3.2 log CFU/well	NM
Lindsay et al., 2022	CDC biofilm bioreactor	NM	TSB	25	24	• NaOCl, 0.04%, 5 min • PAA + H₂O₂, 10%–30%, 5 min • LA, 30%–60%, 5 min • QAC, <10%, 5 min	• NaOCl: >3 log CFU/cm² • PAA + H₂O₂: >3 log CFU/cm² • LA: >3 log CFU/cm² • QAC: >3 log CFU/cm²	Yes

(Continues)

TABLE 3 (Continued)

References	Surface type	Serotype	Biofilm growth media	Biofilm incubation temperature (°C)	Age of biofilm (h)	Treatment conditions (biocide type, concentration, and contact time)	Log reductions	Applied at MRC/MRCT
Liu et al., 2017	Polystyrene	1/2c	BHI	15	96, 168	- LA, 1%, 60 min > 96 h biofilm > 168 h biofilm	- LA > ~4 log CFU/mL > ~1 log CFU/mL	NM
Mørtrø et al., 2013	SS	1/2a, 1/2b, 1/2c, and 4b	BHI	12, 20	48, 168	QAC, 1%, 5 min	> 2 log CFU/mL	Yes
Nguyen Trang et al., 2023	Polystyrene	NM	BHI	30	168	- NaOCl, 100 mg/mL, 10 min - SAHW, 40 mg/mL, 10 min	- NaOCl: 3.2 log CFU/mL - SAHW: 5.3 log CFU/mL	NM
Olszewska et al., 2016	Polystyrene	4b	TSB	37	72	- NaOCl, 200 ppm, 60 min - QAC > 195 ppm, 60 min > 660 ppm, 60 min	- NaOCl: 1.1 log CFU/mL - QAC > 1.4 log CFU/mL > 3.7 log CFU/mL	Yes
Olszewska & Diez-Gonzalez, 2021	Polystyrene	1/2a, 1/2c, and 4b	BPW	34	24	NaOCl, 50 ppm, 1 min	2.9 log CFU/well	NM
Pang et al., 2019	SS	1/2a	dTSB (1:20)	4, 15	24, 96, 168, 336	- QAC, 200 ppm, 1 min—dTSB > 24 h biofilm (4°C) > 96 h biofilm (4°C) > 168 h biofilm (4°C) > 336 h biofilm (4°C) > 24 h biofilm (15°C) > 96 h biofilm (15°C) > 168 h biofilm (15°C) > 336 h biofilm (15°C)	- QAC > 3.8 log CFU/cm² > 4 log CFU/cm² > 3.6 log CFU/cm² > 3.3 log CFU/cm² > 3.9 log CFU/cm² > 4.3 log CFU/cm² > 3.4 log CFU/cm² > 3.6 log CFU/cm²	Yes
Pang et al., 2019	SS	1/2a	Salmon broth (SB)	4, 15	24, 96, 168, 336	- QAC, 200 ppm, 1 min—SB > 24 h biofilm (4°C) > 96 h biofilm (4°C) > 168 h biofilm (4°C) > 336 h biofilm (4°C) > 24 h biofilm (15°C) > 96 h biofilm (15°C) > 168 h biofilm (15°C) > 336 h biofilm (15°C)	- QAC > 3.8 log CFU/cm² > 4.7 log CFU/cm² > 2.2 log CFU/cm² > 0.3 log CFU/cm² > 2.6 log CFU/cm² > 0.5 log CFU/cm² > 0.5 log CFU/cm² > 0.4 log CFU/cm²	Yes

(Continues)

TABLE 3 (Continued)

References	Surface type	Serotype	Biofilm growth media	Biofilm incubation temperature (°C)	Age of biofilm (h)	Treatment conditions (biocide type, concentration, and contact time)	Log reductions	Applied at MRC/MRCT
Pang & Yuk, 2019	SS	1/2a	TSB	15	48	• BAC, 100 µg/mL, 15 min	• 5.42 log CFU/cm²	NM
Papaioannou et al., 2018	SS	4b	Fish juice broth	15	72, 168	• NaOCl, 10%, 30 min • PAA, 1%, 30 min	• <4 log CFU/cm² • <4 log CFU/cm²	Yes
Park et al., 2012	PVC	NM	TSB	25	144	• PAA (aerosolized) ➢ 200 ppm, 5 min ➢ 200 ppm, 10 min ➢ 200 ppm, 30 min • NaOCl (aerosolized) ➢ 100 ppm, 5 min ➢ 100 ppm, 10 min ➢ 100 ppm, 30 min	• PAA ➢ 1.01 log CFU/coupon ➢ 1.49 log CFU/coupon ➢ 6.58 log CFU/coupon • NaOCl ➢ 0.1 log CFU/coupon ➢ 0.2 log CFU/coupon ➢ 2.21 log CFU/coupon	Yes
Park et al., 2012	SS	NM	TSB	25	144	• PAA (aerosolized) ➢ 200 ppm, 5 min ➢ 200 ppm, 10 min ➢ 200 ppm, 30 min • NaOCl (aerosolized) ➢ 100 ppm, 5 min ➢ 100 ppm, 10 min ➢ 100 ppm, 30 min	• PAA ➢ 1.04 log CFU/coupon ➢ 1.94 log CFU/coupon ➢ 7.02 log CFU/coupon • NaOCl ➢ 0.13 log CFU/coupon ➢ 0.74 log CFU/coupon ➢ 1.23 log CFU/coupon	Yes
Piercey et al., 2016	SS	1/2a	TSB	15	24, 48	• BAC, 0.15 mg/mL, 1 min ➢ 24 h biofilm ➢ 48 h biofilm • H₂O₂, 0.375% (w/v), 1 min ➢ 24 h biofilm ➢ 48 h biofilm	• BAC ➢ 2.8 log CFU/cm² ➢ 0.1 log CFU/cm² • H₂O₂ ➢ 0.4 log CFU/cm² ➢ 0.1 log CFU/cm²	NM
Poimenidou et al., 2016	SS and polystyrene	1/2a, 1/2c, and 4b	TSB	20	72	• PAA ➢ 2000 ppm, 5 min, SS ➢ 2000 ppm, 5 min, PVC • QAC ➢ 500 ppm, 5 min, SS ➢ 500 ppm, 5 min, PVC	• PAA ➢ 1.3–4.8 log CFU/cm² ➢ 1.5–3.6 log CFU/cm² • QAC ➢ 2.5–4.9 log CFU/cm² ➢ 1.4–3.7 log CFU/cm²	Yes

(Continues)

TABLE 3 (Continued)

References	Surface type	Serotype	Biofilm growth media	Biofilm incubation temperature (°C)	Age of biofilm (h)	Treatment conditions (biocide type, concentration, and contact time)	Log reductions	Applied at MRC/MRCT
Quan et al., 2017	SS	NM	TSB	37	192	SAHW, 30 mg/L, 3 min	3.22 log CFU/cm²	Yes
Rahman et al., 2022	Polystyrene	1/2b and 4b	TSBYE	37	24, 48, 72	NaOCl, 32 ppm, 5 min	5.44 log CFU/mL	NM
Rodríguez-López et al., 2022	SS	NM	TSB	25	48	BAC, 47.35 µg/mL, 10 min	2.57 log CFU/cm²	NM
Sekhoni et al., 2022	SS	NM	BHI	37	72	- NaOCl, 100 ppm, 1 min (No gas) - NaOCl, 100 ppm, 1 min (gas) - NaOCl + air (UFB) - NaOCl + CO₂ (UFB) - NaOCl + N₂ (UFB) - NaOCl, 200 ppm, 1 min (No gas) - NaOCl, 200 ppm, 1 min (gas) - NaOCl + air (UFB) - NaOCl + CO₂ (UFB) - NaOCl + N₂ (UFB)	3.7 log CFU/cm² - NaOCl (gas) > 5 log CFU/cm² > 4.9 log CFU/cm² > 4 log CFU/cm² > 5.5 log CFU/cm² - NaOCl (gas) > 5.8 log CFU/cm² > 5.8 log CFU/cm² > 5.6 log CFU/cm²	NM

(Continues)

TABLE 3 (Continued)

References	Surface type	Serotype	Biofilm growth media	Biofilm incubation temperature (°C)	Age of biofilm (h)	Treatment conditions (biocide type, concentration, and contact time)	Log reductions	Applied at MRC/MRCT
Skowron et al., 2018	SS and PP	NM	BHI	37	72	• SS	• SS	NM
						➤ QAC, 0.5%, 1 min	➤ 4.06 log CFU/cm ²	
						➤ NaOCl, 0.5%, 1 min	➤ 1.97 log CFU/cm ²	
						➤ PAA + H ₂ O ₂ , 0.5%, 1 min	➤ 6.63 log CFU/cm ²	
						➤ QAC, 0.5%, 5 min	➤ 5.01 log CFU/cm ²	
						➤ NaOCl, 0.5%, 5 min	➤ 3.55 log CFU/cm ²	
						➤ PAA + H ₂ O ₂ , 0.5%, 5 min	➤ 6.63 log CFU/cm ²	
						• PP	• PP	
						➤ QAC, 0.5%, 1 min	➤ 2.57 log CFU/cm ²	
						➤ NaOCl, 0.5%, 1 min	➤ 2.74 log CFU/cm ²	
						➤ PAA + H ₂ O ₂ , 0.5%, 1 min	➤ 6.62 log CFU/cm ²	
						➤ QAC, 0.5%, 5 min	➤ 3.50 log CFU/cm ²	
Srey, Park, Jahid, & Ha, 2014	Lettuce and cabbage surface	1/2a, 3b, and 4b	TSB	4	24	• NaOCl, 200 ppm, 5 min	• NaOCl	Yes
						➤ Lettuce	➤ 0.68 log CFU/cm ²	
						➤ Cabbage	➤ 3.35 log CFU/cm ²	
						• CA, 2%, 5 min	• CA	
						➤ Lettuce	➤ 0.14 log CFU/cm ²	
						➤ Cabbage	➤ 2.68 log CFU/cm ²	
						• MA, 2%, 5 min	• MA	
						➤ Lettuce	➤ 0.56 log CFU/cm ²	
						➤ Cabbage	➤ 3.41 log CFU/cm ²	
						• LA, 2%, 5 min	• LA	
						➤ Lettuce	➤ 0.56 log CFU/cm ²	
						➤ Cabbage	➤ 3.4 log CFU/cm ²	
Srey, Park, Jahid, Oh, et al., 2014	Polystyrene	1/2a, 3b, and 4b	TSB	37	48	• NaOCl	• NaOCl	NM
						➤ 50 mg/L, 1 min	➤ 3.27 log CFU/well	
						➤ 100 mg/L, 1 min	➤ 3.96 log CFU/well	
						➤ 200 mg/L, 1 min	➤ 4.61 log CFU/well	

(Continues)

TABLE 3 (Continued)

References	Surface type	Serotype	Biofilm growth media	Biofilm incubation temperature (°C)	Age of biofilm (h)	Treatment conditions (biocide type, concentration, and contact time)	Log reductions	Applied at MRC/MRCT
Svoboda et al., 2016	Melon surfaces	NM	BHI	22	1	• QAC, 300 ppm, 2 min • H₂O₂ + PAA, 100 ppm, 2 min • H₂O₂ + PAA: 1.8 log CFU/mL • ClO₂, 5 ppm, 2 min • ClO₂: 0.8 log CFU/mL • H₂O₂, 5%, 2 min • H₂O₂: 2.4 log CFU/mL • NaOCl, 200 ppm, 2 min • NaOCl: 1.9 log CFU/mL	• QAC: 0.8 log CFU/mL • H₂O₂ + PAA: 2 • H₂O₂: 0.8 log CFU/mL • H₂O₂: 2.4 log CFU/mL • NaOCl: 1.9 log CFU/mL	NM
Vaid et al., 2010	SS	1/2a and 4b	TSB	22	96	• ClO₂, 7 mg/L, 10 min • NaOCl, 50 mg/L, 10 min	• ClO₂: 3.74 log CFU/cm² • NaOCl: 3.09 log CFU/cm²	NM
Vázquez-Sánchez et al., 2018	SS and polystyrene	NM	TSB	25	24	• PAA, 3% (v/v), 30 min	• 4 log CFU/cm²	NM
Winkelströter & Martinis, 2015	SS	NM	BHI	37	24	• PAA, 1:40 v/v solution, 3 min • NaOCl, 140 ppm, 3 min	• PAA: 2.83 log CFU/cm² • NaOCl: 1.2 log CFU/cm²	NM
Yamakawa et al., 2018	Polystyrene	NM	Todd–Hewitt broth	10	120	• NaOCl, 125 µg/mL	• 5.8 log CFU/well	NM
Yun et al., 2012	GFFs	1/2a, 1/2b, and 4b	TSBYE	25	24	• H₂O₂, 6%, 10 min • H₂O₂, 10%, 10 min	• 2–6 log CFU/GFF • 3.2–9 log CFU/GFF	NM

Abbreviations: BAC, benzalkonium chloride; BHI (YE), brain heart infusion (yeast extract); BPW, buffered peptone water; C&D, cleaning and disinfection; CA, citric acid; CFU/GFF, colony-forming unit/glass fiber filter; ClO₂, chlorine dioxide; (d)TSB, (diluted) tryptone soy broth; GA, gallic acid; H₂O₂, hydrogen peroxide; HDPE, high-density polyethylene; LA, lactic acid; LB, Luria Bertani; LDPE, low-density polyethylene; MIC, minimum inhibitory concentration; MRC/MRCT, manufacturers' recommended concentration and contact times; MWB, modified Welshimer's broth; NaOCl, sodium hypochlorite; NM, not mentioned; PAA, peroxyacetic acid; PET, polyester; PVC, polyvinyl chloride; QAC, quaternary ammonium compounds; SAHW, slightly acidic hypochlorous water; SB, salmon broth; SS, stainless steel; TSBYE, tryptone soy broth extract; UFB, ultrafine bubbles.

^aBatch system.

^bFed-batch system.

TABLE 4 Summary of studies on biocide efficacy against multispecies biofilms containing *Listeria monocytogenes*.

References	Co-specie(s)	Interactions in mixed biofilms	Growth media	Surface type	Culture type	Temperature (°C)	Age of biofilm (h)	Treatment conditions (biocide type, concentration, and contact time)	Log reductions	Applied at MRC/MRCT
Bonneville et al., 2021	<i>L. plantarum</i>	Competitive	TSBYE and dTSBYE (1:10)	SS	Batch	11, 25	24, 168	• H ₂ O ₂ + PAA + AA: 0.5%, 5 min; 1%, 10 min; 2%, 10 min	• 4 log CFU/cm ²	Yes
da Silva Fernandes et al., 2015	<i>Enterococcus faecalis</i> and <i>Enterococcus faecium</i>	Competitive	BHI	SS	Fed-batch	25, 39	24, 192	• NaOCl, 0.2%, 10 min • PAA, 0.2%, 10 min • QAC, 3%, 10 min	• NaOCl: 3 log CFU/cm ² • PAA: 5 log CFU/cm ² • QAC: 5 log CFU/cm ²	Yes
de Oliveira et al., 2012	<i>E. coli</i>	Cooperative	TSB	SS	Batch	37	48	• QAC, 0.5%, 20 min • NaOCl, 0.01%, 20 min • H ₂ O ₂ , 3%, 20 min	• QAC: 5.25 log CFU/cm ² • NaOCl: 0.38 log CFU/cm ² • H ₂ O ₂ : 5.25 log CFU/cm ²	Yes
Fagerlund et al., 2017	16 background microbiota, including <i>Acinetobacter</i> and <i>Pseudomonas</i> spp.	Cooperative	BHI	PVC	Fed-batch	12	96	• QAC, 1%, 1 min • PAA, 1%, 1 min	• QAC: <1 log CFU/coupon • PAA: <1 log CFU/coupon	NM
Giaouris et al., 2013	<i>P. putida</i>	Competitive	BHI	SS	Fed-batch	18	240	• BAC, 50 ppm, 6 min	• 2 log CFU/cm ²	NM
Haddad et al., 2021	<i>L. plantarum</i> (1p) and <i>P. fluorescens</i> (Pf) competitive	Cooperative,	TSB	SS, HDPE	Batch	20	288	• BAC, 5–150 µg/mL, 10 min	• 5.9 log CFU/peg	NM
Ibuzquiza et al., 2012	<i>P. putida</i>	Competitive	MCI and IWM	SS, PP	Batch	25	96	• BAC, 155.6 mg/L, 6 min	• 1 log CFU/mL	NM
Kocot et al., 2021	<i>E. coli</i>	Competitive	TSB	SS	Batch and Fed-batch	25	48	• BAC, 200 ppm, 10 min • Batch • Fed-batch	• BAC • > 2.18 log CFU/coupon • > 1.72 log CFU/coupon	Yes
Lourenco et al., 2011	<i>P. aeruginosa</i>	Competitive	TSB	SS	Batch	12, 37	24, 168	• NaOCl, 12.5 %, 15 min • H ₃ PO ₄ , 4 %, 15 min	• NaOCl: 6 log CFU/peg • H ₃ PO ₄ : 6 log CFU/peg	Yes
Oliszewska & Diez-gonzalez, 2021	<i>L. fermentum</i> (Lf), <i>L. bavaricus</i> (Lb), and <i>L. plantarum</i> (1p)	Cooperative, competitive	BPW	Polystyrene	Batch	34	24	• NaOCl, 50 ppm, 1 min	• 1.9 log CFU/well	NM
Oxaran et al., 2018	<i>Klebsiella</i> spp., <i>E. coli</i> , <i>Comamonas</i> spp., <i>Acinetobacter</i> spp.	Cooperative, competitive	BHI	SS	Batch	25	72	• PAA, 0.04%, 20 min	• 2 log CFU/cm ²	MIC
Pang & Yuk, 2019	<i>P. fluorescens</i>	Competitive	TSB	SS	Batch	15	48	• BAC, 100 µg/mL, 15 min	• 1.9 log CFU/cm ²	NM
Papaioannou et al., 2018	<i>Pseudomonas</i> spp. (<i>fragi</i> , <i>savastanoi</i> , <i>putida</i> , and <i>fluorescens</i>)	Competitive	TSB	SS	Fed-batch	15	168	• NaOCl, 10%, 15 min • PAA, 1%, 15 min	• 1 log CFU/cm ² • 1 log CFU/cm ²	Yes

(Continues)

TABLE 4 (Continued)

References	Co-specie(s)	Interactions in mixed biofilms	Growth media	Surface type	Culture type	Temperature (°C)	Age of biofilm (h)	Treatment conditions (biocide type, concentration, and contact time)	Log reductions	Applied at MRC/MRCT
Rodríguez-López et al., 2022	<i>P. aeruginosa</i> , <i>P. fluorescens</i> , or <i>P. putida</i>	Competitive	TSB	SS	Batch	25	48	• BAC, 44 µg/mL, 10 min	• 1.88 log CFU/cm ²	NM
Van der Veen & Abee, 2011	<i>L. plantarum</i> (Lp)	Competitive	BHI	Polystyrene	Batch	20	48	• BAC, 100 µg/mL, 15 min	• <2 log CFU/well	NM
Yamakawa et al., 2018	<i>P. aeruginosa</i>	Competitive	Todd–Hewitt broth	Polystyrene	Batch	10	120	• NaOCl, 500 µg/mL	• 5.6 log CFU/well	NM

Abbreviations: AA, acetic acid; BAC, benzalkonium chloride; BHI, brain heart infusion; BPW, buffered peptone water; (d)TSB, (diluted) tryptone soy broth; H₂O₂, hydrogen peroxide; H₃PO₄, phosphoric acid; HDPE, high-density polyethylene; IWM, intervalvar water of mussel; LD₅₀, lethal dose 90; MCI, mussel cooking juice; MRC/MRCT, manufacturers' recommended concentration and contact times; NaOCl, sodium hypochlorite; NEW, neutral electrolyzed water; NM, not mentioned; PAA, peroxyacetic acid; SS, stainless steel; TSBYE, tryptone soy broth yeast extract.

3.4 | Influence of *L. monocytogenes* strain characteristics on biofilm formation and disinfectant tolerance

3.4.1 | Persistent and transient/sporadic strains

The concept of “persistence” is a commonly discussed phenomenon in relation to pathogens found in food production environments. It pertains to bacterial pathogens adhering to specific harborage sites within a food-processing facility, capable of withstanding routine cleaning and disinfection (Rodríguez-Campos et al., 2019; Taylor & Stasiewicz, 2019). Consequently, persistent strains of *L. monocytogenes* pose a continuous risk of contamination in food-processing environments and have been associated with numerous listeriosis outbreaks (Ortiz et al., 2016). Research has demonstrated that facility design and operational practices significantly contribute to the persistence of *L. monocytogenes* in the food industry (Barnett-Neefs et al., 2022; Ferreira et al., 2014). For example, hard-to-clean areas, equipment, and inadequate hygienic zoning were identified as key contributors to persistent *Listeria*, underscoring the critical role of facility design and sanitation protocols in bacterial persistence (Belias et al., 2022; Ferreira et al., 2014). Moreover, the resident microbiota has been shown to play a significant role in the persistence of *L. monocytogenes* in food-processing environments (Lake et al., 2023; Møretro & Langsrud, 2017). For instance, symbiosis with other bacterial species like *Escherichia coli* O157:H7, *Pseudomonas fluorescens*, and *Ralstonia insidiosa* significantly increased the survival of *L. monocytogenes* in four-species mixed biofilms by up to 1.9 log CFU/cm² compared to mono-species biofilms (Gu et al., 2024). This suggests that microbial interactions can create favorable conditions for *L. monocytogenes*’ survival and persistence in food-processing environments.

According to EFSA, microbial “persistence” is defined as “the ability of a given organism to be established in niches (or harborage sites) within the food and feed-processing environment for a long term, spanning months or years, despite the frequent application of cleaning and disinfection” (EFSA BIOHAZ, 2024). Persistence may also refer to the duration of survival of microorganisms irrespective of sanitation practices (Taylor & Stasiewicz, 2019). Sporadic or transient strains, on the other hand, refer to those isolated intermittently or as single occurrences (Taylor & Stasiewicz, 2019). It has been suggested that persistent strains of *L. monocytogenes* may exhibit enhanced initial bacterial attachment, biofilm formation, and increased tolerance to disinfectants (Nakamura et al., 2013; Nowak et al., 2017; Poimenidou et al., 2016; Rodríguez-Campos et al., 2019). Nakamura

et al. (2013) found that persistent strains produced relatively greater levels of biofilm and EPS, resulting in greater disinfectant tolerance than transient strains. The minimal biofilm inhibitory concentration (MBIC) of BAC required to decrease biofilm by 50% was about 150-fold higher for persistent strains than for transient strains (Nakamura et al., 2013). Similarly, Wang et al. (2015) demonstrated that persistent strains exhibited a higher propensity for biofilm formation on polystyrene surfaces compared to transient strains, with this effect being contingent on the biofilm maturation stage. Notably, on the first day, persistent strains developed significantly denser biofilms than their transient counterparts. However, by the third and fifth day, no significant differences in biofilm formation capacity were observed between the two groups (Wang et al., 2015). Regarding their tolerance to QAC disinfectants, both persistent and transient strains exhibited similar levels of tolerance under varying nutrient conditions, regardless of the biofilm age (Wang et al., 2015). Additionally, findings from the work done by Rodríguez-Campos et al. (2019) indicated the absence of a significant relationship between persistent strains and enhanced disinfectant tolerance. Similarly, Luque-Sastre et al. (2018) found no significant differences in the tolerance levels between persistent and sporadic *L. monocytogenes* strains, regardless of whether in biofilm or planktonic state. Putting all this together, these findings suggest that the tolerance of *L. monocytogenes* biofilm to disinfectants is strain specific and may not be correlated with persistence.

In contrast to the hypothesis that persistent strains produce significantly more biofilm than transient strains, investigations by Nilsson et al. (2011) reported that many persistent strains formed less biofilm than sporadic strains. Carpentier and Cerf (2011) proposed that “persistence” lacks a precise definition, and no distinct *L. monocytogenes* strains harbor unique properties specifically associated with persistence. Nonetheless, disparities observed among studies may be due to interstrain variability, the different techniques or conditions employed by various researchers when conducting their tests, and the classification method used (Rodríguez-Campos et al., 2019; Taylor & Stasiewicz, 2019). For instance, Assisi et al. (2020) classified a strain as persistent if it was isolated during at least 3 out of a 6 months’ sampling period. Nilsson et al. (2011) considered strains as persistent if they were detected in the same environment during consecutive samplings, with a minimum duration of 12 months and a maximum of 4 years. Fox et al. (2015) classified persistent strains as those detected five or more times over periods of time in excess of 3 months apart in a 23-month study and sporadically isolated strains as those detected between two and four times, on sampling occasions 1 month or more apart. These observations indicate that classification of *L. monocytogenes*

strains as persistent or sporadic is not standardized and has led to conflicting results in literature. Consequently, it is essential to establish a standardized definition of persistence. Specifically, a standardized criterion outlining how many times and for how long a strain must be isolated to be considered persistent would be beneficial. Such criteria would provide clearer guidelines for distinguishing between persistent and sporadic strains, enhancing the ability to accurately assess and manage microbiological risks in food-processing environments.

3.4.2 | *Listeria monocytogenes* strains: Serotype and origin

L. monocytogenes has been classified into distinct genetic lineages (I, II, III, IV) with implications for its pathogenicity and the clinical manifestations of listeriosis (Pohl et al., 2006). The majority of human listeriosis cases are caused by strains from lineages I and II (Bergholz et al., 2018; Gattuso et al., 2022), with specific serotypes such as 1/2a, 1/2b, and 4b being most frequently associated with human illness (Walecka-Zacharska et al., 2013). Epidemic clones, particularly epidemic clone I (ECI), have been linked to several human outbreaks and have been found in ruminants, indicating a potential reservoir for these pathogenic strains (Rocha et al., 2013). *L. monocytogenes* strains linked to foodborne outbreaks and recalls have been identified in food samples (Paduro et al., 2020), as well as in areas such as processing, packing, and surrounding environments in close proximity to the food production area (Hua & Zhu, 2024a; Muhterem-Uyar et al., 2015).

Some studies have investigated the relationship between serotypes of *L. monocytogenes*, biofilm-forming potential, and disinfectant tolerance (Korany et al., 2018; Nakamura et al., 2013; Nilsson et al., 2011; Ribeiro et al., 2023; Russo et al., 2018). Interestingly, variations among studies have been observed. Nakamura et al. (2013) and Nilsson et al. (2011) reported that serotype 1/2a strains produced significantly more biofilm than other serotypes. On the contrary, Russo et al. (2018) observed that serotype 1/2a formed less biofilms than serotype 4b. Moreover, Wang et al. (2017) found that serotype 1/2b strains formed stronger biofilms than serotype 1/2a. However, Korany et al. (2018) reported that there was no clear relationship between biofilm formation and serotype. The seeming differences observed among studies could be attributed to the different environmental setups used in each study, as well as individual strain variability. Thus, variations in substrates for assessing biofilm formation, pH, incubation temperature, nutritive medium, and age of biofilms employed in different experimental designs make it challenging to compare studies. Furthermore, variations in findings may

also result from the use of insufficient number of isolates employed in some studies, limiting statistical power. For instance, Bechtel and Gibbons (2021), who analyzed 504 *L. monocytogenes* isolates, demonstrated the importance of larger sample sizes for drawing robust conclusions in population genomic studies.

In relation to the biofilm-forming potential of different strains of *L. monocytogenes*, Nilsson et al. (2011) suggested that, the strain's origin may impact its ability to form biofilm under the same culture conditions and high-nutrient medium. It was observed in their study that clinical strains of *L. monocytogenes* formed more biofilms at elevated temperatures of 25 and 37°C, while environmental (food factory) strains produced significantly more biofilm at lower temperatures (10°C) (Nilsson et al., 2011). In another study, *L. monocytogenes* isolated from blood, food, and fresh water were tested for their susceptibilities to BAC and desiccation (Piercey et al., 2017). Compared to the clinical (blood) and food-related strains, fresh water isolates were more sensitive to BAC, requiring lower concentrations of BAC to completely inactivate the cells. The absence of genetic markers such as *Listeria* Genomic Island 1 (LGI1) in fresh water isolates may have accounted for the higher susceptibility to BAC and desiccation (Piercey et al., 2017). The LGI1, which has been detected in clinical and food-processing *L. monocytogenes* strains, encodes not only the putative peptidoglycan hydrolase that aids in bacterial adhesion to surfaces but also the multidrug efflux pumps such as the *emrE*, capable of flushing out BAC molecules (Kovacevic et al., 2016; Piercey et al., 2017). Noteworthy, the contribution of genotype versus phenotype in defining biofilm-forming ability is a complex interplay where genetic makeup provides the potential for biofilm formation, while phenotypic expression is influenced by environmental conditions (Chen & Chen, 2020; O'Toole et al., 2000).

3.4.3 | *Listeria monocytogenes* strains: Resistant and sensitive strains

In some studies, repeated exposure of bacteria to subinhibitory concentrations of disinfectants such as QAC-based biocides resulted in the strains developing enhanced tolerance, and this has been linked to the persistence of bacteria in the food industry (Duze et al., 2021; Jennings et al., 2016; Jiang et al., 2017). Studies have indicated that the conditions present in food industries, including the use of biocides and the ability of *L. monocytogenes* to form biofilms, can contribute to the generation of resistant mutants (Camargo et al., 2017). The detection of antimicrobial-resistant strains in food and food-processing environments underscores the potential risk of the emer-

gence and persistence of such mutants, which could compromise food safety and public health (Duze et al., 2021; Rippa et al., 2024).

In the absence of BAC, BAC-resistant strains of *L. monocytogenes* produced significantly less biofilms compared to the BAC-sensitive strains (Ortiz et al., 2014). Here, resistant strains referred to the strains in which the minimum inhibitory concentration (MIC) of BAC was higher than the reference *L. monocytogenes* strain (EGD-e) (Ortiz et al., 2014). In the presence of BAC, a distinct effect was observed: a BAC concentration of 1.25 and 2.5 mg/L (MIC) significantly reduced the biofilm levels on polystyrene for all strains except for two resistant strains. At 5-mg/L BAC, inhibition of biofilm formation was observed for all strains except one BAC-resistant strain that was able to form biofilms at this concentration (Ortiz et al., 2014). This indicates the potential variations in the effect of subinhibitory concentrations of BAC on the formation of biofilms by resistant and sensitive strains. These variations can be observed among strains with similar MIC but possess different genetic determinants of resistance such as Tn6188 (Ortiz et al., 2014). Likewise in another study, the subinhibitory concentrations of BAC resulted in higher tolerance of the resistant strains than with the sensitive strains (Bonneville et al., 2020). These observations may provide a potential reason for the persistence of certain strains within specific areas in the food-processing environment.

3.5 | *Listeria monocytogenes* biofilms on contact surface materials

3.5.1 | *Listeria monocytogenes* biofilms on biotic surfaces



Scientific literature is replete with studies on the biofilm-forming potential of *L. monocytogenes* on both biotic and abiotic surfaces (Dygico et al., 2021; Hua et al., 2019; Lee et al., 2017; Ölmez & Temur, 2010; Srey et al., 2014; Svoboda et al., 2016). Biocides play a significant role in reducing microbial contamination on fruits and vegetables, serving as a critical control measure in the food safety process. Washing with disinfectants is particularly important for removing soil and debris, inactivating biofilms, and preventing cross-contamination between clean and contaminated produce (Mendoza et al., 2022). While most of the included studies focused on biofilm formation on abiotic surfaces, which is discussed in the next section, four articles (Aryal et al., 2023; Ölmez & Temur, 2010; Srey et al., 2014; Svoboda et al., 2016) reported the efficacy of biocides against *L. monocytogenes* biofilms formed on biotic surfaces, including cantaloupe, melon, cabbage, lettuce, and

bell pepper. In a melon and cantaloupe study, the efficacy of a range of disinfectants (QAC, PAA, and H_2O_2) at different concentrations was tested on *L. monocytogenes* grown at 22°C for 1 h (Svoboda et al., 2016). Although it was shown that differences in surface characteristics, such as roughness or smoothness, impact the adherence of bacteria, resulting in either an increased or a decreased tolerance of biofilms to disinfection, no significant differences in biofilm reduction were observed between disinfectants and melon types (Svoboda et al., 2016). In another biotic surface study, the effect of incubation time (biofilm maturation) and biocide efficacy was investigated by Ölmez and Temur (2010). Biofilms grown on the surface of lettuce leaves at 10°C for 6, 24, and 48 h were treated with ozonated water (2 mg/L), total chlorine (100 mg/L), and organic acid (0.25 g/100 g citric acid and 0.50 g/100 g ascorbic acid). The results showed that the biocides' efficacy was reduced with increasing biofilm age. The study found no treatments effective in eliminating bacterial biofilms (Ölmez & Temur, 2010). The lack of effectiveness observed in treatments could be partially attributed to the attachment of bacterial cells to inaccessible sites, such as stomata and cavities, on the lettuce surface. These hard-to-reach areas provide a protective environment for the bacteria, rendering the biocide treatments ineffective in completely removing the biofilms and achieving the desired microbial reduction (Ölmez & Temur, 2010).

Srey et al. (2014) concluded that the efficacy of chemical disinfectant treatments largely depended on the vegetable type. In their study, biofilms of *L. monocytogenes* formed at 4°C for 24 h on lettuce and cabbage surfaces were treated with 200 ppm of chlorine or 2% lactic acid, citric acid, or malic acid for 5 min. Results showed a reduction of less than 1 log CFU/cm² on lettuces for all disinfectants, whereas greater than 2-log reduction was observed on cabbages for all disinfectants (Srey et al., 2014). The smoother surface of the cabbages contributed to the weaker attachment of *L. monocytogenes* than on the lettuce surface, resulting in less biofilm formation on the cabbage surfaces and, consequently, a more significant microbial reduction after disinfection (Srey et al., 2014). Moreover, it has been demonstrated that plants in the Brassica family, such as cabbage, have a cuticular wax on their surfaces (Lee & Suh, 2021; Zhang et al., 2013). This wax forms a hydrophobic barrier that plays an essential role in protecting the plant against various environmental stresses, including pathogen colonization (Lee & Suh, 2021; Zhang et al., 2013). This barrier may have influenced the ability of *L. monocytogenes* to adhere to cabbage surfaces, subsequently affecting its inactivation. These findings highlight the importance of considering the specific characteristics of different fruits and vegetable surfaces in designing effective pathogen control measures to ensure food products' safety.

3.5.2 | *Listeria monocytogenes* biofilms on abiotic surfaces

In the food industry, a variety of contact surfaces including, SS, glass, rubber, Teflon, polypropylene, cement (floors), marble, and granite are prone to bacterial attachment and biofilm formation (Dygico et al., 2020; Russo et al., 2018; Wang, 2019). Recent studies on the efficacy of biocides have incorporated surfaces that closely mimic those commonly encountered in the food-processing environment. SS, a standard surface material in food-processing equipment, is the material of choice in many food industries due to its exceptional mechanical strength, noncorrosive nature, prolonged lifespan, and ease of cleaning (Gómez et al., 2015). Despite its seemingly smooth appearance, the surface of SS bears intricate scores consisting of numerous channels, pores, and irregularities, providing an ideal harborage site for bacteria and posing significant challenges for effective elimination (Gómez et al., 2015). Moreover, its negative charge and hydrophilic nature encourage the rapid adherence of *L. monocytogenes* (Oloketuyi & Khan, 2017). Other frequently used food-grade contact surface materials include polypropylene, polystyrene, polyethylene (PET), PVC, and rubber. Each surface material possesses unique topography, forming specific micro-cracks and harborage sites that provide the potential to serve as shelters for *L. monocytogenes* cells during disinfection (Dygico et al., 2020; Hua & Zhu, 2024b).

Several studies have highlighted the impact of different contact surface materials on biofilm formation and the efficacy of disinfectants in biofilm elimination (Hua et al., 2019; Kim & Park, 2023). The effectiveness of disinfectants against 7-day-old *L. monocytogenes* biofilms differed significantly depending on the surface material, contact time, and the biocide's concentration (Hua et al., 2019). After a 1-min application of QAC and chlorine, the reduction of *L. monocytogenes* biofilms grown on SS, PET, and rubber showed similar results (~2.3- to 2.6-log reduction). However, it was observed that 400-ppm QAC and 200-ppm chlorine with a 5-min exposure time showed greater effectiveness against biofilms on SS (~3.7-log reduction) compared to rubber and LDPE material surfaces (~2.1-log reduction) (Hua et al., 2019). Lee et al. (2016) observed that a 0.5% concentration of PAA showed a complete inactivation of *L. monocytogenes* biofilms on SS. However, the same concentration of PAA and contact time exhibited ineffective removal of the adherent *L. monocytogenes* cells on polystyrene with less than 50% bacterial reduction. Similarly, PAA at 0.5% was less effective in removing biofilms of *L. monocytogenes* from polystyrene surfaces, resulting in less than 2-log removal compared to the 5-log removal observed on SS (Lee et al., 2017). While these studies did not fully explain the apparent differences in bacte-

rial reductions on these surface materials, it is suggested in other studies that the physicochemical properties such as hydrophilicity and roughness of the contact surfaces may have facilitated bacterial attachment on polystyrene compared to SS (Dhowlaghar et al., 2018). Hua and Zhu (2024b) found that surface defects (worn surfaces) showed an increase in the population of 7-day-old *L. monocytogenes* biofilms and a diminished effectiveness of biocides against the biofilms across different surface materials. These findings emphasize the significant challenges that surface defects pose in the eradication of *L. monocytogenes* biofilms in the food industry. To achieve effective disinfection, it is imperative for the food industry to ensure optimal surface conditions and, if possible, change worn out surfaces that may have the potential to harbor bacterial growth and biofilm formation.

Research has demonstrated that the excessive application of disinfectants can lead to physical alterations to contact surfaces, such as increased roughness and the development of cracks and pores. Instances of disinfectant-induced corrosion on SS have been documented, with damage that is not always visible to the naked eye (Truscott, 2017). The continued use of disinfectants on already compromised surfaces can worsen the damage, providing more favorable conditions for bacterial colonization. For example, Jo et al. (2019) found that simulated wiping of surfaces like LDPE and polycarbonate with disinfectants, including QAC-based biocides, over a period of 6 months, can increase surface roughness, leading to changes in surface properties such as wettability or water contact angle. This finding suggests that the repeated use of certain disinfectants may accelerate surface deterioration. It is important to note that surface materials with different water contact angles may facilitate similar bacterial attachment due to similar surface roughness. For instance, Dhowlaghar et al. (2018) showed that attachment of *L. monocytogenes* cells on SS, PET, and polyurethane was similar even though the water angle indicated that SS was more hydrophilic with an angle of 34°, whilst PET and polyurethane had water contact angles of 67° and 72°, respectively, making them less hydrophilic. However, the surface roughness between the materials indicated that PET and polyurethane exhibited a similar rough surface to SS. Hence, the attachment of bacteria cells was similar on all the surfaces (Dhowlaghar et al., 2018). Therefore, when assessing bacterial adherence on surface materials and the potential efficacy of chemical agents, it is essential to consider parameters such as surface energy, water contact angles, and surface roughness. These factors collectively contribute to the degree of bacterial attachment, providing insights into effective strategies for bacterial control and removal.

3.5.3 | Organic matter or soiling

In recent years, the impact of organic matter or soiling on cleaning and disinfection efficacy has been extensively studied and incorporated into regulatory guidelines, such as the European Standard EN 13697 (European Standard, 2015), and documents from the European Chemicals Agency (ECHA) (ECHA, 2018), and EPA (EPA, 2012, 2018). Organic soiling is integrated into biocide efficacy studies to simulate real-world conditions where disinfectants are used, ensuring the practicality of the test results. In laboratory-scale testing of disinfectant's efficacy against *L. monocytogenes* biofilms, the organic matter employed included bovine serum albumen (Dygico et al., 2021; Paluszak et al., 2021), catfish mucus extract (Dhowlaghar et al., 2018), diluted milk and apple juice (Hua et al., 2019; Korany et al., 2018), and food soils media, including mussel cooking juice and intervalvar water of mussel (Ibusquiza et al., 2011). In food-processing facilities, the presence of organic matter such as proteins, fat, minerals, and carbohydrates on food-contact surfaces alters the physicochemical properties of surfaces and affects the efficacy of biocides (Abban et al., 2012; Hua & Zhu, 2024b; Hua et al., 2019; Korany et al., 2018).

In the presence of food residues from apple juice or milk, Korany et al. (2018) reported a decrease in the effectiveness of biocides against 2- and 7-day-old *L. monocytogenes* biofilms formed on polystyrene. Specifically, for 7-day-old biofilms, 160-ppm PAA, 5-ppm ClO₂, and 200-ppm QAC reduced *L. monocytogenes* biofilms by 3.4, 1.3, and 0.9 log CFU/coupon, respectively, and by 3.6, 2.2, and 1.9 log CFU/coupon, respectively, for 2-day-old biofilms in the presence of apple juice. In contrast, reductions of 4.3, 1.7, and 1.4 log CFU/coupon (7-day-old biofilm) and 4.8, 3.6, and 2.4 log CFU/coupon (2-day-old biofilm) were observed in the absence of organic soiling (Korany et al., 2018). Similarly, Hua et al. (2019) reported a decrease in the efficacy of QAC, ClO₂, and NaOCl against 7-day-old *L. monocytogenes* biofilms formed on SS in the presence of diluted apple juice or milk.

Although studies have shown that the efficacy of various biocides against *L. monocytogenes* biofilms is significantly reduced in the presence of organic matter, NaOCl is particularly susceptible to this effect, with its listericidal activity severely impaired by low concentrations of organic material. For instance, Dygico et al. (2021) found that the application of NaOCl on a 3-day-old *L. monocytogenes* biofilm resulted in a 1.7-log CFU/cm² reduction of biofilms grown in the presence of organic matter, compared to a reduction of 4.7 log CFU/cm² without any organic matter. Nonetheless, the other biocides containing active agents such as QAC, glutaraldehyde, BAC,

and peroxides exhibited comparable biofilm inactivation rate with or without the presence of organic soiling (Dygico et al., 2021). In another study, Nyati et al. (2012) demonstrated that the presence of bovine serum albumin (0.3% [w/v]) inhibited the complete inactivation of 3-day-old *L. monocytogenes* biofilm by NaOCl, while QAC maintained its listericidal activity even in high organic loads (3.0% [w/v]). The diminished effectiveness of NaOCl in the presence of organic matter may be attributed to the interaction between the biocide's active agents and the organic matter, potentially reducing the concentration and, consequently, their efficacy (Chowdhury et al., 2019; Dygico et al., 2021). Organic materials can also act as a protective barrier for microorganisms, inhibiting the contact between the biocidal agents and the bacteria (Hua et al., 2019; Lim et al., 2023). These observations indicate the importance of effective cleaning before disinfection.

While the above studies reported the potential for organic matter to significantly reduce the activity of biocides to inactivate *L. monocytogenes* biofilms, other studies have shown that certain biocides retain their effectiveness in the presence of organic matter. For instance, Hua et al. (2019) found that organic matter impaired the efficacy of sanitizers such as ClO₂ against *L. monocytogenes* biofilms but had less influence on the efficacy of PAA. Specifically, PAA at 160 or 200 ppm for 5 min resulted in a greater biofilm reduction of 3–3.7 log CFU/coupon compared to ClO₂ at 5 ppm for 5 min, which showed a 1- to 2-log CFU/coupon reduction in the presence of organic matter, regardless of surface type (Hua et al., 2019). Similarly, Kragh et al. (2024) reported that soiled conditions did not affect the MIC values for PAA, indicating that it maintained its efficacy against *L. monocytogenes* biofilms even when organic residues were present. These findings suggest that the impact of organic matter on biocide effectiveness can vary depending on the specific biocidal agent employed.

3.6 | Biocide: Type, concentration, contact time, and application

3.6.1 | Biocide type

The assessment of disinfectants' efficacy against *L. monocytogenes* biofilms typically involves treating the biofilms with known concentrations of a specific biocide for a given contact time. The majority of studies have primarily investigated the effect of factors such as the type of active agents, concentration, exposure time, mode of application, and pH of biocide on the inactivation of *L. monocytogenes* biofilms on the predominant contact surfaces found in food-processing plants (Belessi et al., 2011; Dhowlaghar et al., 2018; Lee et al., 2016, 2017).

Among the commonly used biocides in the food industry, QAC, PAA, and NaOCl have been frequently employed in studies. Comparing the effectiveness of biocides against *L. monocytogenes* is challenging due to the different proportions of active ingredients and their differing modes of action, as well as the differing influence of extrinsic factors such as soiling, temperature, and so forth. However, under the same biofilm formation and treatment conditions, some studies have attempted to compare the efficacy of disinfectants with different active agents based on bacterial logarithmic reductions. For instance, in a comparison to QAC, chlorine, and chlorine dioxide, Hua et al. (2019) reported that PAA was the most effective disinfectant against 7-day-old *L. monocytogenes* biofilms, achieving ~1.5-log CFU/coupon greater reduction than the other biocides irrespective of the surface type. The reasons suggested for this could be due to PAA's high reactivity, oxidizing capacity, rapid decomposition, and relatively low molecular weight, which facilitated the penetration into the biofilm matrix (Hua et al., 2019). Similarly, Barroso et al. (2019) reported that PAA exhibited greater efficacy compared to a BAC biocide on SS.

3.6.2 | Biocide concentration

The correlation between a biocide's concentration and bacterial inactivation is often characterized by a sigmoidal curve rather than a linear curve (Tamburro et al., 2015). Microbial populations show increased tolerance at lower concentrations of disinfectants, making it challenging to effectively eradicate them (Das & Lalitha, 2022; Rahman & Sahoo, 2022). However, a gradual increase in disinfectant concentration is associated with a greater reduction of *L. monocytogenes* populations until a threshold is reached, where further concentration increments may not cause any significant bacterial inactivation (Das & Lalitha, 2022; Dhowlaghar et al., 2018; Henriques & Fraqueza, 2017; Hua et al., 2019; Tamburro et al., 2015). For instance, in a study with a 5-min contact time with 10 ppm of chlorine, the population of *L. monocytogenes* biofilms on a glass surface was reduced from 5.89 to 1.54 log CFU/cm² (Das & Lalitha, 2022). Further increasing the concentration to 20 ppm reduced biofilms to 1.1 log CFU/cm². In that same study, exposure of *L. monocytogenes* biofilms to 50-, 100-, 150-, and 200-ppm BAC for 5 min resulted in reductions of 1.28, 2.34, 2.98, and >3 log CFU/cm², respectively. Surprisingly, treatment with 950 ppm BAC for 30 s did not completely inactivate biofilms, with recovery of the bacterial cells of approximately 1 log CFU/cm². Observed recoverable cells after treatment could be attributed to injuries sustained by portions of bacterial cells due to exposure to sublethal concentrations of the disinfectants (Das & Lalitha, 2022).

Positive correlations between increased concentrations of biocides and significant reduction of *L. monocytogenes* biofilms have been reported in many studies (Dhowlaghar et al., 2018; Korany et al., 2018; Purkrťová et al., 2010; Silva et al., 2017). Conversely, divergent findings have been reported, suggesting that increased biocide concentrations may not consistently result in substantial biofilm inactivation (Lee et al., 2016, 2017). Certain studies have suggested that achieving significant reductions in *L. monocytogenes* biofilms may require biocide applications beyond the manufacturer's recommended concentrations (Cruz & Fletcher, 2012; Vázquez-Sánchez et al., 2017). Factors such as the age of biofilms, strain type, growth temperatures, and the active agents in the biocide may contribute to the contrasting findings reported in these studies (Cruz & Fletcher, 2012; Dygico et al., 2021). Cruz and Fletcher (2012) observed significant differences in biofilm inactivation potential among different commercial disinfectants with the same concentrations of BAC. The presence of other active agents in the biocides potentially influenced the observed differences. Dygico et al. (2021) found that two QAC biocides showed a 4.5-log reduction of *L. monocytogenes* biofilms on concrete surfaces with different contact times. One of the QAC biocides required a 5-min contact time, while the other required 60 min to achieve a similar reduction, with no significant difference in their overall efficacy. This disparity was attributed to the variation in glutaraldehyde concentration (537 ppm) in one of the QAC biocides compared to the other biocide (1430 ppm) (Dygico et al., 2021). This emphasizes the importance of taking into account the concentrations and formulations of active agents present in biocides. Moreover, it is crucial to adhere to optimum storage conditions recommended by manufacturers, as biocides may undergo degradation of active agents, potentially resulting a decrease in concentration and effectiveness (Dygico et al., 2021).

3.6.3 | Biocide contact time

Adequate contact time between biocides and *L. monocytogenes* biofilms is one of the vital factors for evaluating biocidal efficacy. Extended exposure times have been positively associated with a significant reduction of *L. monocytogenes* biofilms on contact surfaces relevant to the food industry (Dhowlaghar et al., 2018; Hua et al., 2019; Kim et al., 2023; Korany et al., 2018). A 1-min treatment with NaOCl (144 ppm) on 4-day-old *L. monocytogenes* biofilms resulted in a reduction of ~3.6 log CFU/coupon on SS. Extending the treatment duration to 3 min achieved a more substantial reduction of ~4.8 log CFU/coupon (Dhowlaghar et al., 2018). This trend was consistent when using different biocides and concentrations. For instance, applying NaOCl at

72 ppm for 1 and 3 min reduced *L. monocytogenes* biofilms by approximately 2.9 and 4.1 log CFU/coupon, respectively. Similarly, using a mixture of peroxide- and octanoic acid-based biocides at 95 ppm for the same contact times led to reductions of roughly 4 and 6.2 log CFU/coupon, respectively, against 4-day-old *L. monocytogenes* biofilms (Dhowlaghar et al., 2018).

Hua et al. (2019) found that extending treatment durations enhanced the efficacy of disinfectants against 7-day-old *L. monocytogenes* biofilms on SS, LDPE, PVC, PET, and rubber surfaces; however, this effect was biocide dependent. QAC, NaOCl, and ClO₂ showed significantly increased efficacy with a 5-min contact time compared to a 1-min exposure. In contrast, PAA exhibited similar reduction rates, regardless of the treatment duration (Hua et al., 2019). This could be attributed to PAA's oxidative potential, which enables rapid and efficient microbial penetration and inactivation within shorter time periods compared to other disinfectants (Hua & Zhu, 2024a). This observation indicates that when decontaminating *L. monocytogenes* biofilms on FCS, the choice of disinfectant and the duration of treatment should be carefully considered to ensure effective sanitation practices.

3.6.4 | Biocide application

In most experimental designs, the standard mode of biocide application in lab-based studies involves immersing the coupons containing the biofilms into liquid solutions of the disinfectant (Dygico et al., 2021; Hua et al., 2019; Korany et al., 2018; Skowron et al., 2019). In contrast, the application of biocide as a fine mist in air (i.e., aerosolization) to control bacteria and its associated biofilms has been reported in recent studies (Kim & Park, 2023; Kim et al., 2023; Park et al., 2012). This technique, commonly applied in whole room disinfection, is reported to effectively penetrate biofilm, as well as reach inaccessible environmental surfaces (Kim et al., 2023; Møretro et al., 2019; Park et al., 2012). In research by Park et al. (2012), 6-day-old *L. monocytogenes* biofilms formed on SS and PVC coupons were treated with aerosolized NaOCl (100 ppm) and PAA (100, 200, and 400 ppm) for exposure durations of 5, 10, 30, and 50 min. It was observed that higher concentrations and extended contact times positively correlated with greater biofilm reduction. Treatment with 100-ppm NaOCl and PAA significantly reduced *L. monocytogenes* biofilms by 2.46 and >4.58 log CFU/coupon, respectively, after 50 min of exposure on all surfaces. PAA concentrations of 200 and 400 ppm reduced *L. monocytogenes* biofilms to below the detectable limit (1.48 log CFU/coupon) after 30 min of exposure. The finding suggested that disinfectants applied in fine mist have potential to control

bacterial biofilms when applied at higher concentrations and prolonged contact times. Nonetheless, further studies comparing biocide application methods to control *L. monocytogenes* biofilms are warranted.

Sekhon et al. (2022) investigated the efficacy of incorporating ultrafine bubbles containing various gases, including air, CO₂, or N₂, at a concentration of 1.03×10^9 particles/m for 5 min into NaOCl solutions at concentrations of 100 and 200 ppm to inactivate 3-day-old *L. monocytogenes* biofilms on SS. The results showed that a 1-min treatment of 100-ppm NaOCl incorporated with air and CO₂ significantly reduced *L. monocytogenes* biofilms, respectively, by 5 and 4.9 log CFU/cm², compared to no gas incorporation, which resulted in 3.7-log CFU/cm² reduction. The incorporation of NaOCl with N₂ at 200 ppm resulted in a significant reduction of *L. monocytogenes* biofilm by 1.6 log compared to 100 ppm. However, when incorporated into air or CO₂ at 200 ppm, no significant differences in biofilm inactivation were observed relative to the 100-ppm treatment. The potency of this disinfection approach may be due to the generation of reactive free radicals, controlled supply of gases during the dissolution stage, stability of ultrafine bubbles, and an enhanced efficiency of mass transfer (Sekhon et al., 2022; Unger et al., 2023). However, further research is needed to elucidate the mechanisms underlying the efficacy of incorporating ultrafine bubbles in biocidal products for controlling pathogenic bacteria and their associated biofilms.

The use of biocides at sublethal concentrations over extended periods may lead to increased tolerance of bacterial cells to these biocides. For instance, after exposure to QACs, some *L. monocytogenes* strains showed a two- to fourfold increase in MIC (Kastbjerg & Gram, 2012). Similarly, *L. monocytogenes* adapted to trisodium phosphate (TSP) could grow in the presence of 2.53 times higher concentration than the MIC of unexposed cultures (Rodríguez-Melcón et al., 2023). To combat this, disinfectant rotation has been proposed as an effective strategy to control *L. monocytogenes* biofilms (Aryal & Muriana, 2019; Dhowlaghar et al., 2018; Murugesan et al., 2015). Disinfectant rotation refers to a systematic and methodical practice of scheduled alternation between disinfectants containing different active agents, with the intention of mitigating the potential mutability, preventing the acquisition of genetic elements that may enhance disinfectant tolerance, or reducing the expression of efflux pumps in bacteria (Bland et al., 2022). Biocides are also used in the form of a mixture with different active agents. For instance, Aryal and Muriana (2019) investigated the efficacy of a mixture of biocides (QAC, hydrogen peroxide, and diacetin) in inactivating 7-day-old *L. monocytogenes* biofilms. The “hybrid” biocide, applied at 5% and 10% concentration, resulted in a reduction of >6 log CFU/cm² after

2.5 and 1 min, respectively. In comparison, QAC or chlorine individually achieved a 2-log reduction after 2 h of exposure. Similarly, Dhowlaghar et al. (2018) found that individual disinfectants, such as acids, chlorine, and QAC, led to 2.3–4.6 log CFU/coupon reduction following 1 and 3 min of exposure. However, the mixture of active agents (QAC and hydrogen peroxide or peracetic acid, hydrogen peroxide, and octanoic acid) resulted in the complete inactivation of culturable biofilm cells at concentrations below the manufacturer’s recommended concentrations, irrespective of the contact times (1 and 3 min). These findings indicate that biocidal products containing a mixture of active agents are more effective for controlling *L. monocytogenes* biofilms than individual active agents.

3.7 | Biofilm: Type, maturation, incubation temperature, formation, and enumeration method

3.7.1 | Biofilm type: Batch and fed-batch systems

The batch or static system is a widely employed method for assessing biofilm formation in many experimental setups (Das & Lalitha, 2022; Dygico et al., 2021; Hua et al., 2019; Kocot & Olszewska, 2019; Korany et al., 2018). This method involves the incubation of biofilms in closed systems where no new nutrients are added throughout the experiment. This approach offers reproducibility due to its controlled growth conditions, enabling comparative results across multiple trials (Klopper et al., 2023). Furthermore, static biofilms mimic biofilms in inaccessible contamination sites, where dynamic food fluids pose a challenge for access (Hua & Zhu, 2024b). Additionally, static biofilms are particularly valuable for investigating the initial stages of biofilm formation, and in some instances, biofilm development can be detected within less than 60 min (Merritt et al., 2005). Nonetheless, a major limitation is the inability of the model to better simulate conditions found in food-processing environments such as the constant influx of microorganisms and a continuous flow of organic matter, including fat, carbohydrate, proteins, and minerals, depending on the processing factory (Kocot & Olszewska, 2019; Rodríguez-López & Cabo, 2017). In fed-batch systems, nutrients are added to the culture periodically without removing any culture fluid until the end of the experiment, simulating more real-world scenarios (Kocot et al., 2021). Despite the practical relevance of fed-batch biofilms, limited studies exist on the comparison of *L. monocytogenes* biofilms grown under static and continuous conditions and their disinfection susceptibilities.

Kocot et al. (2021) investigated the comparative effect of 48-h *L. monocytogenes* biofilms on SS grown under static and continuous-flow/fed-batch conditions and their susceptibility to BAC. The study demonstrated significantly higher biofilm growth under static conditions compared to a fed-batch system, with a difference of 2 log CFU/coupon. The authors posited that the higher biofilm growth in static conditions was due to cell adhesion influenced by the force of gravity, while under fed-batch conditions, cell adhesion was impeded by the vertical placement of the coupons in the biofilm reactor, along with constant agitation creating more stressful growth conditions. Nonetheless, in terms of their BAC susceptibility, *L. monocytogenes* biofilms from fed-batch conditions exhibited less susceptibility compared to batch conditions. The authors hypothesized that the “adhesion effort” by bacteria influenced the higher tolerance in cells in the fed-batch conditions. Moreover, the continuous addition of fresh media in the fed-batch conditions potentially influenced metabolic activity, increased EPS production, regulated gene expression, and synthesized molecules essential for quorum-sensing mechanisms (Kocot et al., 2021).

3.7.2 | Biofilm maturation (biofilm age)

It is hypothesized that planktonic cultures are more susceptible to disinfectants than their biofilm counterparts (Andrade et al., 2020; Fagerlund et al., 2017; Nakamura et al., 2013). Furthermore, longer incubation periods for biofilms have been associated with increased tolerance of biofilms to disinfectants (Belessi et al., 2011; Fagerlund et al., 2020; Ibusquiza et al., 2011; Kim & Park, 2023; Korany et al., 2018). As biofilms mature, they undergo structural and physiological adaptations that enhance their resistance to disinfectants. Over time, the production of EPS within the biofilm matrix increases, potentially limiting the penetration of antimicrobial agents (Abdallah et al., 2015). Additionally, factors such as nutrient gradients leading to slower bacterial growth, the activation of stress response genes, and the emergence of persister cells and resistant subpopulations contribute to the enhanced tolerance of mature biofilms to disinfectants (Ito et al., 2009).

In a study by Korany et al. (2018), *L. monocytogenes* biofilms formed on polystyrene for 2 and 7 days were treated with different concentrations of QAC, PAA, chlorine, chlorine dioxide, and ozonated water for 1 min. When treated with 100-ppm chlorine, log reductions of 1.91 and 0.98 were observed for 2- and 7-day-old biofilms, respectively. Similar observations were reported for the different disinfectants used in the study, except for PAA, which showed no significant differences in results, irrespective of

concentrations and age of biofilms (Korany et al., 2018). In a separate study, increased tolerance to PAA was observed in 11-day-old biofilms compared to 4-day-old biofilms of *L. monocytogenes* (Ibusquiza et al., 2011). These findings suggest that as *L. monocytogenes* biofilms age, inactivation with disinfectants may become less effective, leading to bacterial persistence on contact surfaces. This indicates the importance of frequent cleaning and disinfection to prevent the buildup and maturation of biofilm.

Interestingly, Korany et al. (2018) observed that biofilms of the same age formed by single and multiple strains (cocktail) of *L. monocytogenes* exhibited different susceptibilities to certain disinfectants. Particularly, a 1-min treatment with 4-ppm ozonated water against 2-day-old biofilms resulted in reductions of 4.1 and ~2.2 log for single and cocktail (six strains) biofilms, respectively. This observation may be due to an increased production of the EPS in the multistrain biofilms compared to the single-strain biofilms. In addition, the strains used in the experiment belonged to different serotypes, with different abilities to attach to material surfaces. Although the specific strain used for the formation of single-strain biofilm was not stated, strain variability (resistant or sensitive, persistent or sporadic) and their potential to form stronger biofilms, as well as susceptibilities to disinfection, may account for such differences (Nilsson et al., 2011; Ortiz et al., 2014; Russo et al., 2018).

3.7.3 | Biofilm incubation temperature

Studies have demonstrated that incubation temperature modulates the biofilm formation process (Kadam et al., 2013; Pang et al., 2019). The correlation between temperature and biofilm formation is often characterized as a bell-shaped curve model, where an optimal temperature threshold facilitates a strong cell attachment on contact surfaces (Das & Lalitha, 2022; Nilsson et al., 2011). In most experimental studies, temperatures ranging between 20 and 37°C have been employed. Some studies have also explored biofilm formation at suboptimal lower temperatures, ranging between 4 and 15°C. It is noteworthy that, in experimental studies, the selection of a particular temperature for biofilm formation normally aims to simulate the temperature prevalent in that food-processing environment. For example, Dygico et al. (2021) formed *L. monocytogenes* biofilms at 25°C before assessing the efficacy of disinfectants used in the mushroom industry to inactivate these biofilms. The selection of 25°C was based on the temperature in the mushroom growth environment in the study's country. Likewise, to simulate the temperatures in the salmon-processing environment, *L. monocytogenes* biofilms were grown at 4 and 15°C, and

their susceptibilities to a QAC disinfectant were assessed (Pang et al., 2019). Oxaran et al. (2018) grew biofilms at 25°C to reproduce the temperature conditions in the dairy-processing environment. Replicating temperatures specific to a food-processing environment is crucial for developing predictive models that enhance our understanding of the biofilm formation process under those conditions, enabling anticipation and effective management of biofilm-related issues.

The parameter commonly employed in studies to assess the effectiveness of biocidal agents against *L. monocytogenes* biofilms is the degree of microbial reduction (Dygico et al., 2021; Hua et al., 2019). Prior to disinfection, it is important that the population of adherent cells surpass a designated threshold. To this end, an initial bacterial population of greater than 4 log is mostly cultivated on surfaces before any biocidal treatment is applied (Dhowlaghar et al., 2018; Korany et al., 2018; Pang et al., 2019). To achieve the ideal bacterial population, biofilm growth parameters are optimized. As a result, the growth of biofilms at lower and higher temperatures may require longer and shorter incubation times, respectively, to facilitate maximal cell attachment to surfaces. In a study by Pang et al. (2019), biofilms grown at lower temperatures (4 and 15°C) were incubated for 14 days to achieve 6 log biofilm population, whereas in a separate study, biofilms incubated for 3 days at 25°C achieved the same population (Dygico et al., 2021). Indeed, some studies have formed biofilms at higher temperatures for longer incubation periods to achieve maximal biofilm population (Andrade et al., 2020; Park et al., 2012). This variation may be due to strain variability, influencing the different degrees of bacterial attachment to surfaces, and the presence of interfering substances in culture media. While some strains may increase metabolic activity, including the biofilm formers at higher temperatures, resulting in faster growth and development, other strains may slow down metabolism and biofilm formation at lower temperatures, depending on the genetic makeup of the strain (Belessi et al., 2011; da Silva et al., 2021; Piercey et al., 2017; Silva et al., 2020).

Observations from studies on the impact of biofilms grown at different temperatures and their susceptibility to disinfection have been conflicting, possibly due to differences in strains and biofilm-forming parameters (Belessi et al., 2011; Nilsson et al., 2011; Ölmez & Temur, 2010). To evaluate the impact of temperature on disinfectant tolerance of *L. monocytogenes* biofilms on SS surface, Belessi et al. (2011) found that biofilms formed at 20°C showed reduced efficacy in comparison to those formed at 5°C when treated with 2% PAA. Similar trends have been reported in studies where the effectiveness of disinfectants significantly reduced when biofilms were grown at higher temperatures employed in studies (Olszewska et al., 2016;

Pang et al., 2019; Skowron et al., 2019). Conversely, the effectiveness of disinfectants against biofilms of *L. monocytogenes* formed at 12 and 37°C showed that the former temperature produced biofilms that exhibited higher tolerance than the latter temperature (Lourenco et al., 2011). These variations reported in studies may potentially be due to the specific temperatures chosen, differences in biocide type, strains, biofilm maturation, and biofilm type.

3.7.4 | Biofilm formation and enumeration method

In biofilm–biocide efficacy assays, different culture media and concentrations have been used to prepare cultures, thereby making comparative evaluation across studies challenging. To cultivate *L. monocytogenes* biofilms on surfaces, different nutrient-limiting (diluted media) or nutrient-rich broths have been employed. These include, for example, tryptone soy broth (TSB) (Lee et al., 2016; Olszewska et al., 2016; Poimenidou et al., 2016), modified Welshimer's broth (MWB) (Hua et al., 2019; Korany et al., 2018), brain heart infusion (BHI) broth (Skowron et al., 2019; Stoller et al., 2019), buffered peptone water (BPW) (Andrade et al., 2020; Henriques & Fraqueza, 2017), salmon broth (Pang et al., 2019), and catfish mucus extracts (Dhowlaghar et al., 2018). Some studies have supplemented broths with yeast extract (YE), such as TSBYE (Barroso et al., 2019; Fagerlund et al., 2017; Ortiz et al., 2014) and BHIYE (Dygico et al., 2021). The addition of yeast extracts to broth is known to enhance biofilm formation by providing essential nutrients, including amino acids, vitamins, minerals, and growth-stimulating compounds that support the metabolic activities of cells during biofilm formation (Solihin et al., 2021).

The comparative impact of different culture media/broths on biofilm formation and disinfectant tolerance has not been extensively studied. Korany et al. (2018) formed *L. monocytogenes* biofilms in MWB and TSB media and found that biofilm density in MWB was significantly higher than in TSB. This observation highlights the challenge of comparing the efficacy of biocides across studies, since growth media play a pivotal role in biofilm formation. In another study, Pang et al. (2019) highlighted that culture medium influenced both biofilm formation and disinfectant tolerance, with a greater biofilm population formed in salmon broth compared to diluted TSB (1:20). The increased cell density in salmon broth resulted in limited biocide penetration, reducing its efficacy against *L. monocytogenes* biofilms (Pang et al., 2019). Skowron et al. (2019) investigated disinfectants' efficacy against *L. monocytogenes* biofilms formed in nutrient-rich or nutrient-depleted medium.

They observed that biofilms formed with nutrients (BHI) at 50% of the manufacturer's recommendation exhibited greater tolerance to disinfectants than those formed in nutrient-abundant environments, where nutrient levels were maintained at either 100% or 150% of the manufacturer's recommendation. A contrasting observation was reported by Kyoui et al. (2016). In their study, *L. monocytogenes* biofilms formed in MWB containing three different glucose concentrations (0.1%, 1.0%, and 2.0%) showed that, when treated with NaOCl, the biofilms formed with glucose concentrations at 1.0% and 2.0% exhibited enhanced tolerance than at 0.1%. These findings suggest that variability in growth media—whether nutrient-rich or nutrient-limiting—and the addition of other components in growth media may influence the tolerance or susceptibility of *L. monocytogenes* biofilms during disinfection. This phenomenon is observed in controlled lab-based studies and is likely to be exacerbated in real-world settings.

Belessi et al. (2011) investigated the impact of salt concentrations (0.5%, 7.5%, and 9.5%) on biofilm formation and its tolerance to PAA (2% for 1, 2, 3, and 6 min) and warm water (60°C for 20 min). To achieve comparable biofilm populations (10^4 to 10^5 CFU/cm²) on SS before treatment, *L. monocytogenes* biofilms grown at 5 and 20°C were incubated for 19 and 12 days, respectively. Results indicated that biofilms formed at 20°C had a higher population than those at 5°C, possibly due to increased flagella production at 20°C, enhancing attachment to SS surface (Belessi et al., 2011). Adding to that, the growth media (TSBYE) with a 9.5% salt concentration exhibited increased biofilm production compared with the 0.5% salt concentration, highlighting the effect of increasing ionic strength on biofilm formation. When treated with warm water, a reduction of <1 log CFU/cm² was observed, suggesting that using warm water alone for cleaning SS utensils like blades and knives may not be an effective disinfection approach. Conversely, treatment with 2% PAA significantly reduced *L. monocytogenes* biofilms, and this reduction increased with extended exposure times. Treatment with 2% PAA for 1 and 2 min reduced biofilms formed at 20°C by 2–2.9 log CFU/cm² and below the LOD (0.3 log CFU/cm²), respectively. Complete elimination of biofilms was observed after 6 min of exposure. The authors also investigated the effect of pH (5 and 7) of growth media on biofilm formation before biocide application. For this, *L. monocytogenes* biofilms were formed on SS at 4°C for 4, 8, and 12 days of incubation. After 12 days of incubation, the biofilm population for growth media with pH 5 and 7 was 3.6 and 4.6 log CFU/cm², respectively. However, statistical analysis showed no significant impact ($p > .05$) of pH on biofilm formation. Following this, a 6-min treatment with 1% QAC demonstrated that

matured *L. monocytogenes* biofilms (aged 8 and 12 days) exhibited greater tolerance to the biocide (Belessi et al., 2011). Putting all this together, the results from their study highlight the multifactorial effect of pH, salt concentration, and biofilm maturation in dictating biocide efficacy against *L. monocytogenes* biofilms.

The biofilm formation assay typically involves inoculating surfaces with bacterial suspensions and subsequent incubation over a given period. Inoculating coupon surfaces with droplets of the cell suspension and allowing the coupons to air-dry may enhance bacterial tolerance during disinfection, as has been reported by Park et al. (2012). In their study, two biofilm formation methods were compared in terms of their disinfectant susceptibilities. The first method involved immersing coupons completely into bacterial suspensions with immediate incubation at 4°C for 24 h to enhance cell attachment. PVC and SS coupons were then washed gently in distilled water with gentle agitation for 5 s. Following this, the washed coupons were immersed in TSB media with further incubation at 25°C for 6 days (Park et al., 2012). In the second method, coupons were inoculated with droplets of the cell suspension, air-dried for 3 h before immersing into distilled water, and then incubated at 25°C for 6 days (Park et al., 2012). When treated with aerosolized disinfectants, biofilms formed using the second method exhibited greater tolerance to aerosolized disinfectants, particularly on PVC surfaces, compared to those formed on SS. While the authors did not explain this observation, a plausible reason might be the enhanced initial attachment during the air-drying period in the second method, which was lacking in the first (Silva et al., 2020). This suggests that the biofilm formation method employed in studies may dictate the level of bacterial attachment on surfaces, thereby influencing the removal efficacy of disinfectants.

Thus far, the factors herein discussed as impacting the assessment of biocide efficacy against *L. monocytogenes* biofilms may be termed as “direct factors.” However, additional “indirect factors” that must be taken into account include the growth media used for enumerating surviving cells, addition of biocide neutralizer, and the detachment of surviving cells on contact surfaces after biocidal treatment. Moreover, verifying that all cells have been effectively removed from the surface is a critical aspect of the enumeration. Nonoptimized enumeration methods, including incomplete cell detachment, may lead to inaccurate estimation of the actual adherent viable cells on surfaces, thereby affecting data interpretation and the assessment of a biocide's efficacy. Consequently, a critical step in enumeration is the effective detachment of cells from contact surfaces into appropriate solutions before serial dilutions and plating. To do this, sonication has been employed in some studies (Dygico et al., 2021; Korany et al.,

2018; Pang et al., 2019). Korany et al. (2018) conducted a comparative assessment between mechanical scraping and 1-min sonication to dislodge 2-day-old *L. monocytogenes* biofilms on polystyrene surfaces. The results indicated comparable detachment efficiencies for both methods. In addition, sonication durations ranging from 1 to 5 min resulted in similar quantification of *L. monocytogenes* biofilms. However, beyond 5 min of sonication, a decrease in viable cells was observed, suggesting cell damage during extended sonication durations (Korany et al., 2018).

3.8 | *Listeria monocytogenes* in mixed-species biofilms

In the food-processing environment, *L. monocytogenes* cohabit and interact with a diverse array of bacteria. The ability of *L. monocytogenes* to form mixed-species biofilms with other bacteria has been linked to its persistence in the food industry (Heir et al., 2018). Mixed-species biofilms can provide a protective environment for *L. monocytogenes*, which can contribute to its survival against cleaning and disinfection processes (Chen et al., 2024). The interaction in mixed-species biofilms can manifest as cooperative, where the secretion of enzymes or metabolic cross-feeding between the different microbial species benefits one or all interacting species. Alternatively, it can be competitive, with one member of the multispecies community dominating by competing for space and nutrients, often using an indirect strategy such as secreting antimicrobials (Olszewska & Diez-Gonzalez, 2021). Mixed-species biofilms, formed in such interactions, are often stable and exhibit enhanced tolerance toward disinfectants (Giaouris et al., 2013; Ibusquiza et al., 2012). Consequently, there is a growing interest in recent times to assess the effectiveness of disinfectants on mixed-species biofilms containing *L. monocytogenes*. To examine their disinfection susceptibilities, *L. monocytogenes* has been co-cultured with various bacteria, including *Lactobacillus* spp. (Bonnevill et al., 2021; Haddad et al., 2021; Olszewska & Diez-Gonzalez, 2021; Van der Veen & Abee, 2011), *Pseudomonas* spp. (Giaouris et al., 2013; Ibusquiza et al., 2012; Lourenco et al., 2011; Pang & Yuk, 2019; Papaioannou et al., 2018; Rodríguez-López et al., 2022; Thomassen et al., 2023; Yamakawa et al., 2018), *Enterococcus* spp. (da Silva Fernandes et al., 2015), *Salmonella* spp. (Kostaki et al., 2012), and *E. coli* (de Oliveira et al., 2012; Kocot et al., 2021). Table 4 provides a summary of studies on biocide efficacy against multispecies biofilms containing *L. monocytogenes*.

Similar to the previously described factors, conflicting observations have been reported regarding the susceptibility of *L. monocytogenes* to disinfection in mixed-species

biofilms. Oxaran et al. (2018) tested the efficacy of PAA against multispecies biofilms formed with two distinct background communities, either for *L. monocytogenes* or for *Staphylococcus aureus*. Their results showed that susceptibility of sessile *L. monocytogenes* cells to PAA at a concentration of 0.0375% resulted in a 3-log bacterial reduction in either mono- or multispecies biofilms. At that same concentration of PAA, a reduction of 4.6 and 1.1 log was observed in *S. aureus* mono- and multispecies biofilms, respectively, indicating that the multispecies biofilms protected *S. aureus*. A similar observation was reported when the background communities were exchanged for each pathogen. Specifically, when the background community for *S. aureus* was interchanged with *L. monocytogenes*, treatment with 0.075% PAA resulted in 3.4- and 1.9-log reductions for mono- and multispecies *L. monocytogenes* biofilm, respectively. This finding suggests that biocide efficacy against bacteria in multispecies is dependent on the microbial community in which the bacteria is found, rather than the individual pathogen itself (Oxaran et al., 2018). This may provide insights into the apparently contradictory observations reported in various studies.

It has been shown that the initial inoculum concentrations can influence species dominance in dual-species biofilms. In their study, de Grandi et al. (2018) examined how different initial concentrations of *E. coli* impacted the attachment and biofilm formation of *L. monocytogenes* on SS surfaces. When *L. monocytogenes* was inoculated at 10^2 CFU/mL in both monoculture or in dual culture with *E. coli* at 10^6 CFU/mL, *L. monocytogenes* exhibited significantly higher attachment in monoculture (de Grandi et al., 2018). Conversely, when the initial concentrations were reversed or equal (10^6 CFU/mL), no significant differences were observed between mono- and dual-species biofilms, with *L. monocytogenes* consistently reaching around 5 log CFU/cm² under the tested conditions (de Grandi et al., 2018). This observation suggests a possible interference from *E. coli* in the dual-species biofilms. Thus, the study underscores the need to account for initial inoculum levels and microbial interactions in developing effective sanitation strategies for *L. monocytogenes* biofilms.

In multispecies biofilms involving *L. monocytogenes* and *Pseudomonas* spp., a dominance of the latter was observed (Lourenco et al., 2011; Pang & Yuk, 2019; Papaioannou et al., 2018; Thomassen et al., 2023). For instance, Pang and Yuk (2019) reported that *P. fluorescens* dominated in dual-species biofilms with *L. monocytogenes*, achieving a cell density of 1 log CFU/cm² higher than *L. monocytogenes* and accounting for 90% of the total population. Moreover, *L. monocytogenes* demonstrated reduced sensitivity to BAC treatment in co-culture biofilms compared

to single-species biofilms, with a reduction of 1.9 and 5.42 log CFU/cm², respectively (Pang & Yuk, 2019). Furthermore, in dual-biofilms of *L. monocytogenes* and *P. aeruginosa* cultured at 12 and 37°C for 1 and 7 days, *P. aeruginosa* consistently dominated the community, constituting 99% of the total biofilm population, irrespective of biofilm formation parameters (Lourenco et al., 2011). In addition, mixed-species biofilms and biofilms produced at 12°C exhibited less susceptibility to disinfection compared to mono-species biofilms of *L. monocytogenes* and biofilms formed at 37°C, respectively (Lourenco et al., 2011). Similar findings of reduced susceptibility to disinfectant in dual-species biofilms containing *L. monocytogenes* with either *P. aeruginosa*, *P. putida*, or *P. fluorescens* have been reported in other studies (Ibusquiza et al., 2012; Rodríguez-López et al., 2022; Van der Veen & Abee, 2011). The compact architectural structure of mixed-species biofilms may hinder the diffusion of biocides into the inner structures of the matrix, affecting the overall efficacy of the biocide to completely inactivate the biofilms (Rodríguez-López et al., 2022). Conversely, when co-cultured with *Salmonella enterica* on SS at 15°C, *L. monocytogenes* produced similar biofilm populations with comparable biocide susceptibilities both in mono- and in dual-species biofilms (Kostaki et al., 2012). The contradictory observations could be attributed to variations in experimental designs, particularly differences in biocides, bacterial species, surface type, biofilm structure, and environmental growth conditions employed. Nonetheless, despite the complexity of these studies, it is difficult to simulate real-world conditions for diverse sectors, given the diversity in the microbial populations present.

4 | EFFICACY OF BIOCIDES AGAINST *L. monocytogenes* BIOFILMS: CONSISTENT PATTERNS AND DISINFECTION CHALLENGES

4.1 | Consistent observations and patterns

Comparing the efficacy results of biocides against *L. monocytogenes* biofilms across studies is often challenging, owing to the variability in factors encompassing surface physicochemistry and topography, strain type (persistent or sporadic), incubation temperature, the biocide's active ingredient and concentration, contact time, organic matter, growth media, the presence of mixed species, and biofilm maturation. Nevertheless, certain recurrent patterns have been consistently observed in scientific studies. As *L. monocytogenes* biofilms age, they tend to exhibit increased tolerance to disinfectants, a phenomenon that

appears to be influenced by the specific substratum they colonize. Surfaces with cracks, pores, and organic matter enhance stronger bacterial attachment, significantly limiting the biocidal activity (Belessi et al., 2011; Dhowlaghar et al., 2018; Dygico et al., 2020; Fagerlund et al., 2020; Hua & Zhu, 2024b; Hua et al., 2019; Ibusquiza et al., 2011; Kim & Park, 2023; Korany et al., 2018; Nilsson et al., 2011; Papaioannou et al., 2018). Studies have consistently shown that certain strains of *L. monocytogenes* can persist in food-processing environments, with varying tolerance to different biocides, highlighting the need for systematic biocide rotation for greater biofilm reduction (Dhowlaghar et al., 2018; Murugesan et al., 2015). Additionally, biocides with multiple active ingredients are more effective in inactivating *L. monocytogenes* biofilms compared to those with a single active ingredient (Aryal & Muriana, 2019; Dhowlaghar et al., 2018; Dygico et al., 2021). Furthermore, using biocides at sublethal concentrations or for shorter contact times can lead to ineffective biofilm inactivation (Rodríguez-López & Cabo, 2017; Rodríguez-Melcón et al., 2019). Moreover, mixed-species biofilms containing *L. monocytogenes* generally exhibit enhanced tolerance to disinfectants compared to their single-species counterparts (Giaouris et al., 2013; Ibusquiza et al., 2012; Kocot & Olszewska, 2019).

4.2 | Challenges associated with the use of biocides against *L. monocytogenes* biofilms

The primary concern regarding the use of biocides for controlling *L. monocytogenes* biofilms is the survival of bacteria after cleaning and disinfection. Biofilm-associated bacteria can develop responses when exposed to sublethal disinfectant concentrations, leading to their persistence in the food-processing environment. This persistence is often due to the expression of specific genes that enable the bacteria to counteract the effects of disinfectants. Various studies have extensively reviewed these bacterial genes and their resistance or tolerance mechanisms toward disinfectants (Azizoglu et al., 2015; Bland et al., 2022; Rodríguez-López et al., 2018), and thus, this aspect will not be elaborated on in the current review.

The majority of the studies examined in this review assessed the effectiveness of biocides mainly through the use of traditional culturing on agar plates. In some studies, complete elimination of *L. monocytogenes* biofilms was observed, indicating the high effectiveness of the biocide (Aryal & Muriana, 2019; Dhowlaghar et al., 2018). However, recent investigations suggest that treatment of *L. monocytogenes* biofilms with biocides may induce stressful conditions, favoring the formation of the viable

but nonculturable (VBNC) state (Brauge et al., 2020; Truchado et al., 2021), although apparent complete elimination (by plate count) was observed. The VBNC state is a stress response mechanism in nonsporulating organisms, including *L. monocytogenes*, triggered by adverse environmental conditions, such as nutrient limitation, temperature upshifts, oxidative stress, and exposure to disinfectants, that inhibit active growth and culturability (Dong et al., 2020; Truchado et al., 2020; Wu et al., 2020). Generally, bacteria in the VBNC state exhibit reduced metabolic activity but do not form colonies on standard agar plates, which in turn may lead to an overestimation of the biocidal treatment (Truchado et al., 2021). Brauge et al. (2020) reported that treatment of *L. monocytogenes* biofilms on SS with QAC and H₂O₂ induced the VBNC state. Using the propidium monoazide quantitative polymerase chain reaction (PMA-qPCR) method, the authors detected VBNC cells after a 20-min treatment with QAC and H₂O₂ biocides, showing 2.12 and 2.95 log CFU/cm² of VBNC cells, respectively (Brauge et al., 2020). This observation demonstrated the incomplete inactivation of biofilms after cleaning and disinfection. Similar observations regarding the induction of the VBNC state following the application of biocides have been reported in other studies (Brauge et al., 2018; Overney et al., 2017). This is concerning as bacteria might resuscitate under favorable conditions, such as nutrient supplementation, stress removal, temperature changes, or the addition of resuscitation-promoting substances. For instance, Zhao et al. (2024) demonstrated that *L. monocytogenes* cells in the VBNC state resuscitated in TSB medium supplemented with Tween80 and also regained viability in pasteurized milk. Such cells pose a significant threat to food safety and public health.

5 | CONCLUSION AND FUTURE CONSIDERATIONS

The persistence of *L. monocytogenes* in the food-processing environment is closely linked to its ability to form biofilms, leading to the potential for cross-contamination events. Biocides remain a primary choice in the food industry for controlling *L. monocytogenes* and its associated biofilms. Ineffective cleaning and disinfection practices, often resulting from applying disinfectants at sublethal concentrations and for shorter contact times, can contribute to the persistence of *L. monocytogenes* in the food industry. The scientific data reviewed in our study indicate that the effectiveness of biocidal products against *L. monocytogenes* is influenced by various factors. Contradictory observations in studies are, in part, attributed

to variations in experimental designs and methodologies. Key factors including strain type, surface characteristics, incubation temperature, biofilm age, presence of organic matter, mono- or multispecies biofilm, biocide's active ingredients and concentration, contact time, and detachment and enumeration methods play pivotal roles in biofilm response and need to be considered in assessing the efficacy of a biocidal product. In spite of these variations, certain consistent patterns were observed among studies, setting the grounds for further investigations.

In future studies on the assessment of *L. monocytogenes* biofilm and biocide efficacy, it is recommended to explore the potential relationship between serotype and biofilm-forming ability. Certain serotypes may exhibit enhanced biofilm formation, leading to increased persistence in food-processing environments and increased contamination risks. Therefore, such investigations will improve our current understanding on *L. monocytogenes* persistence in the food industry. Moreover, this knowledge will inform targeted decontamination strategies, facilitate risk assessment, and guide food industry practices to safeguard public health and ensure product quality.

Complex multispecies biofilms involving *L. monocytogenes* and other bacteria present significant challenges in biocide efficacy due to enhanced tolerance and complex interactions. The presence of diverse microbial communities often results in synergistic or antagonistic interactions that may alter the structural integrity of biofilms and decrease their susceptibility to biocide. Future research must address the intricate dynamics of mixed-species biofilms, as understanding these interactions is essential for developing targeted and effective control strategies.

In addition, the inclusion of soiled conditions in biocide efficacy assays is encouraged to better simulate real-world scenarios. However, it is important to emphasize that thorough cleaning of contaminated surfaces to minimize organic matter should precede the application of disinfectants in order to enhance biocide efficacy. Furthermore, optimization of detachment and enumeration methods is crucial to accurately estimate the number of resistant/tolerant cells remaining on surfaces after disinfection. To fully understand the bacterial attachment on FCS and NFCS, it is advisable to examine the surface water contact angle and surface roughness. If possible, exploring the VBNC state after disinfection should be considered. Finally, authors are also encouraged to specify whether biocides are applied following the manufacturer's instructions, providing transparency for the scientific community to evaluate the effectiveness of biocides. This review provides a detailed examination of how *L. monocytogenes* biofilms react to biocides under various treatment conditions, aiding in the formulation of effective

tive cleaning and disinfection protocols within the food industry.

AUTHOR CONTRIBUTIONS

Michael Arthur: Conceptualization; writing—original draft; data curation; investigation. **Edmund Larbi Afari:** Writing—original draft; investigation; data curation. **Elena-Alexandra Alexa:** Writing—review and editing; data curation. **Mei-Jun Zhu:** Supervision; writing—review and editing. **Michael T. Gaffney:** Project administration; supervision; writing—review and editing. **Jesus Maria Frias Celayeta:** Writing—review and editing; supervision; project administration; funding acquisition. **Catherine M. Burgess:** Writing—review and editing; supervision; project administration; funding acquisition; resources.

ACKNOWLEDGMENTS

This work was funded by the Department of Agriculture, Food and Marine (project 2019R424) and the Teagasc Walsh Scholarship Programme (Ref. 2021055).

Open access funding provided by IReL.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

ORCID

Michael Arthur  <https://orcid.org/0000-0002-1481-1694>

Edmund Larbi Afari  <https://orcid.org/0000-0001-7768-3397>

Jesus Maria Frias Celayeta  <https://orcid.org/0000-0001-5424-5167>

Catherine M. Burgess  <https://orcid.org/0000-0003-3722-5935>

REFERENCES

- Abban, S., Jakobsen, M., & Jespersen, L. (2012). Attachment behaviour of *Escherichia coli* K12 and *Salmonella* Typhimurium P6 on food contact surfaces for food transportation. *Food Microbiology*, 31(2), 139–147. <https://doi.org/10.1016/j.fm.2012.04.003>
- Abdallah, M., Benoliel, C., Dhulster, P. F.-T., Drider, P. D., & Chihib, N.-E. (2015). Effect of culture conditions on the resistance of *Pseudomonas aeruginosa* biofilms to disinfecting agents. *Biofouling*, 31(1), 49–59. <https://doi.org/10.1080/08927014.2014.993390>
- Alvarez-Ordóñez, A., Leong, D., Hunt, K., Scollard, J., Butler, F., & Jordan, K. (2018). Production of safer food by understanding risk factors for *L. monocytogenes* occurrence and persistence in food processing environments. *Journal of Food Safety*, 38(6), Article e12516. <https://doi.org/10.1111/jfs.12516>
- Andrade, J. C., João, A. L., Alonso, C. D. S., Barreto, A. S., & Henriques, A. R. (2020). Genetic subtyping, biofilm-forming ability and biocide susceptibility of *Listeria monocytogenes* strains isolated from a ready-to-eat food industry. *Antibiotics*, 9(7), Article 416. <https://doi.org/10.3390/antibiotics9070416>
- Arachchi, G. G. J., Cridge, A. G., Dias-Wanigasekera, B. M., Cruz, C. D., McIntyre, L., Liu, R., Flint, S. H., & Mutukumira, A. N. (2013). Effectiveness of phages in the decontamination of *Listeria monocytogenes* adhered to clean stainless steel, stainless steel coated with fish protein, and as a biofilm. *Journal of Industrial Microbiology and Biotechnology*, 40(10), 1105–1116. <https://doi.org/10.1007/s10295-013-1313-3>
- Aryal, J., Chhetri, V. S., & Adhikari, A. (2023). Survival and attachment of *Listeria monocytogenes* on bell peppers and influence of attachment time on efficacy of chlorine. *LWT - Food Science and Technology*, 173, Article 114278. <https://doi.org/10.1016/j.lwt.2022.114278>
- Aryal, M., & Muriana, P. M. (2019). Efficacy of commercial sanitizers used in food processing facilities for inactivation of *Listeria monocytogenes*, *E. coli* O157:H7, and *Salmonella* biofilms. *Foods*, 8(12), Article 639. <https://doi.org/10.3390/foods8120639>
- Assisi, C., Forauer, E., Oliver, H. F., & Etter, A. J. (2020). Genomic and transcriptomic analysis of biofilm formation in persistent and transient *Listeria monocytogenes* isolates from the retail deli environment does not yield insight into persistence mechanisms. *Foodborne Pathogens and Disease*, 18(3), 179–188. <https://doi.org/10.1089/fpd.2020.2817>
- Avila-Novoa, M. G., González-Torres, B., González-Gómez, J. P., Guerrero-Medina, P. J., Martínez-Chávez, L., Martínez-González, N. E., Chaidez, C., & Gutiérrez-Lomelí, M. (2023). Genomic insights into *Listeria monocytogenes*: Organic acid interventions for biofilm prevention and control. *International Journal of Molecular Sciences*, 24(17), Article 13108. <https://doi.org/10.3390/ijms241713108>
- Azizoglu, R. O., Dutta, V., Breidt, F., & Kathariou, S. (2015). Resistance of *Listeria monocytogenes* biofilms to sanitizing agents. In A. L. Pometto III & A. Demirci (Eds.), *Biofilms in the food environment* (2nd ed., pp. 51–83). John Wiley & Sons. <https://doi.org/10.1002/9781118864036.ch3>
- Bae, Y. M., & Lee, S. Y. (2012). Inhibitory effects of UV treatment and a combination of UV and dry heat against pathogens on stainless steel and polypropylene surfaces. *Journal of Food Science*, 77(1), 61–64. <https://doi.org/10.1111/j.1750-3841.2011.02476.x>
- Ban, G. H., & Kang, D. H. (2016). Effect of sanitizer combined with steam heating on the inactivation of foodborne pathogens in a biofilm on stainless steel. *Food Microbiology*, 55, 47–54. <https://doi.org/10.1016/j.fm.2015.11.003>
- Ban, G. H., Park, S. H., Kim, S. O., Ryu, S., & Kang, D. H. (2012). Synergistic effect of steam and lactic acid against *Escherichia coli* O157:H7, *Salmonella* Typhimurium, and *Listeria monocytogenes* biofilms on polyvinyl chloride and stainless steel. *International Journal of Food Microbiology*, 157(2), 218–223. <https://doi.org/10.1016/j.ijfoodmicro.2012.05.006>
- Barnett-Neefs, C., Wiedmann, M., & Ivanek, R. (2022). Examining patterns of persistent *Listeria* contamination in packinghouses using agent-based models. *Journal of Food Protection*, 85(12), 1824–1841. <https://doi.org/10.4315/JFP-22-119>
- Barroso, I., Maia, V., Cabrita, P., Martínez-Suárez, J. V., & Brito, L. (2019). The benzalkonium chloride resistant or sensitive phenotype of *Listeria monocytogenes* planktonic cells did not dictate the susceptibility of its biofilm counterparts. *Food Research International*, 123, 373–382. <https://doi.org/10.1016/j.foodres.2019.05.008>
- Bas, S., Kramer, M., & Stopar, D. (2017). Biofilm surface density determines biocide effectiveness. *Frontiers in Microbiology*, 8, Article 2443. <https://doi.org/10.3389/fmicb.2017.02443>
- Bechtel, T. D., & Gibbons, J. G. (2021). Population genomic analysis of *Listeria monocytogenes* from food reveals substrate-specific

- genome variation. *Frontiers in Microbiology*, 12, Article 620033. <https://doi.org/10.3389/fmicb.2021.620033>
- Belessi, C. E. A., Gounadaki, A. S., Psomas, A. N., & Skandamis, P. N. (2011). Efficiency of different sanitation methods on *Listeria monocytogenes* biofilms formed under various environmental conditions. *International Journal of Food Microbiology*, 145(Suppl. 1), S46–S52. <https://doi.org/10.1016/j.ijfoodmicro.2010.10.020>
- Belias, A., Sullivan, G., Wiedmann, M., & Ivanek, R. (2022). Factors that contribute to persistent *Listeria* in food processing facilities and relevant interventions: A rapid review. *Food Control*, 133, Article 108579. <https://doi.org/10.1016/j.foodcont.2021.108579>
- Bergholz, T. M., Shah, M. K., Burall, L. S., Rakic-Martinez, M., & Datta, A. R. (2018). Genomic and phenotypic diversity of *Listeria monocytogenes* clonal complexes associated with human listeriosis. *Applied Microbiology and Biotechnology*, 102(8), 3475–3485. <https://doi.org/10.1007/s00253-018-8852-5>
- Bernbom, N., Vogel, B. F., & Gram, L. (2011). *Listeria monocytogenes* survival of UV-C radiation is enhanced by presence of sodium chloride, organic food material and by bacterial biofilm formation. *International Journal of Food Microbiology*, 147(1), 69–73. <https://doi.org/10.1016/j.ijfoodmicro.2011.03.009>
- Bland, R., Brown, S. R. B., Waite-Cusic, J., & Kovacevic, J. (2022). Probing antimicrobial resistance and sanitizer tolerance themes and their implications for the food industry through the *Listeria monocytogenes* lens. *Comprehensive Reviews in Food Science and Food Safety*, 21(2), 1777–1802. <https://doi.org/10.1111/1541-4337.12910>
- Bolocan, A. S., Oniciuc, E. A., Alvarez-Ordóñez, A., Wagner, M., Rychli, K., Jordan, K., & Nicolau, A. I. (2015). Putative cross-contamination routes of *Listeria monocytogenes* in a meat processing facility in Romania. *Journal of Food Protection*, 78(9), 1664–1674. <https://doi.org/10.4315/0362-028X.JFP-14-539>
- Bonneville, L., Maia, V., Barroso, I., Martínez-Suárez, J. V., & Brito, L. (2021). *Lactobacillus plantarum* in dual-species biofilms with *Listeria monocytogenes* enhanced the anti-*Listeria* activity of a commercial disinfectant based on hydrogen peroxide and peracetic acid. *Frontiers in Microbiology*, 12, Article 631627. <https://doi.org/10.3389/fmicb.2021.631627>
- Bonneville, L., Ortiz, S., Maia, V., Brito, L., & Martínez-Suárez, J. V. (2020). Strain and growth conditions may regulate resistance of *Listeria monocytogenes* biofilms to benzalkonium chloride. *Applied Sciences*, 10(3), Article 988. <https://doi.org/10.3390/app10030988>
- Bosilevac, J. M., Guragain, M., Barkhouse, D. A., Velez, S. E., Katz, T. S., Lu, G., & Wang, R. (2024). Impact of intense sanitization procedures on bacterial communities recovered from floor drains in pork processing plants. *Frontiers in Microbiology*, 15, Article 1379203. <https://doi.org/10.3389/fmicb.2024.1379203>
- Boucher, C., Waite-Cusic, J., Stone, D., & Kovacevic, J. (2021). Relative performance of commercial citric acid and quaternary ammonium sanitizers against *Listeria monocytogenes* under conditions relevant to food industry. *Food Microbiology*, 97, Article 103752. <https://doi.org/10.1016/j.fm.2021.103752>
- Bramer, W. M., Rethlefsen, M. L., Kleijnen, J., & Franco, O. H. (2017). Optimal database combinations for literature searches in systematic reviews: A prospective exploratory study. *Systematic Reviews*, 6(1), Article 245. <https://doi.org/10.1186/s13643-017-0644-y>
- Brauge, T., Faille, C., Leleu, G., Denis, C., Hanin, A., & Midelet, G. (2020). Treatment with disinfectants may induce an increase in viable but non culturable populations of *Listeria monocytogenes* in biofilms formed in smoked salmon processing environments. *Food Microbiology*, 92, Article 103548. <https://doi.org/10.1016/j.fm.2020.103548>
- Brauge, T., Faille, C., Sadovskaya, I., Charbit, A., Benezech, T., Shen, Y., Loessner, M. J., Bautista, J. R., & Midelet-bourdin, G. (2018). The absence of N-acetylglucosamine in wall teichoic acids of *Listeria monocytogenes* modifies biofilm architecture and tolerance to rinsing and cleaning procedures. *PLoS ONE*, 13(1), Article e0190879. <https://doi.org/10.1371/journal.pone.0190879>
- Buchanan, R. L., Gorris, L. G. M., Hayman, M. M., Jackson, T. C., & Whiting, R. C. (2017). A review of *Listeria monocytogenes*: An update on outbreaks, virulence, dose-response, ecology, and risk assessments. *Food Control*, 75, 1–13. <https://doi.org/10.1016/j.foodcont.2016.12.016>
- Buchholz, A. L., Davidson, G. R., Marks, B. P., Todd, E. C. D., & Ryser, E. T. (2012). Quantitative transfer of *Escherichia coli* O157:H7 To equipment during small-scale production of fresh-cut leafy greens. *Journal of Food Protection*, 75(7), 1184–1197. <https://doi.org/10.4315/0362-028X.JFP-11-489>
- Cabeça, T. K., Pizzolitto, A. C., & Pizzolitto, E. L. (2012). Activity of disinfectants against foodborne pathogens in suspension and adhered to stainless steel surfaces. *Brazilian Journal of Microbiology*, 43(3), 1112–1119. <https://doi.org/10.1590/S1517-83822012000300038>
- Camargo, A. C., Woodward, J. J., Call, D. R., & Nero, L. A. (2017). *Listeria monocytogenes* in food-processing facilities, food contamination, and human listeriosis: The Brazilian scenario. *Foodborne Pathogens and Disease*, 14(11), 623–636. <https://doi.org/10.1089/fpd.2016.2274>
- Carpentier, B., & Cerf, O. (2011). Persistence of *Listeria monocytogenes* in food industry equipment and premises. *International Journal of Food Microbiology*, 145(1), 1–8. <https://doi.org/10.1016/j.ijfoodmicro.2011.01.005>
- Castellini, G., & Graffigna, G. (2022). Assessing involvement with food: A systematic review of measures and tools. *Food Quality and Preference*, 97, Article 104444. <https://doi.org/10.1016/j.foodqual.2021.104444>
- Castro-Ibáñez, I., Gil, M. I., & Allende, A. (2017). Ready-to-eat vegetables: Current problems and potential solutions to reduce microbial risk in the production chain. *LWT - Food Science and Technology*, 85, 284–292. <https://doi.org/10.1016/j.lwt.2016.11.073>
- Caurio, D. L., da Silva, D. Q., Dornelles, L., Arechavaleta, N. N., & da Motta, A. D. S. (2022). Influence of the food matrix on dual-species biofilm of *Listeria monocytogenes* and *Pseudomonas fluorescens*. *Acta Scientiarum Biological Sciences*, 44, Article e61925. <https://doi.org/10.4025/actascibiols.v44i1.61925>
- Centers for Disease Control and Prevention (CDC). (2014). *Multistate outbreak of listeriosis linked to commercially produced, prepackaged caramel apples*. <https://www.cdc.gov/listeria/outbreaks/caramel-apples-12-14/index.html>
- Centers for Disease Control and Prevention (CDC). (2015). *Whole-some Soy Products, Inc. sprouts and investigation of human listeriosis cases (Final update)*. <https://www.cdc.gov/listeria/outbreaks/bean-sprouts-11-14/index.html>
- Centers for Disease Control and Prevention (CDC). (2016). *Multistate outbreak of Listeriosis linked to packaged salads produced at Springfield, Ohio Dole processing facility (Final update)*. <https://www.cdc.gov/listeria/outbreaks/bagged-salads-01-16/index.html#:~:text=Outbreaks-,Multistate%20Outbreak%20>

- 20of%20Listeriosis%20Linked%20to%20Packaged%20Salads%20Produced%20at,Dole%20Processing%20Facility%20(Final%20Update)&text=This%20outbreak%20appears%20to%20be, illness%20in%20the%20United%20States
- Centers for Disease Control and Prevention (CDC). (2023). *Listeria outbreak, peaches, nectarines, and plums—November 2023*. https://www.cdc.gov/listeria/outbreaks/details-peaches-11-23.html?CDC_AAref_Val=https://www.cdc.gov/listeria/outbreaks/peaches-11-23/details.html
- Centers for Disease Control and Prevention (CDC). (2012). *Multi-state outbreak of Listeriosis linked to whole cantaloupes from Jensen Farms, Colorado (Final update)*. <https://www.cdc.gov/listeria/outbreaks/cantaloupes-jensen-farms/index.html>
- Chaitiemwong, N., Hazeleger, W. C., & Beumer, R. R. (2010). Survival of *Listeria monocytogenes* on a conveyor belt material with or without antimicrobial additives. *International Journal of Food Microbiology*, 142(1–2), 260–263. <https://doi.org/10.1016/j.ijfoodmicro.2010.06.021>
- Chaitiemwong, N., Hazeleger, W. C., & Beumer, R. R. (2014). Inactivation of *Listeria monocytogenes* by disinfectants and bacteriophages in suspension and stainless steel carrier test. *Journal of Food Protection*, 77(12), 2012–2020. <https://doi.org/10.4315/0362-028X.JFP-14-151>
- Chaves, R. D., Kumazawa, S. H., Khaneghah, A. M., Alvarenga, V. O., Hungaro, H. M., & Sant'Ana, A. S. (2024). Comparing the susceptibility to sanitizers, biofilm-forming ability, and biofilm resistance to quaternary ammonium and chlorine dioxide of 43 *Salmonella enterica* and *Listeria monocytogenes* strains. *Food Microbiology*, 117, Article 104380. <https://doi.org/10.1016/j.fm.2023.104380>
- Chen, D., Zhao, T., & Doyle, M. P. (2015). Control of pathogens in biofilms on the surface of stainless steel by levulinic acid plus sodium dodecyl sulfate. *International Journal of Food Microbiology*, 207, 1–7. <https://doi.org/10.1016/j.ijfoodmicro.2015.04.026>
- Chen, Q., Zhang, X., Wang, Q., Jingxian, Y., & Zhong, Q. (2024). The mixed biofilm formed by *Listeria monocytogenes* and other bacteria: Formation, interaction and control strategies. *Critical Reviews in Food Science and Nutrition*, 64(24), 8570–8586. <https://doi.org/10.1080/10408398.2023.2200861>
- Chen, S. N., Yap, M. L., Kuan, C. H., Son, R., & Saw, S. H. (2021). Could food or food contact surfaces be the favourable hideouts for *Listeria monocytogenes* in Perak, Malaysia? *Food Research*, 5(3), 174–182. [https://doi.org/10.26656/fr.2017.5\(3\).596](https://doi.org/10.26656/fr.2017.5(3).596)
- Chen, Y.-C., & Chen, S.-J. (2020). *Gene expression and phenotypic traits*. IntechOpen. <https://doi.org/10.5772/intechopen.82955>
- Chowdhury, D., Rahman, A., Hu, H., Jensen, S. O., Deva, A. K., & Vickery, K. (2019). Effect of disinfectant formulation and organic soil on the efficacy of oxidizing disinfectants against biofilms. *Journal of Hospital Infection*, 103(1), e33–e41. <https://doi.org/10.1016/j.jhin.2018.10.019>
- Clayton, G. E., Thorn, R. M. S., & Reynolds, D. M. (2019). Comparison of trihalomethane formation using chlorine-based disinfectants within a model system; applications within point-of-use drinking water treatment. *Frontiers in Environmental Science*, 7, Article 35. <https://doi.org/10.3389/fenvs.2019.00035>
- Clayton, G. E., Thorn, R. M. S., & Reynolds, D. M. (2021). The efficacy of chlorine-based disinfectants against planktonic and biofilm bacteria for decentralised point-of-use drinking water. *npj Clean Water*, 4(1), Article 48. <https://doi.org/10.1038/s41545-021-00139-w>
- Conrad, A. R., Tubach, S., Cantu, V., Webb, L. M., Stroika, S., Moris, S., Davis, M., Hunt, D. C., Bradley, K. K., Kucerova, Z., Strain, E., Doyle, M., Fields, A., Neil, K. P., Gould, L. H., Jackson, K. A., Wise, M. E., Griffin, P. M., & Jackson, B. R. (2022). *Listeria monocytogenes* illness and deaths associated with ongoing contamination of a multiregional brand of ice cream products, United States, 2010–2015. *Clinical Infectious Diseases*, 76(1), 89–95. <https://doi.org/10.1093/cid/ciac550>
- Cruz, C. D., & Fletcher, G. C. (2012). Assessing manufacturers' recommended concentrations of commercial sanitizers on inactivation of *Listeria monocytogenes*. *Food Control*, 26(1), 194–199. <https://doi.org/10.1016/j.foodcont.2012.01.041>
- Daeschel, D., Pettengill, J. B., Wang, Y., Chen, Y., Allard, M., & Snyder, A. B. (2022). Genomic analysis of *Listeria monocytogenes* from US food processing environments reveals a high prevalence of QAC efflux genes but limited evidence of their contribution to environmental persistence. *BMC Genomics*, 23(1), Article 488. <https://doi.org/10.1186/s12864-022-08695-2>
- Das, S., & Lalitha, K. V. (2022). *Listeria monocytogenes* biofilms on glass surface: Survival and efficacy of two sanitizing agents for inactivation. *Fishery Technology*, 59, 49–55. <https://krishi.icar.gov.in/jspui/bitstream/123456789/69899/1/Final%20published%20article%20in%20Fishery%20Technology.pdf>
- da Silva, D. A. L., de Melo Tavares, R., Camargo, A. C., Yamatogi, R. S., De Martinis, E. C. P., & Nero, L. A. (2021). Biofilm growth by *Listeria monocytogenes* on stainless steel and expression of biofilm-related genes under stressing conditions. *World Journal of Microbiology and Biotechnology*, 37(7), Article 119. <https://doi.org/10.1007/s11274-021-03092-5>
- da Silva Fernandes, M., Kabuki, D. Y., & Kuaye, A. Y. (2015). Behavior of *Listeria monocytogenes* in a multi-species biofilm with *Enterococcus faecalis* and *Enterococcus faecium* and control through sanitation procedures. *International Journal of Food Microbiology*, 200, 5–12. <https://doi.org/10.1016/j.ijfoodmicro.2015.01.003>
- Davin-Regli, A., & Pages, J. M. (2012). Cross-resistance between biocides and antimicrobials: An emerging question. *OIE Revue Scientifique et Technique*, 31(1), 89–104. <https://doi.org/10.20506/rst.31.1.2101>
- de Grandi, A. Z., Pinto, U. M., & Destro, M. T. (2018). Dual-species biofilm of *Listeria monocytogenes* and *Escherichia coli* on stainless steel surface. *World Journal of Microbiology and Biotechnology*, 34(4), Article 61. <https://doi.org/10.1007/s11274-018-2445-4>
- de Oliveira, M. M. M., Brugnera, D. F., do Nascimento, J. A., Batista, N. N., & Piccoli, R. H. (2012). Cinnamon essential oil and cinnamaldehyde in the control of bacterial biofilms formed on stainless steel surfaces. *European Food Research and Technology*, 234(5), 821–832. <https://doi.org/10.1007/s00217-012-1694-y>
- Dhowlaghar, N., De Abrew Abeyesundara, P., Nannapaneni, R., Schilling, M. W., Chang, S., Cheng, W. H., & Sharma, C. S. (2018). Growth and biofilm formation by *Listeria monocytogenes* in catfish mucus extract on four food contact surfaces at 22 and 10c and their reduction by commercial disinfectants. *Journal of Food Protection*, 81(1), 59–67. <https://doi.org/10.4315/0362-028X.JFP-17-103>
- Dinger, E. (2024). Impact of lactic acid bacteria strains against *Listeria monocytogenes* biofilms on various food-contact surfaces. *Archives of Microbiology*, 206(2), Article 80. <https://doi.org/10.1007/s00203-023-03811-6>
- Dojjad, S. P., Barbuddhe, S. B., Garg, S., Poharkar, K. V., Kalorey, D. R., Kurkure, N. V., Rawool, D. B., & Chakraborty, T. (2015). Biofilm-forming abilities of *Listeria monocytogenes* serotypes isolated from different sources. *PLoS ONE*, 10(9), Article e0137046. <https://doi.org/10.1371/journal.pone.0137046>

- Donaghy, J. A., Jagadeesan, B., Goodburn, K., Grunwald, L., Jensen, O. N., Jespers, A., Kanagachandran, K., Lafforgue, H., Seefelder, W., & Quentin, M. C. (2019). Relationship of sanitizers, disinfectants, and cleaning agents with antimicrobial resistance. *Journal of Food Protection*, 82(5), 889–902. <https://doi.org/10.4315/0362-028X.JFP-18-373>
- Dong, K., Wang, Y., Pan, H., Yang, D., Rao, L., Zhao, L., & Liao, X. (2020). Induction, detection, formation, and resuscitation of viable but non-culturable state microorganisms. *Comprehensive Reviews in Food Science and Food Safety*, 19(1), 149–183. <https://doi.org/10.1111/1541-4337.12513>
- Dumitru, A. E., Gica, N., Botezatu, R., Gica, C., Demetrian, M., Ciobanu, A. M., Cimpoca-Raptis, B. A., Peltecu, G., & Panaitescu, A. M. (2021). *Listeria monocytogenes* infection during pregnancy. *Romanian Journal of Infectious Diseases*, 24, 46–49. <https://doi.org/10.37897/RJID.2021.S.10>
- Duze, S. T., Marimani, M., & Patel, M. (2021). Tolerance of *Listeria monocytogenes* to biocides used in food processing environments. *Food Microbiology*, 97, Article 103758. <https://doi.org/10.1016/j.fm.2021.103758>
- Dygico, L. K., Gahan, C. G. M., Grogan, H., & Burgess, C. M. (2020). The ability of *Listeria monocytogenes* to form biofilm on surfaces relevant to the mushroom production environment. *International Journal of Food Microbiology*, 317, Article 108385. <https://doi.org/10.1016/j.ijfoodmicro.2019.108385>
- Dygico, L. K., Gahan, C. G. M., Grogan, H., & Burgess, C. M. (2021). Examining the efficacy of mushroom industry biocides on *Listeria monocytogenes* biofilm. *Journal of Applied Microbiology*, 130(4), 1106–1116. <https://doi.org/10.1111/jam.14681>
- Environmental Protection Agency (EPA). (2012). *Product performance test guidelines: OCSPP 810.2300 Sanitizers for use on hard surfaces — Efficacy data recommendations*. <https://www.regulations.gov/document/EPA-HQ-OPPT-2009-0150-0022>
- Environmental Protection Agency (EPA). (2018). *Product performance test guidelines: OCSPP 810.2200 Disinfectants for use on hard surfaces—Efficacy data recommendations*. <https://www.regulations.gov/document?D=EPA-HQ-OPPT-2009-0150-0021>
- Estrada, E. M., Hamilton, A. M., Sullivan, G. B., Wiedmann, M., Critzer, F. J., & Strawn, L. K. (2020). Prevalence, persistence, and diversity of *Listeria monocytogenes* and *Listeria* species in produce packinghouses in Three U.S. States. *Journal of Food Protection*, 83(2), 277–286. <https://doi.org/10.4315/0362-028X.JFP-19-411>
- European Centre for Disease Prevention and Control (ECDC). (2022). *Listeriosis: Annual Epidemiological Report for 2021*. <https://www.ecdc.europa.eu/en/publications-data/listeriosis-annual-epidemiological-report-2021>
- European Chemical Agency (ECHA). (2018). *Guidance on the biocidal products regulation (Volume II Efficacy—Assessment and Evaluation (Parts B + C), Version 3.0)*. https://echa.europa.eu/documents/10162/2324906/bpr_guidance_assessment_evaluation_part_vol_ii_part_bc_en.pdf/2c42983a-ee0b-9e35-c596-b172fee61115?t=1644567032606
- European Commission. (2009). Directive 98/8/EC of the European Parliament and of the Council of 16 February 1998 concerning the placing of biocidal products on the market. *Official Journal of the European Union*, 71, 1–83. <https://eur-lex.europa.eu/LexUriServ/LexUriServ.do?uri=CONSLEG:1998L0008:20080926:EN:PDF>
- European Food Safety Authority (EFSA) Panel on Biological Hazards (BIOHAZ). (2024). Persistence of microbiological hazards in food and feed production and processing environments. *EFSA Journal*, 22(1), Article e8521. <https://doi.org/10.2903/j.efsa.2024.8521>
- European Food Safety Authority (EFSA), & European Centre for Disease Prevention and Control (ECDC). (2021). The European Union One Health 2020 Zoonoses Report. *EFSA Journal*, 19(12), Article e06971. <https://doi.org/10.2903/j.efsa.2021.6971>
- European Standard. (2015). *Chemical disinfectants and antiseptics—Quantitative non-porous surface test for the evaluation of bactericidal and yeasticidal and/or fungicidal activity of chemical disinfectants used in food, industrial, domestic and institutional areas without mechanical action - Test method and requirements without mechanical action (phase 2, step 2)*. https://www.sls.se/globalassets/sls/sls/remissvar/remisser/2022/pren-13697_41_e_stf.pdf
- Fagerlund, A., Heir, E., Møretro, T., & Langsrud, S. (2020). *Listeria monocytogenes* biofilm removal using different commercial cleaning agents. *Molecules*, 25(4), Article 792. <https://doi.org/10.3390/molecules25040792>
- Fagerlund, A., Møretro, T., Heir, E., Briandet, R., & Langsrud, S. (2017). Cleaning and disinfection of biofilms composed of *Listeria monocytogenes* and background microbiota from meat processing surfaces. *Applied and Environmental Microbiology*, 83(17), Article e01046–17. <https://doi.org/10.1128/AEM.01046-17>
- Faille, C., Cunault, C., Dubois, T., & Bénézech, T. (2018). Hygienic design of food processing lines to mitigate the risk of bacterial food contamination with respect to environmental concerns. *Innovative Food Science and Emerging Technologies*, 46, 65–73. <https://doi.org/10.1016/j.ifset.2017.10.002>
- Ferreira, V., Wiedmann, M., Teixeira, P., & Stasiewicz, M. J. (2014). *Listeria monocytogenes* persistence in food-associated environments: Epidemiology, strain characteristics, and implications for public health. *Journal of Food Protection*, 77(1), 150–170. <https://doi.org/10.4315/0362-028X.JFP-13-150>
- Food and Drug Administration (FDA). (2023). *Outbreak investigation of Listeria monocytogenes: Enoki mushrooms (November case count map provided by CDC)*. Author.
- Fox, E. M., Wall, P. G., & Fanning, S. (2015). Control of *Listeria* species food safety at a poultry food production facility. *Food Microbiology*, 51, 81–86. <https://doi.org/10.1016/j.fm.2015.05.002>
- Gattuso, A., Cella, E., Fillo, S., Ortoffi, M. F., Angeletti, S., Ciccozzi, M., De Medici, D., Lista, F., & Fiore, A. (2022). Phylogenetic and evolutionary genomic analysis of *Listeria monocytogenes* clinical strains in the framework of foodborne listeriosis risk assessment. *Frontiers in Microbiology*, 13, Article 816880. <https://www.frontiersin.org/journals/microbiology/articles/10.3389/fmicb.2022.816880>
- Giaouris, E., Chorianopoulos, N., Doulgeraki, A., & Nychas, G. (2013). Co-culture with *Listeria monocytogenes* within a dual-species biofilm community strongly increases resistance of *Pseudomonas putida* to benzalkonium chloride. *PLoS ONE*, 8(10), Article e77276. <https://doi.org/10.1371/journal.pone.0077276>
- Gil, M. I., Gómez-López, V. M., Hung, Y. C., & Allende, A. (2015). Potential of electrolyzed water as an alternative disinfectant agent in the fresh-cut industry. *Food and Bioprocess Technology*, 8(6), 1336–1348. <https://doi.org/10.1007/s11947-014-1444-1>
- Gnanadhas, D. P., Marathe, S. A., & Chakravorty, D. (2013). Biocides—Resistance, cross-resistance mechanisms and assessment. *Expert Opinion on Investigational Drugs*, 22(2), 191–206. <https://doi.org/10.1517/13543784.2013.748035>

- Gómez, C. N., Abriouel, H., Grande, M. J., Pérez Pulido, R., & Gálvez, A. (2012). Effect of enterocin AS-48 in combination with biocides on planktonic and sessile *Listeria monocytogenes*. *Food Microbiology*, 30(1), 51–58. <https://doi.org/10.1016/j.fm.2011.12.013>
- Gómez, D., Iguácel, L. P., Rota, M. C., Josécarramiñana, J., Ariño, A., & Yangüela, J. (2015). Occurrence of *Listeria monocytogenes* in ready-to-eat meat products and meat processing plants in Spain. *Foods*, 4(3), 271–282. <https://doi.org/10.3390/foods4030271>
- Gómez, N. C., Ramiro, J. M. P., Quecan, B. X. V., & de Melo Franco, B. D. G. (2016). Use of potential probiotic lactic acid bacteria (LAB) biofilms for the control of *Listeria monocytogenes*, *Salmonella Typhimurium*, and *Escherichia coli*O157:H7 biofilms formation. *Frontiers in Microbiology*, 7, Article 863. <https://doi.org/10.3389/fmicb.2016.00863>
- Gounadaki, A. S., Skandamis, P. N., Drosinos, E. H., & Nychas, G. J. E. (2008). Microbial ecology of food contact surfaces and products of small-scale facilities producing traditional sausages. *Food Microbiology*, 25(2), 313–323. <https://doi.org/10.1016/j.fm.2007.10.001>
- Gu, T., Luo, Y., Jia, Z., Meesrison, A., Lin, S., Ventresca, I. J., Brooks, S. J., Sharma, A., Sriram, S., Yang, M., Pearlstein, A. J., Millner, P. D., Schneider, K. R., & Zhang, B. (2024). Surface topography and chemistry of food contact substances, and microbial nutrition affect pathogen persistence and symbiosis in cocktail *Listeria monocytogenes* biofilms. *Food Control*, 161, Article 110391. <https://doi.org/10.1016/j.foodcont.2024.110391>
- Guan, J., Lacombe, A., Rane, B., Tang, J., Sablani, S., & Wu, V. C. H. (2021). A review: Gaseous interventions for *Listeria monocytogenes* control in fresh apple cold storage. *Frontiers in Microbiology*, 12, Article 782934. <https://doi.org/10.3389/fmicb.2021.782934>
- Haddad, S., Elliot, M., Savard, T., Deschênes, L., Smith, T., & Ells, T. (2021). Variations in biofilms harbouring *Listeria monocytogenes* in dual and triplex cultures with *Pseudomonas fluorescens* and *Lactobacillus plantarum* produced under a model system of simulated meat processing conditions, and their resistance. *Food Control*, 123, Article 107720. <https://doi.org/10.1016/j.foodcont.2020.107720>
- Hanson, H., Whitfield, Y., Lee, C., Badiani, T., Minielly, C., Fenik, J., Makrostergios, T., Kopko, C., Majury, A., Hillyer, E., Fortuna, L., Maki, A., Murphy, A., Lombos, M., Zittermann, S., Yu, Y., Hill, K., Kong, A., Sharma, D., & Warshawsky, B. (2019). *Listeria monocytogenes* associated with pasteurized chocolate milk, Ontario, Canada. *Emerging Infectious Diseases*, 25(3), 581–584. <https://doi.org/10.3201/eid2503.180742>
- Hao, J., Zhang, J., Zheng, X., & Zhao, D. (2022). Bactericidal efficacy of slightly acidic electrolyzed water (SAEW) against *Listeria monocytogenes* planktonic cells and biofilm on food-contact surfaces. *Food Quality and Safety*, 6, Article fyab038. <https://doi.org/10.1093/fqsafe/fyab038>
- Harada, A. M. M., & Nascimento, M. S. (2021). Efficacy of dry sanitizing methods on *Listeria monocytogenes* biofilms. *Food Control*, 124, Article 107897. <https://doi.org/10.1016/j.foodcont.2021.107897>
- Havelaar, A. H., Kirk, M. D., Torgerson, P. R., Gibb, H. J., Hald, T., Lake, R. J., Praet, N., Bellinger, D. C., de Silva, N. R., Gargouri, N., Speybroeck, N., Cawthorne, A., Mathers, C., Stein, C., Angulo, F. J., Devleesschauwer, B., & World Health Organization Foodborne Disease Burden Epidemiology Reference Group. (2015). World Health Organization Global Estimates and Regional Comparisons of the Burden of Foodborne Disease in 2010. *PLoS Medicine*, 12(12), Article e1001923. <https://doi.org/10.1371/journal.pmed.1001923>
- Heir, E., Møretrø, T., Simensen, A., & Langsrud, S. (2018). *Listeria monocytogenes* strains show large variations in competitive growth in mixed culture biofilms and suspensions with bacteria from food processing environments. *International Journal of Food Microbiology*, 275, 46–55. <https://doi.org/10.1016/j.ijfoodmicro.2018.03.026>
- Henriques, A. R., & Fraqueza, M. J. (2017). Biofilm-forming ability and biocide susceptibility of *Listeria monocytogenes* strains isolated from the ready-to-eat meat-based food products food chain. *LWT - Food Science and Technology*, 81, 180–187. <https://doi.org/10.1016/j.lwt.2017.03.045>
- Hua, Z., Korany, A. M., El-Shinawy, S. H., & Zhu, M. J. (2019). Comparative evaluation of different sanitizers against *Listeria monocytogenes* biofilms on major food-contact surfaces. *Frontiers in Microbiology*, 10, Article 2462. <https://doi.org/10.3389/fmicb.2019.02462>
- Hua, Z., & Zhu, M. J. (2024a). Comprehensive strategies for controlling *Listeria monocytogenes* biofilms on food-contact surfaces. *Comprehensive Reviews in Food Science and Food Safety*, 23(3), Article e13348. <https://doi.org/10.1111/1541-4337.13348>
- Hua, Z., & Zhu, M. J. (2024b). Unlocking the hidden threat: Impacts of surface defects on the efficacy of sanitizers against *Listeria monocytogenes* biofilms on food-contact surfaces in tree fruit packing facilities. *Journal of Food Protection*, 87(2), Article 100213. <https://doi.org/10.1016/j.jfp.2023.100213>
- Ibusquiza, P. S., Herrera, J. J. R., & Cabo, M. L. (2011). Resistance to benzalkonium chloride, peracetic acid and nisin during formation of mature biofilms by *Listeria monocytogenes*. *Food Microbiology*, 28(3), 418–425. <https://doi.org/10.1016/j.fm.2010.09.014>
- Ibusquiza, P. S., Herrera, J. J. R., Vázquez-sánchez, D., & Cabo, M. L. (2012). Adherence kinetics, resistance to benzalkonium chloride and microscopic analysis of mixed biofilms formed by *Listeria monocytogenes* and *Pseudomonas putida*. *Food Control*, 25(1), 202–210. <https://doi.org/10.1016/j.foodcont.2011.10.002>
- Ito, A., Taniuchi, A., May, T., Kawata, K., & Okabe, S. (2009). Increased antibiotic resistance of *Escherichia coli* in mature biofilms. *Applied and Environmental Microbiology*, 75(12), 4093–4100. <https://doi.org/10.1128/AEM.02949-08>
- Jennings, M. C., Minbiole, K. P. C., & Wuest, W. M. (2016). Quaternary ammonium compounds: An antimicrobial mainstay and platform for innovation to address bacterial resistance. *ACS Infectious Diseases*, 1(7), 288–303. <https://doi.org/10.1021/acsinfecdis.5b00047>
- Jiang, X., Yu, T., Liu, L., Li, Y., Zhang, K., Wang, H., & Shi, L. (2017). Examination of quaternary ammonium compound resistance in *Proteus mirabilis* isolated from cooked meat products in China. *Frontiers in Microbiology*, 8, Article 2417. <https://doi.org/10.3389/fmicb.2017.02417>
- Jo, H., West, A. M., Teska, P. J., Oliver, H. F., & Howarter, J. A. (2019). Assessment of early onset surface damage from accelerated disinfection protocol. *Antimicrobial Resistance & Infection Control*, 8(1), Article 24. <https://doi.org/10.1186/s13756-019-0467-9>
- John, J., Joy, W. C., & Jovana, K. (2020). Prevalence of *Listeria* spp. in produce handling and processing facilities in the Pacific Northwest. *Food Microbiology*, 90, Article 103468. <https://doi.org/10.1016/j.fm.2020.103468>
- Kadam, S. R., den Besten, H. M. W., van der Veen, S., Zwietering, M. H., Moezelaar, R., & Abbe, T. (2013). Diversity assessment of *Listeria monocytogenes* biofilm formation: Impact of growth condition, serotype and strain origin. *International Journal*

- of *Food Microbiology*, 165(3), 259–264. <https://doi.org/10.1016/j.jfoodmicro.2013.05.025>
- Kastbjerg, V. G., & Gram, L. (2012). Industrial disinfectants do not select for resistance in *Listeria monocytogenes* following long term exposure. *International Journal of Food Microbiology*, 160(1), 11–15. <https://doi.org/10.1016/j.jfoodmicro.2012.09.009>
- Kim, S. Y., Kim, S. H., & Park, S. H. (2023). Inactivation of food-borne pathogen biofilm cells using a combination treatment with gaseous chlorine dioxide and aerosolized sanitizers. *Journal of Food Protection*, 86(7), Article 100105. <https://doi.org/10.1016/j.jfp.2023.100105>
- Kim, S. Y., & Park, S. (2023). Maturation of pathogenic biofilms induces enhanced resistance against gaseous chlorine dioxide. *Letters in Applied Microbiology*, 76(3), Article ovad016. <https://doi.org/10.1093/lambio/ovad016>
- Klopper, K. B., Bester, E., & Wolfaardt, G. M. (2023). *Listeria monocytogenes* biofilms are planktonic cell factories despite peracetic acid exposure under continuous flow conditions. *Antibiotics*, 12(2), Article 209. <https://doi.org/10.3390/antibiotics12020209>
- Kocot, A. M., & Olszewska, M. A. (2019). Interaction and inactivation of *Listeria* and *Lactobacillus* cells in single and mixed species biofilms exposed to different disinfectants. *Journal of Food Safety*, 39(6), Article e12713. <https://doi.org/10.1111/jfs.12713>
- Kocot, A. M., Wróblewska, B., & Cabo, M. L. (2021). Operational culture conditions determinate benzalkonium chloride resistance in *L. monocytogenes*-*E. coli* dual species biofilms. *International Journal of Food Microbiology*, 360, Article 109441. <https://doi.org/10.1016/j.jfoodmicro.2021.109441>
- Korany, A. M., Hua, Z., Green, T., Hanrahan, I., El-Shinawy, S. H., El-Kholy, A., Hassan, G., & Zhu, M. J. (2018). Efficacy of ozonated water, chlorine, chlorine dioxide, quaternary ammonium compounds and peroxyacetic acid against *Listeria monocytogenes* biofilm on polystyrene surfaces. *Frontiers in Microbiology*, 9, Article 2296. <https://doi.org/10.3389/fmicb.2018.02296>
- Kostaki, M., Chorianopoulos, N., Braxou, E., Nychas, G., & Giaouris, E. (2012). Differential biofilm formation and chemical disinfection resistance of sessile cells of *Listeria monocytogenes* strains under monospecies and dual-species (with *Salmonella enterica*) conditions. *Applied and Environmental Microbiology*, 78, 2586–2595. <https://doi.org/10.1128/AEM.07099-11>
- Kovacevic, J., Wałęcka-Zacharska, E., Kitts, D. D., Ziegler, J., Gilmour, M. W., Wałęcka-Zacharska, E., Reimer, A., & Gilmour, M. W. (2016). Tolerance of *Listeria monocytogenes* to quaternary ammonium sanitizers is mediated by a novel efflux pump encoded by *emrE*. *Applied and Environmental Microbiology*, 82(3), 939–953. <https://doi.org/10.1128/AEM.3741-15>
- Kragh, M. L., Ivanova, M., Dalgaard, P., Leekitcharoenphon, P., & Truelstrup Hansen, L. (2024). Sensitivity of 240 *Listeria monocytogenes* isolates to common industrial biocides is more dependent on the presence of residual organic matter or biofilm than on genetic determinants. *Food Control*, 158, Article 110244. <https://doi.org/10.1016/j.foodcont.2023.110244>
- Kübler, F., Uhlemann, T. H. J., Dill, J., & Steinhilper, R. (2015). Energy efficiency and productivity optimization of industrial cleaning equipment. *Energy Efficiency in Strategy of Sustainable Production*, 805, 265–272. <https://doi.org/10.4028/www.scientific.net/AMM.805.265>
- Kyoui, D., Hirokawa, E., Takahashi, H., Kuda, T., & Kimura, B. (2016). Effect of glucose on *Listeria monocytogenes* biofilm formation, and assessment of the biofilm's sanitation tolerance. *Biofouling*, 32(7), 815–826. <https://doi.org/10.1080/08927014.2016.1198953>
- Lake, F. B., Chen, J., van Overbeek, L. S., Baars, J. J. P., Abee, T., & den Besten, H. M. W. (2024). Biofilm formation and desiccation survival of *Listeria monocytogenes* with microbiota on mushroom processing surfaces and the effect of cleaning and disinfection. *International Journal of Food Microbiology*, 411, Article 110509. <https://doi.org/10.1016/j.jfoodmicro.2023.110509>
- Lake, F. B., van Overbeek, L. S., Baars, J. J. P., Abee, T., & den Besten, H. M. W. (2023). Growth performance of *Listeria monocytogenes* and background microbiota from mushroom processing environments. *International Journal of Food Microbiology*, 395, Article 110183. <https://doi.org/10.1016/j.jfoodmicro.2023.110183>
- Lee, B. H., Cole, S., Badel-Berchoux, S., Guillier, L., Felix, B., Krezdorn, N., Hébraud, M., Bernardi, T., Sultan, I., & Piveteau, P. (2019). Biofilm formation of *Listeria monocytogenes* strains under food processing environments and pan-genome-wide association study. *Frontiers in Microbiology*, 10, Article 2698. <https://doi.org/10.3389/fmicb.2019.02698>
- Lee, B. H., Hébraud, M., & Bernardi, T. (2017). Increased adhesion of *Listeria monocytogenes* strains to abiotic surfaces under cold stress. *Frontiers in Microbiology*, 8, Article 2221. <https://doi.org/10.3389/fmicb.2017.02221>
- Lee, S. B., & Suh, M. C. (2021). Regulatory mechanisms underlying cuticular wax biosynthesis. *Journal of Experimental Botany*, 73(9), 2799–2816. <https://doi.org/10.1093/jxb/erab509>
- Lee, S. H. I., Barancelli, G. V., Corassin, C. H., Rosim, R. E., Coppa, C. F. S. C., & de Oliveira, C. A. F. (2017). Effect of peracetic acid on biofilms formed by *Listeria monocytogenes* strains isolated from a Brazilian cheese processing plant. *Brazilian Journal of Pharmaceutical Sciences*, 53(3), Article e00071. <https://doi.org/10.1590/s2175-97902017000300071>
- Lee, S. H. I., Cappato, L. P., Corassin, C. H., Cruz, A. G., & Oliveira, C. A. F. (2016). Effect of peracetic acid on biofilms formed by *Staphylococcus aureus* and *Listeria monocytogenes* isolated from dairy plants. *Journal of Dairy Science*, 99(3), 2384–2390. <https://doi.org/10.3168/jds.2015-10007>
- Leong, D., Alvarez-Ordóñez, A., & Jordan, K. (2014). Monitoring occurrence and persistence of *Listeria monocytogenes* in foods and food processing environments in the Republic of Ireland. *Frontiers in Microbiology*, 5, Article 436. <https://doi.org/10.3389/fmicb.2014.00436>
- Leong, D., NicAogáin, K., Luque-Sastre, L., McManamon, O., Hunt, K., Alvarez-Ordóñez, A., Scollard, J., Schmalenberger, A., Fanning, S., O'Byrne, C., & Jordan, K. (2017). A 3-year multi-food study of the presence and persistence of *Listeria monocytogenes* in 54 small food businesses in Ireland. *International Journal of Food Microbiology*, 249, 18–26. <https://doi.org/10.1016/j.jfoodmicro.2017.02.015>
- Liberati, A., Altman, D. G., Tetzlaff, J., Mulrow, C., Gøtzsche, P. C., Ioannidis, J. P. A., Clarke, M., Devereaux, P. J., Kleijnen, J., & Moher, D. (2009). The PRISMA statement for reporting systematic reviews and meta-analyses of studies that evaluate healthcare interventions: Explanation and elaboration. *BMJ*, 339, Article b2700. <https://doi.org/10.1136/bmj.b2700>
- Lim, K., Li, W. Y., Dinata, A., & Ho, E. T. (2023). Comparing the antibacterial efficacy and functionality of different commercial alcohol-based sanitizers. *PLoS ONE*, 18(3), Article e0282005. <https://doi.org/10.1371/journal.pone.0282005>

- Lindsay, D., Killington, A., Fouhy, K., Loh, M., & Malakar, P. (2022). The CDC biofilm bioreactor is a suitable method to grow biofilms, and test their sanitiser susceptibilities, in the dairy context. *International Dairy Journal*, 126, Article 105264. <https://doi.org/10.1016/j.idairyj.2021.105264>
- Liu, F., Du, L., Zhao, T., Zhao, P., & Doyle, M. P. (2017). Effects of phenyllactic acid as sanitizing agent for inactivation of *Listeria monocytogenes* biofilms. *Food Control*, 78, 72–78. <https://doi.org/10.1016/j.foodcont.2017.02.050>
- Lourenco, A., Machado, H., & Brito, L. (2011). Biofilms of *Listeria monocytogenes* produced at 12°C either in pure culture or in co-culture with *Pseudomonas aeruginosa* showed reduced susceptibility to sanitizers. *Journal of Food Science*, 76(2), 143–148. <https://doi.org/10.1111/j.1750-3841.2010.02011.x>
- Luo, L., Zhang, Z., Wang, H., Wang, P., Lan, R., Deng, J., Miao, Y., Wang, Y., Wang, Y., Xu, J., Zhang, L., Sun, S., Liu, X., Zhou, Y., Chen, X., Li, Q., & Ye, C. (2017). A 12-month longitudinal study of *Listeria monocytogenes* contamination and persistence in pork retail markets in China. *Food Control*, 76, 66–73. <https://doi.org/10.1016/j.foodcont.2016.12.037>
- Luque-Sastre, L., Fox, E. M., Jordan, K., & Fanning, S. (2018). A comparative study of the susceptibility of *Listeria* species to sanitizer treatments when grown under planktonic and biofilm conditions. *Journal of Food Protection*, 81(9), 1481–1490. <https://doi.org/10.4315/0362-028X.JFP-17-466>
- Mafuna, T., Matle, I., Magwedere, K., Pierneef, R. E., & Reva, O. N. (2021). Whole genome-based characterization of *Listeria monocytogenes* isolates recovered from the food chain in South Africa. *Frontiers in Microbiology*, 12, Article 669287. <https://doi.org/10.3389/fmicb.2021.669287>
- Manville, E., Kaya, E. C., Yucel, U., Boyle, D., & Trinetta, V. (2023). Evaluation of *Listeria monocytogenes* biofilms attachment and formation on different surfaces using a CDC biofilm reactor. *International Journal of Food Microbiology*, 399, Article 110251. <https://doi.org/10.1016/j.ijfoodmicro.2023.110251>
- Mazaheri, T., Cervantes-Huamán, B. R. H., Bermúdez-Capdevila, M., Ripolles-Avila, C., & Rodríguez-Jerez, J. J. (2021). *Listeria monocytogenes* biofilms in the food industry: Is the current hygiene program sufficient to combat the persistence of the pathogen? *Microorganisms*, 9(1), Article 181. <https://doi.org/10.3390/microorganisms9010181>
- McLauchlin, J., Aird, H., Amar, C., Barker, C., Dallman, T., Elviss, N., Jørgensen, F., & Willis, C. (2020). *Listeria monocytogenes* in cooked chicken: Detection of an outbreak in the United Kingdom (2016 to 2017) and analysis of *L. monocytogenes* from unrelated monitoring of foods (2013 to 2017). *Journal of Food Protection*, 83(12), 2041–2052. <https://doi.org/10.4315/JFP-20-188>
- Meireles, A., Ferreira, C., Melo, L., & Simões, M. (2017). Comparative stability and efficacy of selected chlorine-based biocides against *Escherichia coli* in planktonic and biofilm states. *Food Research International*, 102, 511–518. <https://doi.org/10.1016/j.foodres.2017.09.033>
- Meireles, A., Giaouris, E., & Simões, M. (2016). Alternative disinfection methods to chlorine for use in the fresh-cut industry. *Food Research International*, 82, 71–85. <https://doi.org/10.1016/j.foodres.2016.01.021>
- Melero, B., Manso, B., Stessl, B., Hernández, M., Wagner, M., Rovira, J., & Rodríguez-Lázaro, D. (2019). Distribution and persistence of *Listeria monocytogenes* in a heavily contaminated poultry processing facility. *Journal of Food Protection*, 82(9), 1524–1531. <https://doi.org/10.4315/0362-028X.JFP-19-087>
- Mendoza, I. C., Luna, E. O., Pozo, M. D., Vásquez, M. V., Montoya, D. C., Moran, G. C., Romero, L. G., Yépez, X., Salazar, R., Romero-Peña, M., & León, J. C. (2022). Conventional and non-conventional disinfection methods to prevent microbial contamination in minimally processed fruits and vegetables. *LWT - Food Science and Technology*, 165, Article 113714. <https://doi.org/10.1016/j.lwt.2022.113714>
- Merritt, J. H., Kadouri, D. E., & O'Toole, G. A. (2005). Growing and analyzing static biofilms. *Current Protocols in Microbiology*, 00, 00–00. <https://doi.org/10.1002/9780471729259.mc01b01s00>
- Montañez-Izquierdo, V. Y., Salas-Vázquez, D. I., & Rodríguez-Jerez, J. J. (2012). Use of epifluorescence microscopy to assess the effectiveness of phage P100 in controlling *Listeria monocytogenes* biofilms on stainless steel surfaces. *Food Control*, 23(2), 470–477. <https://doi.org/10.1016/j.foodcont.2011.08.016>
- Møretro, T., Fanebust, H., Fagerlund, A., & Langsrud, S. (2019). Whole room disinfection with hydrogen peroxide mist to control *Listeria monocytogenes* in food industry related environments. *International Journal of Food Microbiology*, 292, 118–125. <https://doi.org/10.1016/j.ijfoodmicro.2018.12.015>
- Møretro, T., & Langsrud, S. (2017). Residential bacteria on surfaces in the food industry and their implications for food safety and quality. *Comprehensive Reviews in Food Science and Food Safety*, 16(5), 1022–1041. <https://doi.org/10.1111/1541-4337.12283>
- Møretro, T., Langsrud, S., & Heir, E. (2013). Bacteria on meat abattoir process surfaces after sanitation: Characterisation of survival properties of *Listeria monocytogenes* and the commensal bacterial flora. *Advances in Microbiology*, 3(3), 255–264. <https://doi.org/10.4236/aim.2013.33037>
- Muhterem-Uyar, M., Dalmasso, M., Bolocan, A. S., Hernandez, M., Kapetanakou, A. E., Kuchta, T., Manios, S. G., Melero, B., Minarovičová, J., Nicolau, A. I., Rovira, J., Skandamis, P. N., Jordan, K., Rodríguez-Lázaro, D., Stessl, B., & Wagner, M. (2015). Environmental sampling for *Listeria monocytogenes* control in food processing facilities reveals three contamination scenarios. *Food Control*, 51, 94–107. <https://doi.org/10.1016/j.foodcont.2014.10.042>
- Murugesan, L., Kucerova, Z., Knabel, S. J., & Laborde, L. F. (2015). Predominance and distribution of a persistent *Listeria monocytogenes* clone in a commercial fresh mushroom processing environment. *Journal of Food Protection*, 78(11), 1988–1998. <https://doi.org/10.4315/0362-028X.JFP-15-195>
- Nakamura, H., Takakura, K. I., Sone, Y., Itano, Y., & Nishikawa, Y. (2013). Biofilm formation and resistance to benzalkonium chloride in *Listeria monocytogenes* isolated from a fish processing plant. *Journal of Food Protection*, 76(7), 1179–1186. <https://doi.org/10.4315/0362-028X.JFP-12-225>
- Nguyen Trang, P., Thi, A., Ngoc, T., Masuda, Y., Hohjoh, K. I., & Miyamoto, T. (2023). Biofilm formation from *Listeria monocytogenes* isolated from pangasius fish-processing plants. *Journal of Food Protection*, 86(3), Article 100044. <https://doi.org/10.1016/j.jfp.2023.100044>
- Nilsson, R. E., Ross, T., & Bowman, J. P. (2011). Variability in biofilm production by *Listeria monocytogenes* correlated to strain origin and growth conditions. *International Journal of Food Microbiology*, 150(1), 14–24. <https://doi.org/10.1016/j.ijfoodmicro.2011.07.012>

- Nowak, J., Cruz, C. D., Tempelaars, M., Abee, T., Vliet, A. H. M. V., Fletcher, G. C., Hedderley, D., Palmer, J., & Flint, S. (2017). Persistent *Listeria monocytogenes* strains isolated from mussel production facilities form more biofilm but are not linked to specific genetic markers. *International Journal of Food Microbiology*, 256, 45–53. <https://doi.org/10.1016/j.ijfoodmicro.2017.05.024>
- Nyati, H., Beumer, R., Veen, S. V. D., & Hazeleger, W. (2012). Influence of organic material and biofilms on disinfectant efficacy against *Listeria monocytogenes*. *International Journal of Food Studies*, 1, 76–84. <https://doi.org/10.7455/ijfs/1.1.2012.a8>
- Olanya, O. M., Hoshide, A. K., Ijabadeniyi, O. A., Ukuku, D. O., Mukhopadhyay, S., Niemira, B. A., & Ayeni, O. (2019). Cost estimation of listeriosis (*Listeria monocytogenes*) occurrence in South Africa in 2017 and its food safety implications. *Food Control*, 102, 231–239. <https://doi.org/10.1016/j.foodcont.2019.02.007>
- Ölmez, H., & Temur, S. D. (2010). Effects of different sanitizing treatments on biofilms and attachment of *Escherichia coli* and *Listeria monocytogenes* on green leaf lettuce. *LWT - Food Science and Technology*, 43(6), 964–970. <https://doi.org/10.1016/j.lwt.2010.02.005>
- Oloketuyi, S. F., & Khan, F. (2017). Inhibition strategies of *Listeria monocytogenes* biofilms—Current knowledge and future outlooks. *Journal of Basic Microbiology*, 57(9), 728–743. <https://doi.org/10.1002/jobm.201700071>
- Olszewska, M. A., & Diez-gonzalez, F. (2021). Characterization of binary biofilms of *Listeria monocytogenes* and *Lactobacillus* and their response to chlorine treatment. *Frontiers in Microbiology*, 12, Article 638933. <https://doi.org/10.3389/fmicb.2021.638933>
- Olszewska, M. A., Zhao, T., & Doyle, M. P. (2016). Inactivation and induction of sublethal injury of *Listeria monocytogenes* in biofilm treated with various sanitizers. *Food Control*, 70, 371–379. <https://doi.org/10.1016/j.foodcont.2016.06.015>
- Ortiz, S., López, V., & Martínez-Suárez, J. V. (2014). The influence of subminimal inhibitory concentrations of benzalkonium chloride on biofilm formation by *Listeria monocytogenes*. *International Journal of Food Microbiology*, 189, 106–112. <https://doi.org/10.1016/j.ijfoodmicro.2014.08.007>
- Ortiz, S., López-Alonso, V., Rodríguez, P., & Martínez-Suárez, J. V. (2016). The connection between persistent, disinfectant-resistant *Listeria monocytogenes* strains from two geographically separate Iberian pork processing plants: Evidence from comparative genome analysis. *Applied and Environmental Microbiology*, 82(1), 308–317. <https://doi.org/10.1128/AEM.02824-15>
- O'Toole, G., Kaplan, H. B., & Kolter, R. (2000). Biofilm formation as microbial development. *Annual Review of Microbiology*, 54, 49–79. <https://doi.org/10.1146/annurev.micro.54.1.49>
- Overney, A., Jacques-andré-coquin, J., Ng, P., Carpentier, B., Guillier, L., & Firmesse, O. (2017). Impact of environmental factors on the culturability and viability of *Listeria monocytogenes* under conditions encountered in food processing plants. *International Journal of Food Microbiology*, 244, 74–81. <https://doi.org/10.1016/j.ijfoodmicro.2016.12.012>
- Oxaran, V., Dittmann, K., Lee, S. H. I., Chaul, T., Augusto, C., Oliveira, F. D., Corassin, H., Alves, F., Cristina, E., & Martinis, P. D. (2018). Behavior of foodborne pathogens *Listeria monocytogenes* and *Staphylococcus aureus* in mixed-species biofilms exposed to biocides. *Applied and Environmental Microbiology*, 84(24), Article e02038–18. <https://doi.org/10.1128/AEM.02038-18>
- Paduro, C., Montero, D. A., Chamorro, N., Carreño, L. J., Vidal, M., & Vidal, R. (2020). Ten years of molecular epidemiology surveillance of *Listeria monocytogenes* in Chile 2008–2017. *Food Microbiology*, 85, Article 103280. <https://doi.org/10.1016/j.fm.2019.103280>
- Pagadala, S., Parveen, S., Rippen, T., Luchansky, J. B., Call, J. E., Tamplin, M. L., & Porto-Fett, A. C. S. (2012). Prevalence, characterization and sources of *Listeria monocytogenes* in blue crab (*Callinectes sapidus*) meat and blue crab processing plants. *Food Microbiology*, 31(2), 263–270. <https://doi.org/10.1016/j.fm.2012.03.015>
- Paluszak, Z., Gryń, G., Bauza-Kaszewska, J., Skowron, K. J., Wiktorczyk-Kapischke, N., Korkus, J., Pawlak, M., Szymańska, E., Kraszewska, Z., Buszko, K., & Skowron, K. (2021). Prevalence and antimicrobial susceptibility of *Listeria monocytogenes* strains isolated from a meat processing plant. *Annals of Agricultural and Environmental Medicine*, 28(4), 595–604. <https://doi.org/10.26444/aaem/131799>
- Pang, X., Wong, C., Chung, H., & Yuk, H. (2019). Biofilm formation of *Listeria monocytogenes* and its resistance to quaternary ammonium compounds in a simulated salmon processing environment. *Food Control*, 98, 200–208. <https://doi.org/10.1016/j.foodcont.2018.11.029>
- Pang, X., & Yuk, H. G. (2019). Effects of the colonization sequence of *Listeria monocytogenes* and *Pseudomonas fluorescens* on survival of biofilm cells under food-related stresses and transfer to salmon. *Food Microbiology*, 82, 142–150. <https://doi.org/10.1016/j.fm.2019.02.002>
- Papaioannou, E., Giaouris, E. D., Berillis, P., & Boziaris, I. S. (2018). Dynamics of biofilm formation by *Listeria monocytogenes* on stainless steel under mono-species and mixed-culture simulated fish processing conditions and chemical disinfection challenges. *International Journal of Food Microbiology*, 267, 9–19. <https://doi.org/10.1016/j.ijfoodmicro.2017.12.020>
- Pappelbaum, K., Grif, K., Heller, I., Würzner, R., Hein, I., Ellerbroek, L., & Wagner, M. (2008). Monitoring hygiene on- and at-line is critical for controlling *Listeria monocytogenes* during produce processing. *Journal of Food Protection*, 71(4), 735–741. <https://doi.org/10.4315/0362-028X-71.4.735>
- Parisi, A., Latorre, L., Fraccalvieri, R., Miccolupo, A., Normanno, G., Caruso, M., & Santagada, G. (2013). Occurrence of *Listeria* spp. in dairy plants in Southern Italy and molecular subtyping of isolates using AFLP. *Food Control*, 29(1), 91–97. <https://doi.org/10.1016/j.foodcont.2012.05.036>
- Park, S. H., Cheon, H. L., Park, K. H., Chung, M. S., Choi, S. H., Ryu, S., & Kang, D. H. (2012). Inactivation of biofilm cells of foodborne pathogen by aerosolized sanitizers. *International Journal of Food Microbiology*, 154(3), 130–134. <https://doi.org/10.1016/j.ijfoodmicro.2011.12.018>
- Piercey, M. J., Ells, T. C., Macintosh, A. J., & Truelstrup, L. (2017). Variations in biofilm formation, desiccation resistance and benzalkonium chloride susceptibility among *Listeria monocytogenes* strains isolated in Canada. *International Journal of Food Microbiology*, 257, 254–261. <https://doi.org/10.1016/j.ijfoodmicro.2017.06.025>
- Piercey, M. J., Hingston, P. A., & Truelstrup Hansen, L. (2016). Genes involved in *Listeria monocytogenes* biofilm formation at a simulated food processing plant temperature of 15°C. *International Journal of Food Microbiology*, 223, 63–74. <https://doi.org/10.1016/j.ijfoodmicro.2016.02.009>
- Pietrysiak, E., Smith, S., & Ganjyal, G. M. (2019). Food safety interventions to control *Listeria monocytogenes* in the fresh apple packing industry: A review. *Comprehensive Reviews in Food*

- Science and Food Safety*, 18(6), 1705–1726. <https://doi.org/10.1111/1541-4337.12496>
- Pohl, M. A., Wiedmann, M., & Nightingale, K. K. (2006). Associations among *Listeria monocytogenes* genotypes and distinct clinical manifestations of listeriosis in cattle. *American Journal of Veterinary Research*, 67(4), 616–626. <https://doi.org/10.2460/ajvr.67.4.616>
- Poimenidou, S. V., Chrysadaku, M., Tzakoniati, A., Bikouli, V. C., Nychas, G. J., & Skandamis, P. N. (2016). Variability of *Listeria monocytogenes* strains in biofilm formation on stainless steel and polystyrene materials and resistance to peracetic acid and quaternary ammonium compounds. *International Journal of Food Microbiology*, 237, 164–171. <https://doi.org/10.1016/j.ijfoodmicro.2016.08.029>
- Possas, A., Hernández, M., Esteban-Carbonero, Ó., Valero, A., & Rodríguez-Lázaro, D. (2022). *Listeria monocytogenes* survives better at lower storage temperatures in regular and low-salt soft and cured cheeses. *Food Microbiology*, 104, Article 103979. <https://doi.org/10.1016/j.fm.2022.103979>
- Purkrtová, S., Turonová, H., Pilchová, T., Demnerová, K., & Pazlarová, J. (2010). Resistance of *Listeria monocytogenes* biofilms to disinfectants. *Czech Journal of Food Sciences*, 28(4), 326–332. <https://doi.org/10.17221/153/2010-cjfs>
- Quan, Y., Kim, H.-Y., & Shin, I.-S. (2017). Bactericidal activity of strong acidic hypochlorous water against *Escherichia coli* O157:H7 and *Listeria monocytogenes* in biofilms attached to stainless steel. *Food Science and Biotechnology*, 26(3), 841–846. <https://doi.org/10.1007/s10068-017-0086-2>
- European Commission. (2012). Regulation (EU) No 524/2013 of the European Parliament and of the Council. In O. Radley-Gardner, H. Beale, & R. Zimmermann (Eds.), *Fundamental texts on European private law* (pp. 189–207). Hart Publishing. <https://doi.org/10.5040/9781782258674.0009>
- Rahman, A., & Sahoo, N. (2022). Analysis of sanitizer rotation on the susceptibility, biofilm forming ability and Caco-2 cell adhesion and invasion of *Listeria*. *Pathogens*, 11(9), Article 961. <https://doi.org/10.3390/pathogens11090961>
- Rahman, M. A., Sahoo, N., & Yemmireddy, V. (2022). Analysis of Sanitizer Rotation on the Susceptibility, Biofilm Forming Ability and Caco-2 Cell Adhesion and Invasion of *Listeria*. *Pathogens*, 11(9), 961. <https://doi.org/10.3390/pathogens11090961>
- Ramires, T., Kleinubing, N. R., Iglesias, M. A., Vitola, H. R. S., Nuncio, A. S. P., Kroning, I. S., Moreira, G. M. S. G., Fiorentini, Â. M., & da Silva, W. P. (2021). Genetic diversity, biofilm and virulence characteristics of *Listeria monocytogenes* in salmon sushi. *Food Research International*, 140, Article 109871. <https://doi.org/10.1016/j.foodres.2020.109871>
- Ribeiro, A. C., Almeida, F. A. D., Medeiros, M. M., Miranda, B. R., Pinto, U. M., & Alves, V. F. (2023). *Listeria monocytogenes*: An inconvenient hurdle for the dairy industry. *Dairy*, 4(2), 316–344. <https://doi.org/10.3390/dairy4020022>
- Rippa, A., Bilei, S., Peruzy, M. F., Marrocco, M. G., Leggeri, P., Bossù, T., & Murru, N. (2024). Antimicrobial resistance of *Listeria monocytogenes* strains isolated in food and food-processing environments in Italy. *Antibiotics*, 13(6), Article 525. <https://doi.org/10.3390/antibiotics13060525>
- Rocha, P. R. D., Lomonaco, S., Bottero, M. T., Dalmaso, A., Dondo, A., Grattarola, C., Zuccon, F., Iulini, B., Knabel, S. J., Capucchio, M. T., & Casalone, C. (2013). Ruminant rhombencephalitis-associated *Listeria monocytogenes* strains constitute a genetically homogeneous group related to human outbreak strains. *Applied and Environmental Microbiology*, 79(9), 3059–3066. <https://doi.org/10.1128/AEM.00219-13>
- Rodríguez-Campos, D., Rodríguez-Melcón, C., Alonso-Calleja, C., & Capita, R. (2019). Persistent *Listeria monocytogenes* isolates from a poultry-processing facility form more biofilm but do not have a greater resistance to disinfectants than sporadic strains. *Pathogens*, 8(4), Article 250. <https://doi.org/10.3390/pathogens8040250>
- Rodríguez-López, P., & Cabo, M. L. (2017). Tolerance development in *Listeria monocytogenes*-*Escherichia coli* dual-species biofilms after sublethal exposures to pronase-benzalkonium chloride combined treatments. *Food Microbiology*, 67, 58–66. <https://doi.org/10.1016/j.fm.2017.06.002>
- Rodríguez-López, P., Rodríguez-Herrera, J. J., & Cabo, M. L. (2022). Architectural features and resistance to food-grade disinfectants in *Listeria monocytogenes*-*Pseudomonas* spp. *Frontiers in Microbiology*, 13, Article 917964. <https://doi.org/10.3389/fmicb.2022.917964>
- Rodríguez-López, P., Rodríguez-Herrera, J. J., Vázquez-Sánchez, D., & Cabo, M. L. (2018). Current knowledge on *Listeria monocytogenes* biofilms in Food-Related environments: Incidence, resistance to biocides, ecology and biocontrol. *Foods*, 7(6), Article 85. <https://doi.org/10.3390/foods7060085>
- Rodríguez-Melcón, C., Alonso-Calleja, C., & Capita, R. (2023). Effect of low doses of biocides on the susceptibility of *Listeria monocytogenes* and *Salmonella enterica* to various antibiotics of clinical importance. *Food Control*, 149, Article 109602. <https://doi.org/10.1016/j.foodcont.2023.109602>
- Rodríguez-Melcón, C., Capita, R., Rodríguez-Jerez, J. J., Martínez-Suárez, J. V., & Alonso-Calleja, C. (2019). Effect of low doses of disinfectants on the biofilm-forming ability of *Listeria monocytogenes*. *Foodborne Pathogens and Disease*, 16(4), 262–268. <https://doi.org/10.1089/fpd.2018.2472>
- Russo, P., Hadjilouka, A., Beneduce, L., Capozzi, V., Paramithiotis, S., Drosinos, E. H., & Spano, G. (2018). Effect of different conditions on *Listeria monocytogenes* biofilm formation and removal. *Czech Journal of Food Sciences*, 36(3), 208–214. <https://doi.org/10.17221/199/2017-CJFS>
- Sadekuzzaman, M., Yang, S., Mizan, M. F. R., Kim, H. S., & Ha, S. D. (2017). Effectiveness of a phage cocktail as a biocontrol agent against *L. monocytogenes* biofilms. *Food Control*, 78, 256–263. <https://doi.org/10.1016/j.foodcont.2016.10.056>
- Santos, T., Viala, D., Chambon, C., Esbelin, J., & Hébraud, M. (2019). *Listeria monocytogenes* biofilm adaptation to different temperatures seen through shotgun proteomics. *Frontiers in Nutrition*, 6, Article 89. <https://doi.org/10.3389/fnut.2019.00089>
- Sargeant, J. M., & O'Connor, A. M. (2020). Scoping reviews, systematic reviews, and meta-analysis: Applications in veterinary medicine. *Frontiers in Veterinary Science*, 7, Article 11. <https://doi.org/10.3389/fvets.2020.00011>
- Sekhon, A. S., Unger, P., Singh, A., Yang, Y., & Michael, M. (2022). Impact of gas ultrafine bubbles on the potency of chlorine solutions against *Listeria monocytogenes* biofilms. *Journal of Food Safety*, 42(3), Article e12954. <https://doi.org/10.1111/jfs.12954>
- Silva, D. A. L., Camargo, A. C., Todorov, S. D., & Nero, L. A. (2017). *Listeria* spp. contamination in a butcher shop environment and *Listeria monocytogenes* adhesion ability and sensitivity to food-contact surface sanitizers. *Journal of Food Safety*, 37(2), Article e12313. <https://doi.org/10.1111/jfs.12313>

- Silva, D. A. L., Tavares, R. M., & Nero, L. A. (2020). Interference of sanitizers, NaCl and curing salts on *Listeria monocytogenes* adhesion and subsequent biofilm formation. *Letters in Applied Microbiology*, 71(5), 438–443. <https://doi.org/10.1111/lam.13374>
- Simonetti, T., Peter, K., Chen, Y., Jin, Q., Zhang, G., LaBorde, L. F., & Macarisin, D. (2021). Prevalence and distribution of *Listeria monocytogenes* in three commercial tree fruit packinghouses. *Frontiers in Microbiology*, 12, Article 652708. <https://doi.org/10.3389/fmicb.2021.652708>
- Sinclair, P., Rolon, M. L., Feng, J., Padín-López, A. F., LaBorde, L., & Kovac, J. (2022). Ability of two strains of lactic acid bacteria to inhibit *Listeria monocytogenes* by spot inoculation and in an environmental microbiome context. *Microbiology Spectrum*, 10(4), Article e01018–22. <https://doi.org/10.1128/spectrum.01018-22>
- Skowron, K., Hulisz, K., Gryń, G., Olszewska, H., Wiktorczyk, N., & Paluszak, Z. (2018). Comparison of selected disinfectants efficiency against *Listeria monocytogenes* biofilm formed on various surfaces. *International Microbiology*, 21(1), 23–33. <https://doi.org/10.1007/s10123-018-0002-5>
- Skowron, K., Walecka-Zacharska, E., Grudlewska, K., Gajewski, P., Wiktorczyk, N., Wietlicka-Piszczyk, M., Dudek, A., Skowron, K. J., & Gospodarek-Komkowska, E. (2019). Disinfectant susceptibility of biofilm formed by *Listeria monocytogenes* under selected environmental conditions. *Microorganisms*, 7(9), Article 280. <https://doi.org/10.3390/microorganisms7090280>
- Solihin, J., Waturangi, D. E., & Purwadaria, T. (2021). Induction of amylase and protease as antibiofilm agents by starch, casein, and yeast extract in *Arthrobacter* sp. CW01. *BMC Microbiology*, 21, Article 232. <https://doi.org/10.1186/s12866-021-02294-z>
- Spanu, C., Scarano, C., Ibba, M., Spanu, V., & De Santis, E. P. L. (2015). Occurrence and traceability of *Listeria monocytogenes* strains isolated from sheep's milk cheese-making plants environment. *Food Control*, 47, 318–325. <https://doi.org/10.1016/j.foodcont.2014.07.027>
- Srey, S., Park, S. Y., Jahid, I. K., & Ha, S. D. (2014). Reduction effect of the selected chemical and physical treatments to reduce *L. monocytogenes* biofilms formed on lettuce and cabbage. *Food Research International*, 62, 484–491. <https://doi.org/10.1016/j.foodres.2014.03.067>
- Srey, S., Park, S. Y., Jahid, I. K., Oh, S.-R., Han, N., Zhang, C.-Y., Kim, S.-H., Cho, J.-I., & Ha, S.-D. (2014). Evaluation of the removal and destruction effect of a chlorine and thiamine dilaurylsulfate combined treatment on *L. monocytogenes* biofilm. *Foodborne Pathogens and Disease*, 11(8), 658–663. <https://doi.org/10.1089/fpd.2013.1734>
- Stessl, B., Ruppitsch, W., & Wagner, M. (2022). *Listeria monocytogenes* post-outbreak management—When could a food production be considered under control again? *International Journal of Food Microbiology*, 379, Article 109844. <https://doi.org/10.1016/j.ijfoodmicro.2022.109844>
- Stoller, A., Stevens, M. J. A., Stephan, R., & Guldemann, C. (2019). Characteristics of *Listeria monocytogenes* strains persisting in a meat processing facility over a 4-year period. *Pathogens*, 8(1), Article 32. <https://doi.org/10.3390/pathogens8010032>
- Sullivan, G., & Wiedmann, M. (2020). Detection and prevalence of *Listeria* in U.S. Produce packinghouses and fresh-cut facilities. *Journal of Food Protection*, 83(10), 1656–1666. <https://doi.org/10.4315/JFP-20-094>
- Svoboda, A., Shaw, A., Dzubak, J., Mendonca, A., Wilson, L., & Nair, A. (2016). Effectiveness of broad-spectrum chemical produce sanitizers against foodborne pathogens as in vitro planktonic cells and on the surface of whole cantaloupes and watermelons. *Journal of Food Protection*, 79(4), 524–530. <https://doi.org/10.4315/0362-028X.JFP-15-490>
- Swedish Institute for Standards. (2001). *EN 13697: Chemical disinfectants and antiseptics - Quantitative non-porous surface test for the evaluation of bactericidal and/or fungicidal activity of chemical disinfectants used in food, industrial, domestic and institutional areas - Test method and requirements without mechanical action (phase2/step2)*. Author.
- Tamburro, M., Ripabelli, G., Vitullo, M., Dallman, T. J., Pontello, M., Amar, C. F. L., & Sammarco, M. L. (2015). Gene expression in *Listeria monocytogenes* exposed to sublethal concentration of benzalkonium chloride. *Comparative Immunology, Microbiology and Infectious Diseases*, 40, 31–39. <https://doi.org/10.1016/j.cimid.2015.03.004>
- Taylor, A. J., & Stasiewicz, M. J. (2019). Persistent and sporadic *Listeria monocytogenes* strains do not differ when growing at 37°C, in planktonic state, under different food associated stresses or energy sources. *BMC Microbiology*, 19(1), Article 257. <https://doi.org/10.1186/s12866-019-1631-3>
- Tchatchouang, C. D. K., Fri, J., De Santi, M., Brandi, G., Schiavano, G. F., Amagliani, G., & Ateba, C. N. (2020). Listeriosis outbreak in South Africa: A comparative analysis with previously reported cases worldwide. *Microorganisms*, 8(1), Article 135. <https://doi.org/10.3390/microorganisms8010135>
- Thomassen, G. M. B., Reiche, T., Hjørungnes, M., & Mehli, L. (2023). High disinfectant tolerance in *Pseudomonas* spp. biofilm aids the survival of *Listeria monocytogenes*. *Microorganisms*, 11(6), Article 1414. <https://doi.org/10.3390/microorganisms11061414>
- Tolvanen, R., Lundén, J., Hörman, A., & Korkeala, H. (2009). Pilot-scale continuous ultrasonic cleaning equipment reduces *Listeria monocytogenes* levels on conveyor belts. *Journal of Food Protection*, 72(2), 408–411. <https://doi.org/10.4315/0362-028X-72.2.408>
- Townsend, A., Strawn, L. K., Chapman, B. J., & Dunn, L. L. (2021). A systematic review of *Listeria* species prevalence, persistence, and diversity throughout the fresh produce supply chain. *Foods*, 10(6), Article 1427. <https://doi.org/10.3390/foods10061427>
- Truchado, P., Gil, M. I., & Allende, A. (2021). Peroxyacetic acid and chlorine dioxide unlike chlorine induce viable but non-culturable (VBNC) stage of *Listeria monocytogenes* and *Escherichia coli* O157:H7 in wash water. *Food Microbiology*, 100, Article 103866. <https://doi.org/10.1016/j.fm.2021.103866>
- Truchado, P., Gil, M. I., Larrosa, M., & Allende, A. (2020). Detection and quantification methods for viable but non-culturable (VBNC) cells in process wash water of fresh-cut produce: Industrial validation. *Frontiers in Microbiology*, 11, Article 673. <https://doi.org/10.3389/fmicb.2020.00673>
- Truscott, W. (2017). *Researching the right disinfectant for your facility*. Association for the Health Care Environment. <https://www.ahe.org/researching-right-disinfectant-your-facility>
- Tuyschaever, T., Raes, K., & Sampers, I. (2023). *Listeria monocytogenes* in food businesses: From persistence strategies to intervention/prevention strategies—A review. *Comprehensive Reviews in Food Science and Food Safety*, 22(5), 3910–3950. <https://doi.org/10.1111/1541-4337.13219>

- Unger, P., Sekhon, A. S., Sharma, S., Lampien, A., & Michael, M. (2023). Impact of gas ultrafine bubbles on the efficacy of antimicrobials for eliminating fresh and aged *Listeria monocytogenes* biofilms on dairy processing surfaces. *Journal of Food Safety*, 43(5), Article e13057. <https://doi.org/10.1111/jfs.13057>
- Vaid, R., Linton, R. H., & Morgan, M. T. (2010). Comparison of inactivation of *Listeria monocytogenes* within a biofilm matrix using chlorine dioxide gas, aqueous chlorine dioxide and sodium hypochlorite treatments. *Food Microbiology*, 27(8), 979–984. <https://doi.org/10.1016/j.fm.2010.05.024>
- Van der Veen, S., & Abee, T. (2011). Mixed species biofilms of *Listeria monocytogenes* and *Lactobacillus plantarum* show enhanced resistance to benzalkonium chloride and peracetic acid. *International Journal of Food Microbiology*, 144(3), 421–431. <https://doi.org/10.1016/j.ijfoodmicro.2010.10.029>
- Vassar, M., Yerokhin, V., Sinnett, P. M., Weiher, M., Muckelrath, H., Carr, B., Varney, L., & Cook, G. (2017). Database selection in systematic reviews: An insight through clinical neurology. *Health Information and Libraries Journal*, 34(2), 156–164. <https://doi.org/10.1111/hir.12176>
- Vázquez-Sánchez, D., Galvão, J. A., Ambrosio, C. M. S., Gloria, E. M., & Oetterer, M. (2018). Single and binary applications of essential oils effectively control *Listeria monocytogenes* biofilms. *Industrial Crops and Products*, 121, 452–460. <https://doi.org/10.1016/j.indcrop.2018.05.045>
- Vázquez-Sánchez, D., Galvão, J. A., & Oetterer, M. (2017). Contamination sources, serogroups, biofilm-forming ability and biocide resistance of *Listeria monocytogenes* persistent in tilapia-processing facilities. *Journal of Food Science and Technology*, 54(12), 3867–3879. <https://doi.org/10.1007/s13197-017-2843-x>
- Véghová, A., Minarovičová, J., Koreňová, J., Drahovská, H., & Kacliková, E. (2017). Prevalence and tracing of persistent *Listeria monocytogenes* strains in meat processing facility production chain. *Journal of Food Safety*, 37(2), Article e12315. <https://doi.org/10.1111/jfs.12315>
- Viswanath, P., Murugesan, L., Knabel, S. J., Verghese, B., Chikthimmah, N., & Laborde, L. F. (2013). Incidence of *Listeria monocytogenes* and *Listeria* spp. in a small-scale mushroom production facility. *Journal of Food Protection*, 76(4), 608–615. <https://doi.org/10.4315/0362-028X.JFP-12-292>
- Walecka-Zacharska, E., Kosek-Paszkowska, K., Bania, J., Karpíšková, R., & Stefaniak, T. (2013). Salt stress-induced invasiveness of major *Listeria monocytogenes* serotypes. *Letters in Applied Microbiology*, 56(3), 216–221. <https://doi.org/10.1111/lam.12036>
- Wang, J., Ray, A. J., Hammons, S. R., & Oliver, H. F. (2015). Persistent and transient *Listeria monocytogenes* strains from retail deli environments vary in their ability to adhere and form biofilms and rarely have inLA premature stop codons. *Foodborne Pathogens and Disease*, 12(2), 151–158. <https://doi.org/10.1089/fpd.2014.1837>
- Wang, R. (2019). Biofilms and meat safety: A mini-review. *Journal of Food Protection*, 82(1), 120–127. <https://doi.org/10.4315/0362-028X.JFP-18-311>
- Wang, W., Zhou, X., Suo, Y., Deng, X., Cheng, M., Shi, C., & Shi, X. (2017). Prevalence, serotype diversity, biofilm-forming ability and eradication of *Listeria monocytogenes* isolated from diverse foods in Shanghai, China. *Food Control*, 73, 1068–1073. <https://doi.org/10.1016/j.foodcont.2016.10.025>
- Winkelströter, L. K., & Martinis, E. C. P. D. (2015). Different methods to quantify *Listeria monocytogenes* biofilms cells showed different profile in their viability. *Brazilian Journal of Microbiology*, 235, 231–235. <https://doi.org/10.1590/S1517-838220131071>
- Wu, D., Forghani, F., Daliri, E. B. M., Li, J., Liao, X., Liu, D., Ye, X., Chen, S., & Ding, T. (2020). Microbial response to some nonthermal physical technologies. *Trends in Food Science and Technology*, 95, 107–117. <https://doi.org/10.1016/j.tifs.2019.11.012>
- Yamakawa, T., Tomita, K., & Sawai, J. (2018). Characteristics of biofilms formed by co-culture of *Listeria monocytogenes* with *Pseudomonas aeruginosa* at low temperatures and their sensitivity to antibacterial substances test bacteria. *Biocontrol Science*, 23(3), 107–119. <https://doi.org/10.4265/bio.23.107>
- Yun, H. S., Kim, Y., Oh, S., Jeon, W. M., Frank, J. F., & Kim, S. H. (2012). Susceptibility of *Listeria monocytogenes* biofilms and planktonic cultures to hydrogen peroxide in food processing environments. *Bioscience, Biotechnology and Biochemistry*, 76(11), 2008–2013. <https://doi.org/10.1271/bbb.120238>
- Zhang, H., Que, F., Xu, B., Sun, L., Zhu, Y., Chen, W., Ye, Y., Dong, Q., Liu, H., & Zhang, X. (2021). Identification of *Listeria monocytogenes* contamination in a ready-to-eat meat processing plant in China. *Frontiers in Microbiology*, 12, Article 628204. <https://doi.org/10.3389/fmicb.2021.628204>
- Zhang, H., Wang, J., Chang, Z., Liu, X., Chen, W., Yu, Y., Wang, X., Dong, Q., Ye, Y., & Zhang, X. (2021). *Listeria monocytogenes* contamination characteristics in two ready-to-eat meat plants from 2019 to 2020 in Shanghai. *Frontiers in Microbiology*, 12, Article 729114. <https://doi.org/10.3389/fmicb.2021.729114>
- Zhang, X., Liu, Z., Wang, P., Wang, Q., Yang, S., & Feng, H. (2013). Fine mapping of BrWax1, a gene controlling cuticular wax biosynthesis in Chinese cabbage (*Brassica rapa* L. ssp. *pekinensis*). *Molecular Breeding*, 32(4), 867–874. <https://doi.org/10.1007/s1032-013-9914-0>
- Zhao, Q., Xu, Z., Liu, X., Zhu, H., Li, Z., Liu, Y., Yang, J., & Dong, Q. (2024). Formation and recovery of *Listeria monocytogenes* in viable but nonculturable state under different temperatures combined with low nutrition and high NaCl concentration. *Food Research International*, 192, Article 114774. <https://doi.org/10.1016/j.foodres.2024.114774>
- Zoz, F., Grandvalet, C., Lang, E., Iaconelli, C., Gervais, P., Firmesse, O., Guyot, S., & Beney, L. (2017). *Listeria monocytogenes* ability to survive desiccation: Influence of serotype, origin, virulence, and genotype. *International Journal of Food Microbiology*, 248, 82–89. <https://doi.org/10.1016/j.ijfoodmicro.2017.02.010>

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

How to cite this article: Arthur, M., Afari, E. L., Alexa, E.-A., Zhu, M.-J., Gaffney, M. T., Celayeta, J. M. F., & Burgess, C. M. (2025). Recent advances in examining the factors influencing the efficacy of biocides against *Listeria monocytogenes* biofilms in the food industry: A systematic review. *Comprehensive Reviews in Food Science and Food Safety*, 24, e70083. <https://doi.org/10.1111/1541-4337.70083>